Quantum Information Theory, Algorithms, Channels, and Continuous Variables

Foundations of Quantum Information

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CHAPTER 1

Classical Probability and Information

1.1 Finite probability spaces and stochastic transformations

Definition 1.1 (Discrete state)**.** A finite classical system X with alphabet $\mathcal{X}$ is specified by a probability vector $p_X \in \mathbb{R}^{|\mathcal{X}|}$ with $p_X(x) \geq 0$ and $\sum_x p_X(x) = 1$. Its support is $\operatorname{supp} p_X = \{x : p_X(x) > 0\}$, and its rank is $|\operatorname{supp} p_X|$.

For a joint distribution p_{XY}, the marginal $p_X(x) = \sum_y p_{XY}(x, y)$ and the conditional distribution $p(y \mid x) = p_{XY}(x, y)/p_X(x)$ are defined whenever $p_X(x) > 0$. Values on zero-probability events are arbitrary. Independence means $p_{XY} = p_X \otimes p_Y$.

Definition 1.2 (Classical channel)**.** A channel $W : X \to Y$ is a column-stochastic matrix $W(y \mid x) \geq 0$ with $\sum_y W(y \mid x) = 1$. It acts as $(Wp)(y) = \sum_x W(y \mid x)p(x)$. For two channels V and W, $(VW)(z \mid x) = \sum_y V(z \mid y)W(y \mid x)$.

The binary symmetric channel $\mathrm{BSC}(q)$ flips a bit with probability q. Independent successive flips of probabilities r, s produce

$$q = r(1 - s) + (1 - r)s = r + s - 2rs. \tag{1.1}$$

1.2 Entropy, conditional entropy, and information

All discrete entropies in this book use logarithms to base two and $0 \log 0 = 0$.

Definition 1.3 (Entropy and information)**.** For finite random variables X, Y, Z, define

$$H(X) = -\sum_x p(x) \log p(x), \tag{1.2}$$

$$H(Y \mid X) = -\sum_{x,y} p(x, y) \log p(y \mid x), \tag{1.3}$$

$$I(X : Y) = H(X) + H(Y) - H(X, Y), \tag{1.4}$$

$$I(X : Y \mid Z) = H(X \mid Z) + H(Y \mid Z) - H(X, Y \mid Z). \tag{1.5}$$

The binary entropy function is $h_2(t) = -t \log t - (1 - t) \log(1 - t)$.

Proposition 1.4 (Chain rules)**.** *For arbitrary finite variables,*

$$H(X, Y) = H(X) + H(Y \mid X), \tag{1.6}$$

$$I(X : Y, Z) = I(X : Y) + I(X : Z \mid Y). \tag{1.7}$$

Proof. Insert $p(x, y) = p(x)p(y \mid x)$ into $-\sum_{x,y} p(x, y) \log p(x, y)$ and separate the logarithm. The second identity follows by expanding each mutual information into differences of conditional entropies. □

Theorem 1.5 (Subadditivity and strong subadditivity)**.**

$$H(X,Y) \le H(X) + H(Y), \qquad H(X,Y,Z) + H(Y) \le H(X,Y) + H(Y,Z). \tag{1.8}$$

Equivalently, $I(X:Y) \ge 0$ *and* $I(X:Z \mid Y) \ge 0$.

Remark 1.6. Mutual information itself is not generally subadditive under grouping of its inputs. For independent uniform bits A, B and $C = A \oplus B$, $I(A:C) = I(B:C) = 0$ but $I(AB:C) = 1$.

1.3 Relative entropy and data processing

Definition 1.7 (Kullback–Leibler divergence)**.** For probability vectors p, q, define

$$D(p\|q) = \sum_{x : p(x) > 0} p(x) \log \frac{p(x)}{q(x)}, \tag{1.9}$$

with $D(p\|q) = +\infty$ when $\operatorname{supp} p \not\subseteq \operatorname{supp} q$.

Theorem 1.8 (Classical data processing)**.** *For every stochastic channel* W*,* $D(p\|q) \ge D(Wp\|Wq)$*. Consequently,* $I(X:Y) \ge I(X:Z)$ *whenever* $X \to Y \to Z$ *is a Markov chain.*

Proof. The log-sum inequality gives

$$\sum_x a_x \log \frac{a_x}{b_x} \ge \left(\sum_x a_x \right) \log \frac{\sum_x a_x}{\sum_x b_x}. \tag{1.10}$$

Apply it for each output y with $a_x = W(y \mid x)p(x)$ and $b_x = W(y \mid x)q(x)$, then sum over y. Since $I(X:Y) = D(p_{XY}\|p_X p_Y)$, the mutual-information claim follows by applying a channel locally to Y. □

1.4 Source coding and channel capacity

Definition 1.9 (Prefix code)**.** A binary prefix code assigns to each source symbol x a bit string of length $\ell(x)$ such that no codeword is a prefix of another. Its lengths obey Kraft's inequality $\sum_x 2^{-\ell(x)} \le 1$.

Theorem 1.10 (Noiseless source-coding bounds)**.** *If* $L = \sum_x p(x)\ell(x)$ *is the expected length of a binary prefix code, then*

$$H(X) \le L < H(X) + 1 \tag{1.11}$$

for a suitable code; no uniquely decodable code can improve the lower bound.

Proof. Set $K = \sum_x 2^{-\ell(x)} \le 1$ and $q(x) = 2^{-\ell(x)}/K$. Nonnegativity of $D(p\|q)$ yields $L \ge H(X) - \log K \ge H(X)$. Choosing $\ell(x) = \lceil -\log p(x) \rceil$ and invoking Kraft's inequality gives the upper bound. □

Definition 1.11 (Rate of a discrete memoryless channel)**.** For n independent uses of a channel W, an (n, M, ε) code has M equiprobable messages, an encoder into $\mathcal{X}^n$, and a decoder from $\mathcal{Y}^n$ with average error at most ε. A rate R is achievable if there are codes with $\liminf_{n\to\infty} n^{-1} \log M_n \ge R$ and $\varepsilon_n \to 0$.

Theorem 1.12 (Classical channel coding)**.** *The capacity in bits per use of a discrete memoryless channel is*

$$C(W) = \max_{p_X} I(X : Y), \qquad p_{XY}(x, y) = p_X(x)W(y \mid x). \tag{1.12}$$

For a binary symmetric channel, $C(\mathrm{BSC}(q)) = 1 - h_2(q)$.

Proposition 1.13 (Product-channel additivity)**.** *For memoryless channels* W *and* V*,* $C(W \otimes V) = C(W) + C(V)$.

Proof. For possibly correlated inputs X_1X_2, subadditivity and conditional independence give $I(X_1X_2 : Y_1Y_2) = H(Y_1Y_2) - H(Y_1Y_2 \mid X_1X_2) \leq H(Y_1) + H(Y_2) - H(Y_1 \mid X_1) - H(Y_2 \mid X_2)$. Product capacity-achieving inputs give the reverse inequality. □

Example 1.14 (Parity conditioning)**.** Let A be uniform on $\{0, 1, 2, 3\}$ and $B = A \bmod 2$. Then $H(A) = 2$, $H(B) = 1$, $H(A \mid B) = 1$ and $I(A : B) = 1$ bit. For uniform independent bits A, B and $C = A \oplus B$, $I(A : C) = 0$ while $I(A : C \mid B) = 1$.

1.5 Entropy identities from first principles

1.5.1 Conditional distributions and the chain rule

Let p_{XY} be a probability mass function on finite alphabets $\mathcal{X} \times \mathcal{Y}$. The marginal $p_X(x) = \sum_y p_{XY}(x, y)$ is nonnegative and normalized. For $p_X(x) > 0$, the conditional distribution is $p_{Y|X}(y|x) = p_{XY}(x, y)/p_X(x)$. Its value for a zero-probability conditioning event does not affect any weighted conditional entropy. Consequently

$$H(Y|X) = \sum_x p_X(x) H(p_{Y|X=x}) \tag{1.13}$$

$$= -\sum_{x,y} p_{XY}(x, y) \log p_{Y|X}(y|x). \tag{1.14}$$

Substituting $p_{XY}(x, y) = p_X(x) p_{Y|X}(y|x)$ directly gives

$$H(X, Y) = -\sum_{x,y} p_{XY}(x, y) \log[p_X(x) p_{Y|X}(y|x)] \tag{1.15}$$

$$= H(X) + H(Y|X). \tag{1.16}$$

Iterating establishes the full chain rule

$$H(X_1, \ldots, X_n) = \sum_{j=1}^{n} H(X_j|X_1, \ldots, X_{j-1}). \tag{1.17}$$

Thus the order of decomposition changes intermediate conditional terms but not the sum. In contrast, the naive identity $H(X, Y) = H(X) + H(Y)$ holds exactly when X and Y are independent.

Proposition 1.15 (Conditional information as a divergence)**.** *For finite random variables X, Y, Z,*

$$I(X : Y|Z) = \sum_{z: p_Z(z) > 0} p_Z(z) D(p_{XY|z} \| p_{X|z} p_{Y|z}) \geq 0. \tag{1.18}$$

Equality holds exactly when X and Y are conditionally independent given Z for all z of positive probability.

Proof. Expand the Kullback–Leibler divergence of each conditional joint distribution, then collect logarithms:

$$I(X : Y|Z) = \sum_{x,y,z} p(x, y, z) \log \frac{p(x, y|z)}{p(x|z) p(y|z)}$$

$$= H(X|Z) - H(X|Y, Z).$$

Gibbs' inequality ensures each conditional divergence is nonnegative. If the weighted sum vanishes, every term with $p_Z(z) > 0$ must vanish, forcing the conditional product factorization. □

Example 1.16 (A parity variable: information synergy)**.** Let X, Y be independent unbiased bits and $Z = X \oplus Y$. Because the joint distribution of (X, Z) is uniform on the four possible pairs, $I(X : Z) = 0$; symmetrically $I(Y : Z) = 0$. Nevertheless Z is a deterministic function

of the pair, so

$$I(XY : Z) = H(Z) - H(Z|XY) = 1 - 0 = 1. \tag{1.19}$$

In fact $I(X : Z|Y) = 1$. Knowing Y transforms the otherwise independent variable X into a perfect predictor of Z. This explicitly refutes the inequality $I(XY : Z) \leq I(X : Z)+I(Y : Z)$. A different counterexample rules out the opposite inequality: taking $X = Y = Z$ to be a uniform bit gives $I(XY : Z) = 1 < 2 = I(X : Z) + I(Y : Z)$.

1.5.2 Gibbs' inequality by the log-sum argument

Lemma 1.17 (Log-sum inequality)**.** *For nonnegative families (a_i) and (b_i), with positive totals $a = \sum_i a_i$ and $b = \sum_i b_i$,*

$$\sum_i a_i \log \frac{a_i}{b_i} \geq a \log \frac{a}{b}. \tag{1.20}$$

The conventional interpretation of $a_i \log(a_i/0)$ is $+\infty$ when $a_i > 0$.

Proof. Set $\alpha_i = a_i/a$ and $\beta_i = b_i/b$. Convexity of $-\log t$ gives Jensen's inequality

$$-\sum_i \alpha_i \log \frac{\beta_i}{\alpha_i} \geq -\log \left(\sum_i \alpha_i \frac{\beta_i}{\alpha_i} \right) = 0. \tag{1.21}$$

Multiplication by a and rearrangement yield the claim. Equality occurs when a_i/b_i is constant on the common support. □

Corollary 1.18 (Positivity of relative entropy)**.** *$D(p\|q) \geq 0$, with equality if and only if $p = q$ on the alphabet.*

Proof. Apply log-sum with $a_i = p_i$, $b_i = q_i$ and $a = b = 1$. □

Proposition 1.19 (Data processing for stochastic maps)**.** *Let $W(z|x)$ be a stochastic kernel and set $(Wp)(z) = \sum_x W(z|x)p(x)$. Then*

$$D(Wp\|Wq) \leq D(p\|q). \tag{1.22}$$

Proof. At fixed z, use the log-sum inequality on $a_x = W(z|x)p(x)$ and $b_x = W(z|x)q(x)$:

$$(Wp)(z) \log \frac{(Wp)(z)}{(Wq)(z)} \leq \sum_x W(z|x)p(x) \log \frac{p(x)}{q(x)}. \tag{1.23}$$

Sum in z and use $\sum_z W(z|x) = 1$. This proof also explains why a stochastic map cannot increase distinguishability as measured by relative entropy. □

Example 1.20 (Binary symmetric channel composition)**.** A bit transmitted through independent binary symmetric channels with crossover probabilities r and s is flipped exactly when one of the two channels flips it. The effective crossover probability is

$$t = r(1 - s) + (1 - r)s = r + s - 2rs. \tag{1.24}$$

The same identity follows by multiplying the transition matrices

$$\begin{pmatrix} 1-s & s \\ s & 1-s \end{pmatrix} \begin{pmatrix} 1-r & r \\ r & 1-r \end{pmatrix} = \begin{pmatrix} 1-t & t \\ t & 1-t \end{pmatrix}. \tag{1.25}$$

For $r = s = 0.1$ the resulting crossover is 0.18, not 0.2; two flips cancel. Since $h_2(t)$ increases on $[0, 1/2]$, this cascade cannot improve the channel capacity relative to the first noisy link when $s \leq 1/2$.

1.6 Markov structures, inference, and coding

1.6.1 Markov chains and equality in data processing

The Markov condition $X \to Y \to Z$ means $p(z|x, y) = p(z|y)$. Equivalently $I(X : Z|Y) = 0$. Compute $I(X : YZ)$ in two orders:

$$I(X : YZ) = I(X : Y) + I(X : Z|Y) = I(X : Y), \tag{1.26}$$

$$I(X : YZ) = I(X : Z) + I(X : Y|Z). \tag{1.27}$$

Subtracting yields the exact identity

$$I(X : Y) - I(X : Z) = I(X : Y|Z) \geq 0. \tag{1.28}$$

Therefore $I(X : Y) = I(X : Z)$ precisely when X and Y are conditionally independent given Z as well. This is a condition for no information loss by the channel $Y \mapsto Z$ relative to the specified input ensemble, not for invertibility on every possible distribution.

Example 1.21 (Noisy binary observation)**.** Let $X \sim \text{Bernoulli}(1/2)$ and let $Y = X \oplus N_r$, $Z = Y \oplus N_s$, with independent noises. Then $H(X) = H(Y) = H(Z) = 1$ and

$$I(X : Y) = 1 - h_2(r), \qquad I(X : Z) = 1 - h_2(r + s - 2rs). \tag{1.29}$$

For $0 < r, s < 1/2$ the second value is strictly smaller. At $s = 0$, equality holds, as the second channel is noiseless; at $s = 1/2$, Z is independent of X.

1.6.2 Source codes and Kraft's inequality

A binary prefix code assigns strings $c(x)$ such that no codeword is the prefix of another. Writing $\ell(x)$ for the codeword lengths,

$$\sum_x 2^{-\ell(x)} \leq 1. \tag{1.30}$$

To prove this, regard each binary string $c(x)$ as a node of a complete binary tree at depth $\ell(x)$. Prefix-free codewords select pairwise disjoint subtrees, with normalized leaf measures $2^{-\ell(x)}$. Their total measure is at most one.

Theorem 1.22 (Entropy lower bound and a constructive upper bound)**.** *For a random variable X with finite entropy $H(X)$, the expected length L of any binary prefix code satisfies $L \geq H(X)$. There exists one with $L < H(X) + 1$.*

Proof. Given lengths ℓ_x, put $K = \sum_x 2^{-\ell_x} \le 1$ and $q_x = 2^{-\ell_x}/K$. Then

$$\begin{aligned} D(p\|q) &= \sum_x p_x \log p_x + \sum_x p_x \ell_x + \log K \\ &= -H(X) + L + \log K \ge 0. \end{aligned}$$

It follows that $L \ge H(X) - \log K \ge H(X)$. For the upper bound choose $\ell_x = \lceil -\log p_x \rceil$, for which $\sum 2^{-\ell_x} \le 1$. Kraft's converse constructs a prefix code with these lengths and $L < \sum_x p_x(-\log p_x + 1) = H(X) + 1$. □

Example 1.23 (A four-symbol distribution)**.** For probabilities $(1/2, 1/4, 1/8, 1/8)$ choose the prefix words 0, 10, 110, 111. The expected length is

$$L = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3) = \frac{7}{4}. \tag{1.31}$$

The distribution is dyadic, so its entropy also equals $7/4$ bits and the lower bound is attained.

1.6.3 The two canonical binary channels

Let $X \in \{0,1\}$ have input bias $p = \mathbb{P}(X = 0)$. For BSC(ϵ) the output $Y = X \oplus N_\epsilon$. Since $H(Y|X) = h_2(\epsilon)$ and $H(Y) \le 1$,

$$I(X : Y) \le 1 - h_2(\epsilon). \tag{1.32}$$

Uniform input makes Y uniform and attains equality; hence $C(\mathrm{BSC}(\epsilon)) = 1 - h_2(\epsilon)$. For the binary erasure channel BEC(ϵ), the output alphabet is $\{0, 1, \perp\}$ with an erasure flag that is known to the receiver. Using $H(X|Y) = \epsilon H(X)$ gives

$$I(X : Y) = (1 - \epsilon)H(X) \le 1 - \epsilon, \qquad C(\mathrm{BEC}(\epsilon)) = 1 - \epsilon. \tag{1.33}$$

At identical numerical $\epsilon \in (0, 1/2)$ the erasure channel has the larger capacity because its bad outputs are flagged, whereas a bit flip is not flagged. Comparing the two capacities beyond the common crossover range does not have the same operational interpretation.

Example 1.24 (Explicit mutual information calculation)**.** Consider BEC(1/4) with $p = 1/4$. Then

$$p_Y(0) = 3/16, \quad p_Y(1) = 9/16, \quad p_Y(\perp) = 1/4. \tag{1.34}$$

The conditional input entropy given a non-erasure is zero and given an erasure is $h_2(1/4)$. Hence

$$I(X : Y) = H(X) - H(X|Y) = \frac{3}{4} h_2(1/4), \tag{1.35}$$

strictly below the capacity $3/4$, which requires a uniform input.

1.7 Entropy, one-shot guessing, and operational error

The Shannon entropy $H(X)$ measures expected uncertainty, whereas the min-entropy $H_{\min}(X) = -\log \max_x p_x$ controls the probability of a correct single guess. Since $p_x \le p_{\max}$, averaging $-\log p_x \ge -\log p_{\max}$ yields $H(X) \ge H_{\min}(X)$. One must not replace Shannon entropy by min-entropy in a coding theorem: they govern different operational settings.

Example 1.25 (A diverging separation)**.** Set $p(0) = 1/2$ and $p(j) = 1/(2N)$ for $1 \le j \le N$. Then

$$H_{\min}(X) = 1, \qquad H(X) = 1 + \frac{1}{2}\log N. \tag{1.36}$$

The maximum guessing probability remains one half no matter how large the support becomes, although the Shannon entropy diverges.

Theorem 1.26 (Fano's inequality)**.** *Let $\hat{X} = \hat{X}(Y)$ be an estimator for a random variable X taking values in an alphabet of size $d \ge 2$. Put $P_e = \mathbb{P}(\hat{X} \ne X)$. Then*

$$H(X|Y) \le h_2(P_e) + P_e \log(d-1). \tag{1.37}$$

Proof. Define the error indicator $E = \mathbf{1}\{\hat{X} \ne X\}$. Because E is determined by (X, Y),

$$\begin{aligned} H(X|Y) &= H(E, X|Y) = H(E|Y) + H(X|E, Y) \\ &\le h_2(P_e) + P_e \log(d-1). \end{aligned}$$

When $E = 0$, X is determined by Y; when $E = 1$, at most $d-1$ values are possible after observing Y. This proves the bound. □

Fano's inequality converts a conditional entropy constraint into a decoding-error statement. Combined with $I(X : Y) = H(X) - H(X|Y)$, it is the usual converse tool for finite-message transmission.

1.8 Further coding theorems and finite-block calculations

1.8.1 Deriving the asymptotic equipartition property in the i.i.d. case

Let $X_1, \dots, X_n$ be independent copies of X on a finite alphabet with distribution p. Define the random information content $Z_i = -\log p(X_i)$, which is integrable because only positive-probability outcomes matter. Then

$$-\frac{1}{n}\log p(X^n) = \frac{1}{n}\sum_{i=1}^{n} Z_i. \tag{1.38}$$

By the weak law of large numbers this converges in probability to $\mathbb{E}Z_i = H(X)$. For $\varepsilon > 0$ define the typical set

$$\mathcal{T}_{n,\varepsilon} = \left\{ x^n : 2^{-n(H+\varepsilon)} \le p(x^n) \le 2^{-n(H-\varepsilon)} \right\}. \tag{1.39}$$

It has probability approaching one. Since probabilities in the set are between the displayed bounds, its cardinality satisfies

$$\mathbb{P}(X^n \in \mathcal{T}_{n,\varepsilon})\, 2^{n(H-\varepsilon)} \le |\mathcal{T}_{n,\varepsilon}| \le 2^{n(H+\varepsilon)}. \tag{1.40}$$

The right inequality follows by summing the lower probability bound over the set; the left follows by summing the upper probability bound. Encoding only typical sequences by their indices, with an exceptional code for atypical sequences, yields the intuition for lossless compression near $nH(X)$ bits.

Example 1.27 (A Bernoulli source block)**.** Let $X_i \sim \text{Bernoulli}(1/4)$. A binary sequence with k ones has probability $(1/4)^k(3/4)^{n-k}$, and there are $\binom{n}{k}$ such sequences. A typical block has $k \approx n/4$, so its per-letter self-information is approximately

$$-\frac{1}{4}\log\frac{1}{4} - \frac{3}{4}\log\frac{3}{4} = h_2(1/4) \approx 0.81128. \tag{1.41}$$

The source contains only about $0.81128n$ bits of typical information although its raw representation uses n bits. This is an asymptotic statement, not a license to assign a shorter unique binary word to every one of the 2^n possible sequences.

1.8.2 Mutual information decompositions

Mutual information has several equivalent forms:

$$I(X:Y) = H(X) + H(Y) - H(XY) = H(X) - H(X|Y) = D(p_{XY}\|p_X p_Y). \tag{1.42}$$

The divergence form proves $I(X:Y) \geq 0$, with equality exactly at independence. Conditional mutual information is similarly a nonnegative weighted divergence and satisfies chain rules such as

$$I(X:YZ) = I(X:Y) + I(X:Z|Y). \tag{1.43}$$

For three variables, the signed interaction information

$$I(X:Y:Z) = I(X:Y) - I(X:Y|Z) \tag{1.44}$$

can be positive or negative. In the parity example it equals -1, showing that higher-order correlations cannot be characterized by a naive sum of pairwise mutual informations.

Example 1.28 (Positive interaction information)**.** For $X = Y = Z$ an unbiased bit, $I(X:Y) = 1$ and $I(X:Y|Z) = 0$, so $I(X:Y:Z) = 1$. The positive and negative examples establish that this three-variable quantity does not satisfy an unconditional positivity law.

1.9 A strong converse calculation for a binary noisy observation

For the binary symmetric channel BSC(p) with uniform input, Bayes' theorem yields the posterior

$$\mathbb{P}(X = y \mid Y = y) = 1 - p, \qquad \mathbb{P}(X \neq y \mid Y = y) = p. \tag{1.45}$$

The maximum-likelihood decoder guesses $\hat{X} = Y$ when $p < 1/2$ and has error probability p. Its conditional entropy is $H(X|Y) = h_2(p)$, and Fano's inequality gives $H(X|Y) \leq h_2(P_e)$ for $d = 2$, which is saturated by this symmetric optimum. The one-use error and one-use mutual information should not be confused with the vanishing-error block coding theorem: block codes make reliable transmission possible at any rate below $1 - h_2(p)$ using many independent channel uses.

1.9.1 Capacity additivity for product classical channels

Let $W_1 : Y_1|X_1$ and $W_2 : Y_2|X_2$ be independent channels. Even with correlated inputs X_1, X_2, output independence conditioned on both inputs implies

$$\begin{aligned} I(X_1X_2 : Y_1Y_2) &= H(Y_1Y_2) - H(Y_1Y_2|X_1X_2) && (1.46)\\ &\leq H(Y_1) + H(Y_2) - H(Y_1|X_1) - H(Y_2|X_2) && (1.47)\\ &= I(X_1 : Y_1) + I(X_2 : Y_2) && (1.48)\\ &\leq C(W_1) + C(W_2). && (1.49) \end{aligned}$$

Choose independent capacity-achieving marginals for X_1 and X_2. Then Y_1, Y_2 are independent as well, and both inequalities become equalities. Therefore

$$C(W_1 \otimes W_2) = C(W_1) + C(W_2). \tag{1.50}$$

This clean single-letter additivity of classical memoryless channels contrasts with possible nonadditivity of one-use Holevo information for quantum channels, which motivates regularization later in the text.

CHAPTER 2

Hilbert Spaces and Operator Geometry

2.1 Quantum state spaces

Definition 2.1 (Finite-dimensional Hilbert space)**.** A system A is associated with a complex inner-product space $\mathcal{H}_A \simeq \mathbb{C}^{d_A}$. A normalized pure state is a ray $\{e^{i\theta}\,|\psi\rangle : \theta \in \mathbb{R}\}$ of vectors obeying $\langle\psi|\psi\rangle = 1$.

An orthonormal basis $\{|j\rangle\}_{j=0}^{d-1}$ satisfies $\langle i|j\rangle = \delta_{ij}$ and $\sum_j |j\rangle\langle j| = I$. Thus $|\psi\rangle = \sum_j \psi_j\,|j\rangle$, with $\sum_j |\psi_j|^2 = 1$. A pure state in dimension d requires $2d-2$ independent real parameters after normalization and removal of global phase.

Proposition 2.2 (Cauchy–Schwarz inequality)**.** *For $u, v \in \mathcal{H}$, $|\langle u|v\rangle|^2 \le \langle u|u\rangle\,\langle v|v\rangle$, with equality precisely when u, v are linearly dependent.*

Proof. For $v \neq 0$, the nonnegative norm of $u - \langle v|u\rangle\ v/\left\|v\right\|^2$ expands to $\left\|u\right\|^2 - |\langle v|u\rangle|^2 / \left\|v\right\|^2$. □

2.2 Linear operators and norms

Write $\mathcal{L}(A, B)$ for linear maps $\mathcal{H}_A \to \mathcal{H}_B$. The adjoint $T^\dagger$ satisfies $\langle Tu|v\rangle = \left\langle u\middle|T^\dagger v\right\rangle$. Hermitian maps obey $H = H^\dagger$; unitary maps obey $U^\dagger U = UU^\dagger = I$; isometries obey $V^\dagger V = I$ without requiring surjectivity.

Definition 2.3 (Operator norms)**.** For $M \in \mathcal{L}(A, B)$ with singular values s_i, define

$$\left\|M\right\|_1 = \operatorname{Tr}\sqrt{M^\dagger M} = \sum_i s_i, \quad \left\|M\right\|_2 = \sqrt{\operatorname{Tr}(M^\dagger M)}, \quad \left\|M\right\|_\infty = \max_i s_i. \tag{2.1}$$

The Hilbert–Schmidt inner product is $\langle A, B\rangle_{\mathrm{HS}} = \operatorname{Tr}(A^\dagger B)$.

Proposition 2.4 (Trace norm variational principle)**.** *For a square matrix M, $\left\|M\right\|_1 = \max_{U\ \text{unitary}} |\operatorname{Tr}(UM)|$. More generally, for Hermitian M, $\left\|M\right\|_1 = \max_{-I \le K \le I} \operatorname{Tr}(KM)$.*

Proof. Take a singular-value decomposition $M = V\Sigma W^\dagger$. Then

$$|\operatorname{Tr}(UM)| = \left|\operatorname{Tr}(W^\dagger U V \Sigma)\right| \le \sum_i s_i. \tag{2.2}$$

Equality is achieved with $U = WV^\dagger$. For Hermitian M, choose K to be its spectral sign operator. □

Proposition 2.5 (Unitary invariance)**.** *For unitary U, V, $\left\|UMV\right\|_p = \left\|M\right\|_p$ for $p = 1, 2, \infty$ and $\operatorname{Tr}(UMU^\dagger) = \operatorname{Tr} M$.*

2.3 Tensor products, partial trace, and operator bases

For systems A, B, the composite space is $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$, of dimension $d_A d_B$. The partial trace $\mathrm{Tr}_B : \mathcal{L}(AB) \to \mathcal{L}(A)$ is the unique linear map satisfying

$$\mathrm{Tr}\big[X_A \,\mathrm{Tr}_B(M_{AB})\big] = \mathrm{Tr}\big[(X_A \otimes I_B) M_{AB}\big], \qquad \forall X_A. \tag{2.3}$$

Equivalently, $\mathrm{Tr}_B(M) = \sum_j (I_A \otimes \langle j|) M (I_A \otimes |j\rangle)$.

Definition 2.6 (Pauli operator basis)**.** For $d = 2$, let

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.4}$$

Together with I, these form an orthogonal basis of $\mathcal{L}(\mathbb{C}^2)$ with $\mathrm{Tr}(PQ) = 2\delta_{PQ}$. The n-qubit Pauli strings form an orthogonal basis of $\mathcal{L}((\mathbb{C}^2)^{\otimes n})$.

Every n-qubit matrix has the unique expansion $M = 2^{-n} \sum_{P \in \{I,X,Y,Z\}^{\otimes n}} \mathrm{Tr}(PM) P$. In particular, $|0\rangle\langle 1| = (X + iY)/2$ and $\|X^{\otimes n}\|_1 = 2^n$.

2.4 Spectral calculus and matrix exponentials

If $A = A^\dagger = \sum_i \lambda_i |v_i\rangle\langle v_i|$, the functional calculus defines $f(A) = \sum_i f(\lambda_i) |v_i\rangle\langle v_i|$. A Hamiltonian H generates the unitary $U(t) = e^{-itH}$ (with $\hbar = 1$), and $U(t)U(s) = U(t+s)$ for time-independent H.

Proposition 2.7 (Commutator estimate)**.** *If $[A, B] = 0$, then $e^{A+B} = e^A e^B$. In general,*

$$e^{tA} e^{tB} = e^{t(A+B) + \frac{t^2}{2}[A,B] + O(t^3)}, \tag{2.5}$$

where the expansion is interpreted locally for bounded finite-dimensional operators.

2.5 Matrix decompositions

Every $M \in \mathbb{C}^{m \times n}$ admits a singular-value decomposition $M = U \Sigma V^\dagger$. The nonzero eigenvalues of $MM^\dagger$ and $M^\dagger M$ coincide; their square roots are the singular values of M. This fact will also establish the Schmidt decomposition.

2.6 Coordinate descriptions and physical equivalence

A normalized vector $|\psi\rangle \in \mathbb{C}^d$ has $2d$ real coordinates, one normalization constraint, and an irrelevant global phase. Thus a generic pure physical state has $2d-2$ real degrees of freedom; its natural parameter space is complex projective space $\mathbb{CP}^{d-1}$. A density operator, being Hermitian of trace one, has d^2-1 independent real coordinates before imposing positivity. Positivity cuts out a compact convex body with nonempty interior rather than an affine subspace.

Example 2.8 (Qubit and two-qubit parameter counts)**.** For a qubit, a pure state has $2(2)-2=2$ real parameters and can be written

$$|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{i\phi}\sin(\theta/2)\,|1\rangle\,, \quad 0 \le \theta \le \pi. \tag{2.6}$$

A general qubit density matrix has $2^2-1=3$ real parameters. On two qubits, a pure state has $2\cdot 4-2=6$ real parameters; a general mixed state has $4^2-1=15$.

2.6.1 Inner products in non-real bases

Complex conjugation in the bra matters. In an orthonormal qutrit basis $|0\rangle, |1\rangle, |2\rangle$, take

$$|u\rangle = \frac{1}{2}|0\rangle - \frac{i}{2}|1\rangle + \frac{1}{\sqrt{2}}|2\rangle\,, \qquad |v\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{2}|1\rangle + \frac{1}{2}|2\rangle\,. \tag{2.7}$$

Both are normalized. The inner product is

$$\langle u|v\rangle = \frac{1}{2\sqrt{2}} + \frac{i}{4} + \frac{1}{2\sqrt{2}} = \frac{1}{\sqrt{2}} + \frac{i}{4}. \tag{2.8}$$

By contrast, omitting conjugation of the coefficient $-i/2$ produces an incorrect sign and violates the Hermitian inner-product convention. The transition probability is $|\langle u|v\rangle|^2 = 9/16$.

2.7 Operator inequalities and norms

2.7.1 Positive operators and the spectral calculus

A Hermitian matrix $A = A^\dagger$ satisfies $A \succeq 0$ exactly when $\langle\psi|A|\psi\rangle \ge 0$ for all $|\psi\rangle$. By the spectral theorem $A = \sum_i \lambda_i |u_i\rangle\langle u_i|$ with real eigenvalues, positivity is equivalent to $\lambda_i \ge 0$ for every i. For a function f defined on its spectrum, $f(A) = \sum_i f(\lambda_i)\,|u_i\rangle\langle u_i|$. In particular, when $A \succeq 0$, the positive square root $A^{1/2}$ is unambiguous and

$$\operatorname{Tr} A = \sum_i \lambda_i, \qquad \|A\|_1 = \operatorname{Tr}\sqrt{A^\dagger A}, \qquad \|A\|_\infty = \max_i s_i(A). \tag{2.9}$$

A finite-dimensional operator satisfies $\|A\|_\infty \le \|A\|_2 \le \|A\|_1 \le \sqrt{\operatorname{rank} A}\,\|A\|_2$, where $\|A\|_2 = (\operatorname{Tr} A^\dagger A)^{1/2}$ and $s_i(A)$ are its singular values.

Lemma 2.9 (Variational formula for Hermitian trace norm)**.** *For Hermitian D, one has*

$$\frac{1}{2}(\|D\|_1 + \operatorname{Tr} D) = \max_{0 \preceq M \preceq I} \operatorname{Tr}(MD). \tag{2.10}$$

The maximum is achieved by $M = \Pi_+(D)$, the spectral projector on the strictly positive eigenspace.

Proof. Write $D = D_+ - D_-$ with $D_\pm \succeq 0$ and orthogonal supports. For an effect $0 \preceq M \preceq I$,

$$\operatorname{Tr} MD = \operatorname{Tr} MD_+ - \operatorname{Tr} MD_- \leq \operatorname{Tr} D_+. \tag{2.11}$$

The upper bound is achieved by $\Pi_+(D)$, and $\operatorname{Tr} D_+ = (\|D\|_1 + \operatorname{Tr} D)/2$. □

Example 2.10 (The Pauli trace norm)**.** The tensor power $X^{\otimes n}$ is Hermitian and unitary, so its singular values are all one. It acts on a space of dimension 2^n, giving

$$\|X^{\otimes n}\|_1 = 2^n, \qquad \|X^{\otimes n}\|_\infty = 1, \qquad \|X^{\otimes n}\|_2 = 2^{n/2}. \tag{2.12}$$

This example separates the three norms explicitly.

2.7.2 Trace identities and operator expansions

The Pauli matrices obey

$$\sigma_j\sigma_k = \delta_{jk}I + i\sum_l \epsilon_{jkl}\sigma_l, \qquad \operatorname{Tr}(\sigma_j\sigma_k) = 2\delta_{jk}. \tag{2.13}$$

Every 2×2 operator therefore admits the unique Pauli expansion

$$A = \frac{1}{2}\sum_{\mu=0}^{3} \operatorname{Tr}(\sigma_\mu A)\sigma_\mu, \quad \sigma_0 = I. \tag{2.14}$$

For two qubits the sixteen strings $\sigma_\mu \otimes \sigma_\nu$ form an orthogonal operator basis; hence

$$\rho_{AB} = \frac{1}{4}\sum_{\mu,\nu=0}^{3} \operatorname{Tr}[(\sigma_\mu \otimes \sigma_\nu)\rho_{AB}]\sigma_\mu \otimes \sigma_\nu. \tag{2.15}$$

The coefficient for $(\mu, \nu) = (0, 0)$ equals one and ensures trace normalization. Local Bloch vectors are given by coefficients with exactly one identity factor, while coefficients with $\mu, \nu \geq 1$ describe two-body correlators.

Example 2.11 (Off-diagonal rank-one operators)**.** Using $X = |0\rangle\langle 1| + |1\rangle\langle 0|$ and $Y = -i\,|0\rangle\langle 1| + i\,|1\rangle\langle 0|$ gives

$$|0\rangle\langle 1| = \frac{X + iY}{2}, \qquad |1\rangle\langle 0| = \frac{X - iY}{2}. \tag{2.16}$$

Multiplying the right-hand side of the first identity gives $(X + iY)/2 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$; no phase ambiguity remains.

2.8 Tensor-product operations and reduced states

For product basis vectors the tensor product follows the bilinear rule $(\sum_i a_i\,|i\rangle) \otimes (\sum_j b_j\,|j\rangle) = \sum_{i,j} a_i b_j\,|ij\rangle$. In matrix notation $A \otimes B$ is a block matrix whose (i, j) block is $A_{ij}B$. The

multiplicativity identities

$$(A \otimes B)(C \otimes D) = AC \otimes BD, \qquad \operatorname{Tr}(A \otimes B) = \operatorname{Tr} A \operatorname{Tr} B \tag{2.17}$$

hold whenever the matrix products are defined.

2.8.1 Partial trace by matrix units

Define $\operatorname{Tr}_B : \mathcal{L}(A \otimes B) \to \mathcal{L}(A)$ through

$$\operatorname{Tr}_B(|i\rangle\langle k| \otimes |j\rangle\langle \ell|) = \delta_{j\ell}\, |i\rangle\langle k|\,. \tag{2.18}$$

Equivalently, $\operatorname{Tr}[(M_A \otimes I_B) R_{AB}] = \operatorname{Tr}[M_A \operatorname{Tr}_B R_{AB}]$ for every M_A. This identity determines Tr_B uniquely and makes it basis-independent.

Example 2.12 (Three-body reduction)**.** Let $|\Psi\rangle = (|000\rangle + |111\rangle)/\sqrt{2}$. Expanding $|\Psi\rangle\langle\Psi|$ gives four matrix units, with cross terms containing $|0\rangle\langle 1|$ or $|1\rangle\langle 0|$ on the third subsystem. These vanish upon tracing out the third qubit. Therefore

$$\operatorname{Tr}_C |\Psi\rangle\langle\Psi| = \frac{1}{2}\,|00\rangle\langle 00| + \frac{1}{2}\,|11\rangle\langle 11|\,. \tag{2.19}$$

The reduced state is a separable mixed state, despite the pure global state being entangled across every one-versus-two bipartition.

Proposition 2.13 (Purity and rank of a product)**.** *For positive normalized operators ρ_A, σ_B,*

$$\operatorname{rank}(\rho_A \otimes \sigma_B) = \operatorname{rank}\rho_A \operatorname{rank}\sigma_B, \qquad \operatorname{Tr}[(\rho_A \otimes \sigma_B)^2] = \operatorname{Tr}\rho_A^2 \operatorname{Tr}\sigma_B^2. \tag{2.20}$$

Proof. Diagonalize both factors. Their tensor-product eigenvalues are pairwise products $\lambda_i \mu_j$, so ranks multiply; summing squares of the eigenvalues factorizes. □

2.9 Matrix exponentials, generators, and unitary paths

For a Hermitian Hamiltonian H, $U(t) = e^{-itH}$ is unitary because $U(t)^\dagger = e^{itH} = U(t)^{-1}$. If $H = \sum_j E_j\, |E_j\rangle\langle E_j|$, then $U(t) = \sum_j e^{-itE_j}\, |E_j\rangle\langle E_j|$. The global phase associated with $H \mapsto H + cI$ has no effect on unconditioned probabilities but does affect controlled evolution by introducing a relative phase on the control register.

Lemma 2.14 (Exponential of an involution)**.** *If $P = P^\dagger$ and $P^2 = I$, then*

$$e^{-i\theta P} = \cos\theta\, I - i \sin\theta\, P. \tag{2.21}$$

Proof. Separate the exponential Taylor series into even and odd powers and use $P^{2k} = I$, $P^{2k+1} = P$. □

Example 2.15 (Pauli rotations)**.** $R_z(\theta) = e^{-i\theta Z/2} = \operatorname{diag}(e^{-i\theta/2}, e^{i\theta/2})$. The Hadamard identity $HZH = X$ gives

$$R_x(\theta) = H R_z(\theta) H. \tag{2.22}$$

Moreover $R_z(\pi/2)$ agrees with $S = \operatorname{diag}(1, i)$ up to global phase $e^{-i\pi/4}$; $R_z(\pi/4)$ similarly gives the T gate up to phase.

Remark 2.16 (Noncommutativity matters). In general $e^{A+B} \neq e^A e^B$. Equality follows if $[A, B] = 0$; otherwise the leading discrepancy is proportional to $[A, B]$. More precisely, the Baker–Campbell–Hausdorff expansion begins $\log(e^A e^B) = A + B + \frac{1}{2}[A, B] + \cdots$ where the series is valid. This distinction underlies product-formula Hamiltonian simulation.

2.10 Spectral inequalities used throughout quantum information

Lemma 2.17 (Cauchy–Schwarz for the trace inner product). *For matrices A, B of the same dimensions,*

$$\left|\operatorname{Tr}(A^\dagger B)\right|^2 \leq \operatorname{Tr}(A^\dagger A)\operatorname{Tr}(B^\dagger B). \tag{2.23}$$

Proof. If $B = 0$ the statement is immediate. Otherwise positivity of

$$0 \leq \operatorname{Tr}[(A - tB)^\dagger (A - tB)] \tag{2.24}$$

for every complex t allows us to choose $t = \operatorname{Tr}(B^\dagger A)/\operatorname{Tr}(B^\dagger B)$, yielding the stated inequality after rearrangement. □

Proposition 2.18 (Hölder duality for trace and operator norms). *For a matrix X,*

$$\|X\|_1 = \max_{\|Y\|_\infty \leq 1} \left|\operatorname{Tr}(Y^\dagger X)\right|. \tag{2.25}$$

Proof. Let $X = U\Sigma V^\dagger$ be a singular-value decomposition. For an arbitrary contraction Y the singular-value inequality gives $\left|\operatorname{Tr}(Y^\dagger X)\right| \leq \sum_j s_j(X)$. Equality holds for $Y = UV^\dagger$ on the support of X, extended by zero on the remaining space. □

The trace norm is operationally important because $\frac{1}{2}\|\rho - \sigma\|_1$ is the maximum classical total-variation distance between distributions obtained by applying a POVM to ρ and σ.

Derivation of the last statement. For a measurement $\{M_y\}$, define the Hermitian difference $D = \rho - \sigma$ and the set of outcomes $S = \{y : \operatorname{Tr}(M_y D) \geq 0\}$. The total-variation distance equals

$$\frac{1}{2}\sum_y |\operatorname{Tr}(M_y D)| = \sum_{y \in S} \operatorname{Tr}(M_y D) = \operatorname{Tr}(ED), \tag{2.26}$$

where $E = \sum_{y \in S} M_y$ is an effect. The variational formula for trace norm bounds this by $\frac{1}{2}\|D\|_1$. Measuring the two-outcome spectral POVM $(\Pi_+(D), I - \Pi_+(D))$ attains equality. □

2.11 Spectral projection and compression examples

Example 2.19 (Qutrit projectors and matrix entries). Let $|\psi\rangle = (|0\rangle + i\,|1\rangle + |2\rangle)/\sqrt{3}$ on $\mathbb{C}^3$. Its projector is

$$|\psi\rangle\langle\psi| = \frac{1}{3}\begin{pmatrix} 1 & -i & 1 \\ i & 1 & i \\ 1 & -i & 1 \end{pmatrix}. \tag{2.27}$$

This matrix is Hermitian, its trace is one, and squaring reproduces the same matrix. The off-diagonal entries encode relative phase information that is lost if only computational-basis probabilities are retained.

Example 2.20 (The partial trace of a mixed operator)**.** Let $R_{AB} = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|10\rangle\langle 10|$. Both terms have B in $|0\rangle$, so

$$\mathrm{Tr}_B\, R_{AB} = \frac{I}{2}, \qquad \mathrm{Tr}_A\, R_{AB} = |0\rangle\langle 0|\,. \tag{2.28}$$

The global density matrix factorizes as $I_A/2 \otimes |0\rangle\langle 0|_B$. It is mixed on A and pure on B, with no correlation between the systems.

2.12 The geometry of expectation values

For a qubit $\rho = (I + \boldsymbol{r}\cdot\boldsymbol{\sigma})/2$ and a Hermitian observable $A = a_0 I + \boldsymbol{a}\cdot\boldsymbol{\sigma}$, the expectation is

$$\mathrm{Tr}(A\rho) = a_0 + \boldsymbol{a}\cdot\boldsymbol{r}. \tag{2.29}$$

Its two eigenvalues are $a_0 \pm \|\boldsymbol{a}\|$, while the expectation can attain any value in this interval by varying the Bloch vector over the unit ball. A projective measurement in the direction $\boldsymbol{n}$ gives outcome ± 1 with probabilities $(1 \pm \boldsymbol{n}\cdot\boldsymbol{r})/2$.

Example 2.21 (Rotated measurement axis)**.** For $|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{i\phi}\sin(\theta/2)\,|1\rangle$, the Bloch vector is

$$\boldsymbol{r} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta). \tag{2.30}$$

Thus $p_Z(+) = \cos^2(\theta/2)$ and $p_X(+) = \frac{1}{2}(1 + \sin\theta\cos\phi)$. The Z probability contains no information about the relative phase ϕ, whereas the X and Y measurements are sensitive to it.

CHAPTER 3

Density Operators and Entanglement

3.1 States and convex geometry

Definition 3.1 (Density operator)**.** A quantum state on A is an operator ρ_A such that $\rho_A = \rho_A^\dagger$, $\rho_A \succeq 0$, and $\operatorname{Tr}\rho_A = 1$. Denote the convex set of such operators by $\mathcal{D}(A)$. A state is pure if and only if $\rho = |\psi\rangle\langle\psi|$; equivalently $\operatorname{Tr}(\rho^2) = 1$.

An ensemble $\{p_j, |\psi_j\rangle\}$ induces $\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$. Different ensembles can induce the same ρ: $\frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}(|+\rangle\langle +| + |-\rangle\langle -|) = I/2$.

Definition 3.2 (Bloch representation)**.** Every qubit state is uniquely written as $\rho = (I + \boldsymbol{r}\cdot\boldsymbol{\sigma})/2$, with $\boldsymbol{r} \in \mathbb{R}^3$ and $\|\boldsymbol{r}\|_2 \le 1$. Its eigenvalues are $(1 \pm \|\boldsymbol{r}\|_2)/2$, and

$$\operatorname{Tr}\rho^2 = \frac{1 + \|\boldsymbol{r}\|_2^2}{2}. \tag{3.1}$$

3.2 Schmidt decomposition and purification

Theorem 3.3 (Schmidt decomposition)**.** *For every pure state $|\psi\rangle_{AB}$, there exist orthonormal families $\{|u_j\rangle\}_A, \{|v_j\rangle\}_B$ and positive coefficients λ_j such that*

$$|\psi\rangle_{AB} = \sum_{j=1}^{r} \sqrt{\lambda_j}\, |u_j\rangle_A |v_j\rangle_B, \qquad \lambda_j > 0, \quad \sum_j \lambda_j = 1. \tag{3.2}$$

The nonzero spectra of $\rho_A = \operatorname{Tr}_B |\psi\rangle\langle\psi|$ and ρ_B are both $\{\lambda_j\}$.

Proof. Expand $|\psi\rangle = \sum_{ij} M_{ij} |i\rangle_A |j\rangle_B$ and perform an SVD $M = U\Sigma V^\dagger$. Rotating both local bases yields the Schmidt form. Partial tracing gives $\rho_A = \sum_j \lambda_j |u_j\rangle\langle u_j|$ and similarly for B. □

Corollary 3.4 (Pure-state product criterion)**.** *A bipartite pure state is a product state if and only if its Schmidt rank is one, or equivalently $\operatorname{Tr}\rho_A^2 = 1$.*

Definition 3.5 (Purification)**.** A purification of ρ_A is a vector $|\Psi\rangle_{AE}$ satisfying $\operatorname{Tr}_E |\Psi\rangle\langle\Psi| = \rho_A$. For $\rho_A = \sum_j \lambda_j |u_j\rangle\langle u_j|$, a canonical purification is $|\Psi\rangle = \sum_j \sqrt{\lambda_j}\, |u_j\rangle_A |j\rangle_E$.

Any two purifications are related by an isometry on sufficiently large purifying systems (by a unitary when the purifying spaces have equal dimensions).

Example 3.6 (Purification of a nonorthogonal mixture)**.** For $\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|+\rangle\langle +|$, $\rho = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{smallmatrix}\right)$ and its eigenvalues are $\lambda_\pm = (2 \pm \sqrt{2})/4$. Thus $\operatorname{Tr}\rho^2 = 3/4$, $S(\rho) = h_2((2+\sqrt{2})/4)$, and $\sqrt{\lambda_+}\,|e_+\rangle|0\rangle + \sqrt{\lambda_-}\,|e_-\rangle|1\rangle$ is a purification.

3.3 Mixed-state entanglement and witnesses

Definition 3.7 (Separable states)**.** A bipartite density operator is separable if $\rho_{AB} = \sum_j p_j \rho_A^{(j)} \otimes \rho_B^{(j)}$ for a finite probability distribution p_j. Otherwise it is entangled.

Definition 3.8 (Entanglement witness)**.** A Hermitian W is an entanglement witness if $\mathrm{Tr}(W\sigma) \geq 0$ for every separable state σ, while $\mathrm{Tr}(W\rho) < 0$ for at least one entangled ρ.

For the Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, the operator $W = \frac{1}{2}I - |\Phi^+\rangle\langle\Phi^+|$ is a witness. Its expectation on $\rho_p = p\,|\Phi^+\rangle\langle\Phi^+| + (1-p)I/4$ is $(1-3p)/4$, certifying entanglement for $p > 1/3$.

Theorem 3.9 (Partial-transpose criterion)**.** *If ρ_{AB} is separable, then $\rho_{AB}^{T_B} \succeq 0$. For systems of dimensions 2×2 or 2×3, the converse also holds.*

Remark 3.10. For higher dimensions, positivity under partial transpose is only a necessary condition for separability; PPT-entangled states exist.

3.4 Entropy of entanglement and correlation constraints

Definition 3.11 (Von Neumann entropy)**.** $S(\rho) = -\mathrm{Tr}(\rho \log \rho)$. For pure $|\psi\rangle_{AB}$ with Schmidt coefficients $\{\lambda_j\}$, its entanglement entropy is $E(|\psi\rangle) = S(\rho_A) = S(\rho_B) = H(\{\lambda_j\})$.

For any pure tripartite state ABC, entropy duality gives $S(A) = S(BC)$, $S(B) = S(AC)$, and $S(C) = S(AB)$. Consequently,

$$I(A:B) + I(A:C) = 2S(A). \tag{3.3}$$

This identity quantifies an information-theoretic monogamy constraint without asserting that every conceivable entanglement measure obeys the same equality.

Example 3.12 (Tripartite GHZ state)**.** For $|\mathrm{GHZ}\rangle = (|000\rangle + |111\rangle)/\sqrt{2}$, each single-qubit marginal is maximally mixed, but $\rho_{AB} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$ is separable. In particular, pairwise classical correlations do not imply bipartite entanglement.

3.5 The geometry of mixed quantum states

The three conditions defining a density operator are logically independent: Hermiticity, positive semidefiniteness, and unit trace. For a qubit, the Pauli expansion $\rho = (I + \boldsymbol{r} \cdot \boldsymbol{\sigma})/2$ gives eigenvalues $\lambda_\pm = (1 \pm \|\boldsymbol{r}\|_2)/2$. Consequently positivity is equivalent to $\|\boldsymbol{r}\|_2 \leq 1$. The boundary of the Bloch ball consists of pure states, and its center $I/2$ is maximally mixed.

Example 3.13 (Physical and nonphysical Bloch operators)**.** Consider $\rho_1 = (I + 3X/5 + 4Z/5)/2$. Its Bloch vector has length 1, so its eigenvalues are $1, 0$ and the state is pure. By contrast, the diagonal operator $\rho_2 = \mathrm{diag}(5/4, -1/4)$ is Hermitian and has unit trace but fails positivity. The convex mixture $\rho_3 = \frac{2}{3}|0\rangle\langle 0| + \frac{1}{3}|+\rangle\langle +|$ has Bloch vector $(1/3, 0, 2/3)$ and length $\sqrt{5}/3 < 1$, so it is a valid mixed state.

The purity can be calculated without diagonalizing:

$$\mathrm{Tr}\,\rho^2 = \frac{1}{4}\,\mathrm{Tr}[I + 2\boldsymbol{r} \cdot \boldsymbol{\sigma} + (\boldsymbol{r} \cdot \boldsymbol{\sigma})^2] \tag{3.4}$$

$$= \frac{1}{2}(1 + \|\boldsymbol{r}\|_2^2). \tag{3.5}$$

The identity $(\boldsymbol{r} \cdot \boldsymbol{\sigma})^2 = \|\boldsymbol{r}\|_2^2 I$ follows from the anticommutation relations of the Pauli matrices. Thus $1/2 \leq \mathrm{Tr}\,\rho^2 \leq 1$, with equality at maximally mixed and pure states, respectively.

3.6 Schmidt coefficients, purification, and uniqueness

3.6.1 The singular-value proof of Schmidt decomposition

Write a bipartite state as $|\psi\rangle_{AB} = \sum_{i,j} M_{ij}\,|i\rangle_A\,|j\rangle_B$. The coefficient array M is a matrix. Choose its singular-value decomposition $M = U\Sigma V^\dagger$. Substituting U and V into the two local basis changes gives

$$|\psi\rangle_{AB} = \sum_{k=1}^{r} s_k\,|u_k\rangle_A\,|v_k\rangle_B\,, \qquad \sum_k s_k^2 = 1. \tag{3.6}$$

The number r of nonzero singular values is the Schmidt rank. The reduced states are $\rho_A = MM^\dagger$ and $\rho_B = M^{\mathsf{T}}\overline{M}$, so they possess the same positive eigenvalues s_k^2, possibly with different numbers of zero eigenvalues. This gives an algebraic proof of the equal reduced spectra of a pure bipartite state.

Corollary 3.14 (Schmidt characterization of pure-state separability)**.** *A bipartite pure state is a tensor product if and only if its Schmidt rank equals one, equivalently if and only if either reduced density operator is pure.*

Proof. A product vector has coefficient matrix of rank one. Conversely a single nonzero singular value yields a single Schmidt term. Reduced purity is one precisely when one Schmidt eigenvalue equals one. □

Example 3.15 (Explicit diagonalization and purification)**.** Let $\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|+\rangle\langle +|$, with

$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. In the computational basis,

$$\rho = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}, \qquad \det \rho = \frac{1}{8}, \quad \operatorname{Tr} \rho = 1. \tag{3.7}$$

Solving $\lambda^2 - \lambda + 1/8 = 0$ gives

$$\lambda_\pm = \frac{2 \pm \sqrt{2}}{4}. \tag{3.8}$$

An orthonormal eigenbasis is $|e_+\rangle = \cos(\pi/8)\,|0\rangle + \sin(\pi/8)\,|1\rangle$ and $|e_-\rangle = -\sin(\pi/8)\,|0\rangle + \cos(\pi/8)\,|1\rangle$. One purification is

$$|\Phi\rangle_{SE} = \sqrt{\lambda_+}\,|e_+\rangle_S\,|0\rangle_E + \sqrt{\lambda_-}\,|e_-\rangle_S\,|1\rangle_E\,. \tag{3.9}$$

Tracing over E eliminates the cross terms and reproduces ρ. Its purity is $\lambda_+^2 + \lambda_-^2 = 3/4$, and its entropy is $h_2((2+\sqrt{2})/4)$ bits.

Theorem 3.16 (All purifications differ by an environment isometry)**.** *Let $|\psi\rangle_{AE}$ and $|\phi\rangle_{AF}$ purify the same ρ_A, with $\dim F \geq \dim E \geq \operatorname{rank} \rho_A$ after restricting to their supports. Then there exists an isometry $V : E \to F$ on the support of the first purification such that*

$$|\phi\rangle_{AF} = (I_A \otimes V)\,|\psi\rangle_{AE}\,. \tag{3.10}$$

Proof. The Schmidt decompositions of both purifications have the same nonzero Schmidt coefficients, namely square roots of the nonzero eigenvalues of ρ_A. Within degenerate eigenspaces one first rotates the Schmidt bases on A consistently. Map the corresponding orthonormal environment vectors in E to those in F and extend this map isometrically. The resulting operator produces the second purification term by term. □

3.7 Separability tests and correlation witnesses

A mixed state is separable when it admits $\rho_{AB} = \sum_j p_j \rho_A^{(j)} \otimes \rho_B^{(j)}$. An entangled state may have separable marginals; local state tomography alone cannot certify entanglement without correlation measurements. Conversely, a witness is an observable W for which $\operatorname{Tr}(W\sigma) \geq 0$ on all separable σ but $\operatorname{Tr}(W\rho) < 0$ on at least one entangled state.

Example 3.17 (A two-qubit Bell witness from three correlators)**.** Define $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ and

$$\Pi_{\Phi^+} = \frac{1}{4}(I \otimes I + X \otimes X - Y \otimes Y + Z \otimes Z), \qquad W = \frac{1}{2} I - \Pi_{\Phi^+}. \tag{3.11}$$

For a product pure state $|a\rangle\,|b\rangle$, $|\langle \Phi^+ | a, b\rangle|^2 \leq 1/2$ by the largest Schmidt coefficient; convexity extends the inequality to separable mixtures. Suppose measured correlations are

$$\langle X \otimes X \rangle = \frac{3}{5}, \qquad \langle Y \otimes Y \rangle = -\frac{3}{5}, \qquad \langle Z \otimes Z \rangle = \frac{2}{5}. \tag{3.12}$$

Then $\operatorname{Tr}(\Pi_{\Phi^+}\rho) = \frac{1}{4}(1 + 3/5 + 3/5 + 2/5) = 13/20$, giving $\operatorname{Tr}(W\rho) = -3/20 < 0$. Thus the state is entangled although its complete density matrix was never reconstructed.

3.7.1 Partial transposition in the computational basis

For matrix units, $(|i\rangle\langle k|_A \otimes |j\rangle\langle \ell|_B)^{T_B} = |i\rangle\langle k|_A \otimes |\ell\rangle\langle j|_B$. Consider the Bell projector

$$\rho_\Phi = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|). \tag{3.13}$$

Its partial transpose is

$$\rho_\Phi^{T_B} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{3.14}$$

whose eigenvalues are $1/2, 1/2, 1/2, -1/2$. A negative eigenvalue certifies entanglement. Positivity under partial transpose is necessary for separability in every dimension and also sufficient for 2×2 and 2×3 bipartite systems. In higher dimensions PPT entangled states exist, so a nonnegative partial transpose is not generally an entanglement certificate.

3.8 Entropy, coherent correlations, and strong subadditivity

Proposition 3.18 (Bipartite pure-state entropy)**.** *If $|\psi\rangle_{AB}$ has Schmidt probabilities $\lambda_1, \ldots, \lambda_r$, then*

$$S(A) = S(B) = -\sum_{j=1}^{r} \lambda_j \log \lambda_j. \tag{3.15}$$

Proof. The reduced density operators have the same nonzero spectra, and von Neumann entropy depends only on the eigenvalues. □

The quantum conditional entropy $S(A|B) = S(AB) - S(B)$ can be negative. For a maximally entangled state $|\Phi_d\rangle$, the joint entropy is zero while each marginal is maximally mixed, so $S(A|B) = -\log d$. This is impossible for classical conditional Shannon entropy and reflects entanglement rather than an algebraic inconsistency.

Example 3.19 (GHZ, its marginals, and conditional entropy)**.** For $|\mathrm{GHZ}\rangle = (|000\rangle + |111\rangle)/\sqrt{2}$, the one-qubit state is $I/2$ with entropy 1. Every two-qubit marginal is the mixture $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$, also with entropy 1, and the global entropy is 0. Therefore

$$S(A|BC) = -1, \qquad S(A|B) = 0, \qquad I(A:B) = 1, \qquad I(A:BC) = 2. \tag{3.16}$$

The reductions AB, AC, and BC are separable, even though the pure tripartite state possesses genuine global entanglement.

Theorem 3.20 (Araki–Lieb triangle inequality)**.** *For every finite-dimensional bipartite state,*

$$|S(A) - S(B)| \le S(AB) \le S(A) + S(B). \tag{3.17}$$

Proof. The upper bound is subadditivity. For the lower bound, purify ρ_{AB} with R. Purity implies $S(A) = S(BR)$, $S(B) = S(AR)$, and $S(AB) = S(R)$. Subadditivity on BR gives $S(A) \le S(B) + S(R)$; exchange A, B for the other direction. □

Remark 3.21 (Strong subadditivity and conditional mutual information). For any ABC, $I(A:C|B) = S(AB) + S(BC) - S(B) - S(ABC) \ge 0$. In quantum mechanics this is

a substantive theorem (strong subadditivity), not a consequence of classical conditioning reducing entropy; a negative conditional quantum entropy is perfectly admissible. Equality is the quantum Markov-chain condition and is closely connected to channel recovery developed later.

3.9 Bipartite examples with different kinds of correlations

3.9.1 Bell-diagonal states

A Bell-diagonal two-qubit state has the form

$$\rho = \sum_{j=1}^{4} p_j \, |\beta_j\rangle\langle\beta_j| \,, \qquad p_j \geq 0, \quad \sum_j p_j = 1, \tag{3.18}$$

where $\{\beta_j\}$ are the four Bell vectors. Each Bell vector has maximally mixed one-qubit marginals; therefore $\rho_A = \rho_B = I/2$ for every Bell-diagonal mixture. The reduced states alone cannot determine the p_j, and in particular cannot distinguish an entangled Bell state from the maximally mixed state $I_4/4$. The von Neumann entropy is simply the classical entropy of the Bell weights: $S(AB) = H(p_1, p_2, p_3, p_4)$.

Example 3.22 (A Werner-type line)**.** Consider $\rho_w = w\,|\Psi^-\rangle\langle\Psi^-| + (1-w)I_4/4$ for $0 \leq w \leq 1$. Its Bell eigenvalues are $(1+3w)/4$ on $|\Psi^-\rangle$ and $(1-w)/4$ on the other three Bell vectors. The partial transpose has one potentially negative eigenvalue $(1-3w)/4$, so the state is entangled exactly for $w > 1/3$. At $w = 1$ it is pure and maximally entangled; at $w = 0$ it is completely mixed and separable.

3.9.2 Pure-state entanglement and local uncertainty

For a bipartite pure state in Schmidt form $|\psi\rangle = \sum_j \sqrt{\lambda_j}\,|j\rangle_A\,|j\rangle_B$, the global state is pure but the local entropies satisfy $S(A) = S(B) = H(\lambda)$. The quantum mutual information is therefore

$$I(A:B) = S(A) + S(B) - S(AB) = 2H(\lambda). \tag{3.19}$$

The factor two is characteristic of a pure entangled bipartite state. If the state is separable and pure, then one Schmidt coefficient is one and $I(A:B) = 0$.

Example 3.23 (Partially entangled two-qubit state)**.** Let $|\psi\rangle = \sqrt{3/4}\,|00\rangle + \sqrt{1/4}\,|11\rangle$. The reduced density matrix is $\rho_A = \mathrm{diag}(3/4, 1/4)$, with entropy $h_2(1/4)$; the joint entropy is zero. Hence

$$I(A:B) = 2h_2(1/4) \simeq 1.62256 \text{ bits}. \tag{3.20}$$

The value can exceed one bit because quantum mutual information measures total correlations, including those associated with entanglement, rather than the one-way accessible classical information from a single local measurement.

3.10 Multipartite entanglement and reduced structures

The GHZ state and the W state are inequivalent classes of genuine three-party entanglement under invertible local transformations. Define

$$|\mathrm{GHZ}\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}, \qquad |W\rangle = \frac{|001\rangle + |010\rangle + |100\rangle}{\sqrt{3}}. \tag{3.21}$$

Each one-qubit GHZ marginal is $I/2$. A one-qubit W marginal is $\mathrm{diag}(2/3, 1/3)$ up to a permutation of basis states, with entropy $h_2(1/3)$. The two-qubit marginals behave differently: GHZ has classical correlation but no pairwise entanglement, whereas tracing one qubit out of W gives

$$\rho_{AB}^{W} = \frac{1}{3}|00\rangle\langle 00| + \frac{2}{3}\left|\Psi^+\right\rangle\left\langle\Psi^+\right|. \tag{3.22}$$

The second term retains entanglement. To see it, write the partial transpose in the ordered basis $00, 01, 10, 11$:

$$(\rho_{AB}^{W})^{T_B} = \begin{pmatrix} 1/3 & 0 & 0 & 1/3 \\ 0 & 1/3 & 0 & 0 \\ 0 & 0 & 1/3 & 0 \\ 1/3 & 0 & 0 & 0 \end{pmatrix}. \tag{3.23}$$

The 2×2 corner block has determinant $-1/9 < 0$, so one eigenvalue is negative and the reduction is entangled. This gives a practical way to distinguish GHZ- and W-type distribution of correlations without full tripartite invariants.

3.11 Entanglement witnesses as partial information

An entanglement witness is experimentally economical when it depends on a small set of measurable correlators. The Bell fidelity witness $W = I/2 - |\Phi^+\rangle\langle\Phi^+|$ needs only $\langle X \otimes X\rangle$, $\langle Y \otimes Y\rangle$, and $\langle Z \otimes Z\rangle$, because the Bell projector has a three-Pauli decomposition. The witness certifies entanglement when its expectation is negative, but a nonnegative value is inconclusive: there are entangled states for which this specific witness is nonnegative.

Example 3.24 (An entangled state missed by a particular witness)**.** The Bell state $|\Psi^-\rangle$ is entangled and orthogonal to $|\Phi^+\rangle$. Hence

$$\mathrm{Tr}\left[(I/2 - \left|\Phi^+\right\rangle\left\langle\Phi^+\right|)\,|\Psi^-\rangle\langle\Psi^-|\right] = \frac{1}{2} > 0. \tag{3.24}$$

Therefore this witness does not detect $|\Psi^-\rangle$, illustrating why a witness test is a sufficient condition for entanglement, not a complete classifier of all entangled states.

CHAPTER 4

Measurements, Discrimination, and Tomography

4.1 Projective measurements and observables

Definition 4.1 (Projective measurement)**.** A projective measurement is a family of orthogonal projectors $\{\Pi_m\}$ satisfying $\Pi_m\Pi_n = \delta_{mn}\Pi_m$ and $\sum_m \Pi_m = I$. On input ρ, outcome m occurs with probability $p(m) = \operatorname{Tr}(\Pi_m\rho)$ and the Lüders update is $\rho_m = \Pi_m\rho\Pi_m/p(m)$ when $p(m) > 0$.

An observable $M = \sum_m m\Pi_m$ has expectation $\langle M\rangle_\rho = \operatorname{Tr}(M\rho)$ and variance $\operatorname{Var}_\rho(M) = \operatorname{Tr}(M^2\rho) - \operatorname{Tr}(M\rho)^2$. Degenerate eigenspaces must be treated by their projectors, not by an arbitrary rank-one refinement.

Example 4.2 (A degenerate two-qubit observable)**.** For $M = X \otimes X$, the projectors onto eigenvalues ± 1 are $\Pi_\pm = (I \pm X \otimes X)/2$, both of rank two. For $\rho = |00\rangle\langle 00|$, each outcome has probability $1/2$ and the normalized postmeasurement states are $|\Phi^+\rangle$ and $|\Phi^-\rangle$, respectively.

4.2 POVMs and their dilation

Definition 4.3 (Positive-operator-valued measurement)**.** A POVM with outcome set Ω is a family of effects $\{E_y\}_{y\in\Omega}$ such that $E_y \succeq 0$ and $\sum_y E_y = I$. It defines $p(y) = \operatorname{Tr}(E_y\rho)$. The POVM does not by itself specify a unique postmeasurement state.

Theorem 4.4 (Naimark dilation)**.** *Every finite POVM can be realized as a projective measurement on a larger Hilbert space after an isometric embedding.*

Proof. Take an orthonormal pointer basis $\{|y\rangle\}$ and define $V|\psi\rangle = \sum_y \sqrt{E_y}\,|\psi\rangle \otimes |y\rangle$. Since $V^\dagger V = \sum_y E_y = I$, V is an isometry. Measuring the pointer basis yields $\|(I \otimes \langle y|)V|\psi\rangle\|^2 = \langle\psi|\,E_y\,|\psi\rangle$; convexity extends the formula to mixed states. □

4.3 Optimal binary state discrimination

Problem 4.5 (Minimum-error discrimination)**.** Given states ρ_0, ρ_1 with prior probabilities $p, 1-p$, choose a binary POVM $\{E_0, E_1\}$ maximizing $P_{\text{succ}} = p\operatorname{Tr}(E_0\rho_0) + (1-p)\operatorname{Tr}(E_1\rho_1)$.

Theorem 4.6 (Helstrom bound)**.** *Let $\Delta = p\rho_0 - (1-p)\rho_1$. The optimal value and a realizing POVM are*

$$P^*_{\text{succ}} = \frac{1 + \|\Delta\|_1}{2}, \qquad E_0 = \Pi_+(\Delta), \quad E_1 = I - E_0, \tag{4.1}$$

where the decision on the kernel of Δ is arbitrary. For equal priors, $P^*_{\text{succ}} = \frac{1}{2} + \frac{1}{4}\|\rho_0 - \rho_1\|_1$.

Proof. Substitute $E_1 = I - E_0$: $P_{\text{succ}} = 1 - p + \operatorname{Tr}(E_0\Delta)$. The positive spectral subspace of Δ maximizes the linear objective over $0 \le E_0 \le I$. Since $\operatorname{Tr}\Delta = 2p - 1$, sum of positive eigenvalues equals $(\|\Delta\|_1 + \operatorname{Tr}\Delta)/2$. □

Example 4.7 (Nonorthogonal pure states)**.** Distinguishing $|0\rangle$ from $|+\rangle$ with equal prior gives $\||0\rangle\langle 0| - |+\rangle\langle +|\|_1 = \sqrt{2}$, hence $P^*_{\text{succ}} = (2+\sqrt{2})/4$. For general pure states $|u\rangle, |v\rangle$ with equal prior,

$$P^*_{\text{succ}} = \frac{1}{2}\left(1 + \sqrt{1 - |\langle u|v\rangle|^2}\right). \tag{4.2}$$

Definition 4.8 (Unambiguous discrimination)**.** A POVM $\{E_0, E_1, E_?\}$ distinguishes two states without misidentification when $\operatorname{Tr}(E_0\rho_1) = \operatorname{Tr}(E_1\rho_0) = 0$, allowing an inconclusive outcome ?.

For distinct nonorthogonal pure qubits $|\psi_0\rangle, |\psi_1\rangle$, the conclusive effects must be supported on $\operatorname{span}\{|\psi_1^\perp\rangle\}$ and $\operatorname{span}\{|\psi_0^\perp\rangle\}$, respectively. Exact certainty of identification is traded for a positive failure probability.

4.4 Pauli tomography and estimation

Definition 4.9 (Informational completeness)**.** A POVM or collection of observables is informationally complete if the map $\rho \mapsto \{\operatorname{Tr}(M_a\rho)\}_a$ is injective on density matrices.

For a qubit, measuring X, Y, Z gives the Bloch vector $r_k = \operatorname{Tr}(\sigma_k\rho)$ and $\rho = (I + r_xX + r_yY + r_zZ)/2$. Physicality requires $r_x^2 + r_y^2 + r_z^2 \le 1$. If $r_x = r_y = 1/2$, then $|r_z| \le 1/\sqrt{2}$. For n qubits, $d = 2^n$ and an arbitrary density operator has $d^2 - 1 = 4^n - 1$ real parameters. One physical reconstruction method solves

$$\min_{\rho\succeq 0,\ \operatorname{Tr}\rho=1} \sum_a w_a \big(\operatorname{Tr}(M_a\rho) - \widehat{m}_a\big)^2. \tag{4.3}$$

The positivity constraint distinguishes a valid quantum estimator from unconstrained linear inversion.

4.5 Trace distance and operational distinguishability

Definition 4.10 (Trace distance)**.** $T(\rho,\sigma) = \frac{1}{2}\|\rho - \sigma\|_1$.

For every channel $\mathcal{N}$, $T(\mathcal{N}(\rho), \mathcal{N}(\sigma)) \le T(\rho,\sigma)$. In the binary equal-prior discrimination problem, $P^*_{\text{succ}} = (1 + T(\rho_0,\rho_1))/2$. This identifies trace distance as the optimal bias achievable by a physical measurement.

4.6 State reduction and spectral measurement calculations

4.6.1 Projectors of an involutory observable

Let $M = M^\dagger$ and $M^2 = I$. Its spectrum is contained in $\{+1, -1\}$, with spectral projectors $P_\pm = (I \pm M)/2$. Indeed

$$P_\pm^2 = \frac{1}{4}(I \pm 2M + M^2) = P_\pm, \quad P_+P_- = 0, \quad P_+ + P_- = I. \tag{4.4}$$

For an input pure state $|\psi\rangle$, the outcome probabilities and normalized post-measurement states are

$$p_\pm = \langle\psi| P_\pm |\psi\rangle = \|P_\pm |\psi\rangle\|^2, \qquad |\psi_\pm\rangle = \frac{P_\pm |\psi\rangle}{\sqrt{p_\pm}}. \tag{4.5}$$

The normalized vector is defined only when $p_\pm > 0$. A degenerate measurement preserves coherence within each eigenspace; it does not identify a particular vector in that subspace.

Example 4.11 (A two-qubit $X \otimes X$ measurement)**.** For $M = X \otimes X$ and

$$|\psi\rangle = \frac{|00\rangle + 2|01\rangle + 3|10\rangle + 4|11\rangle}{\sqrt{30}}, \tag{4.6}$$

one has $M|\psi\rangle = (4|00\rangle + 3|01\rangle + 2|10\rangle + |11\rangle)/\sqrt{30}$. Thus

$$P_+ |\psi\rangle = \frac{5}{2\sqrt{30}}(|00\rangle + |01\rangle + |10\rangle + |11\rangle), \tag{4.7}$$

$$P_- |\psi\rangle = \frac{1}{2\sqrt{30}}(-3|00\rangle - |01\rangle + |10\rangle + 3|11\rangle). \tag{4.8}$$

The squared norms are $p_+ = 25/30 = 5/6$ and $p_- = 5/30 = 1/6$. The corresponding normalized states are

$$|\psi_+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = |++\rangle, \tag{4.9}$$

$$|\psi_-\rangle = \frac{-3|00\rangle - |01\rangle + |10\rangle + 3|11\rangle}{\sqrt{20}}. \tag{4.10}$$

The two probabilities sum to one and each normalized vector has unit norm, independently checking the calculation.

4.7 POVMs as operational probability rules

A POVM is a family of positive effects $\{E_y\}$ with $\sum_y E_y = I$. It specifies $p(y|\rho) = \mathrm{Tr}(E_y\rho)$, but effects alone do not determine the conditional output states. An *instrument* includes completely positive maps $\mathcal{I}_y$ satisfying $\mathrm{Tr}\,\mathcal{I}_y(\rho) = \mathrm{Tr}(E_y\rho)$ and $\sum_y \mathcal{I}_y$ trace preserving. Different instruments can share a POVM while producing different back-action.

Example 4.12 (A nonprojective qubit measurement)**.** For $0 < \eta < 1$ consider

$$E_\pm = \frac{1}{2}(I \pm \eta Z). \tag{4.11}$$

Both eigenvalues $(1 \pm \eta)/2$ are nonnegative and $E_+ + E_- = I$. For a state with Bloch

z coordinate r_z, the probabilities are $p_\pm = (1 \pm \eta r_z)/2$. At $\eta = 1$ the POVM becomes projective; at $\eta = 0$ both outcomes are fair independent of the input. The continuous range interpolates between a sharp measurement and a totally uninformative readout.

4.7.1 Constructive Naimark extension

Let $E_1, \ldots, E_m$ be a POVM on A. Introduce an m-dimensional ancilla E and an isometry

$$V : |\psi\rangle_A \longmapsto \sum_{y=1}^{m} \sqrt{E_y}\, |\psi\rangle_A \otimes |y\rangle_E \,. \tag{4.12}$$

Its isometry property follows from

$$\langle V\psi | V\phi \rangle = \sum_y \langle \psi |\, E_y \,| \phi \rangle = \langle \psi | \phi \rangle \,. \tag{4.13}$$

Measuring the ancilla in its computational basis yields $p_y = \langle \psi |\, E_y \,| \psi \rangle$. The isometry can be extended to a unitary after adding any required ancillary dimensions. This establishes the finite-dimensional operational realization of arbitrary POVMs.

4.8 Optimal discrimination as a variational problem

For binary hypotheses ρ_0, ρ_1 with priors $p, 1-p$, choose an effect $0 \preceq E \preceq I$ and guess zero upon the corresponding outcome. The success probability is

$$P_{\text{succ}}(E) = p \operatorname{Tr}(E\rho_0) + (1-p) \operatorname{Tr}[(I-E)\rho_1] \tag{4.14}$$

$$= (1-p) + \operatorname{Tr}[E\Delta], \qquad \Delta = p\rho_0 - (1-p)\rho_1. \tag{4.15}$$

Apply the variational formula for the positive spectral part of Δ. Since $\operatorname{Tr}\Delta = 2p - 1$,

$$\max_E P_{\text{succ}}(E) = \frac{1}{2}(1 + \|\Delta\|_1). \tag{4.16}$$

The optimal effect is the projector onto the positive eigenspace of Δ; the arbitrary decision in its zero eigenspace does not change success probability. Thus Helstrom discrimination is an instance of optimizing a linear functional over operator effects, with a closed-form spectral solution.

Example 4.13 (Two nonorthogonal pure states, equal priors)**.** Let $\rho_0 = |0\rangle\langle 0|$, $\rho_1 = |+\rangle\langle +|$, $p = 1/2$. Then

$$\rho_0 - \rho_1 = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & -1/2 \end{pmatrix} \tag{4.17}$$

has eigenvalues $\pm 1/\sqrt{2}$. Since $\Delta = (\rho_0 - \rho_1)/2$, $\|\Delta\|_1 = 1/\sqrt{2}$ and

$$P^*_{\text{succ}} = \frac{1}{2}\left(1 + \frac{1}{\sqrt{2}}\right) \simeq 0.85355. \tag{4.18}$$

For two pure states with equal priors and overlap $c = |\langle \psi_0 | \psi_1 \rangle|$, this generalizes to $P^*_{\text{succ}} = (1 + \sqrt{1 - c^2})/2$.

Example 4.14 (Commuting states reduce to a classical likelihood test)**.** Take $\rho_0 =$

$\mathrm{diag}(3/4, 1/4)$ and $\rho_1 = \mathrm{diag}(1/4, 3/4)$, with equal prior. Then

$$\Delta = \frac{1}{2}(\rho_0 - \rho_1) = \mathrm{diag}(1/4, -1/4), \qquad \|\Delta\|_1 = 1/2. \tag{4.19}$$

Thus $P^*_{\mathrm{succ}} = 3/4$. The optimal POVM is ordinary computational-basis measurement: outcome zero favors ρ_0, outcome one favors ρ_1. Quantum optimization coincides with the classical maximum-likelihood rule for commuting density operators.

4.8.1 Unambiguous discrimination

For two linearly independent nonorthogonal pure states, an unambiguous POVM has outcomes $0, 1, ?$ and must satisfy

$$\langle\psi_1| E_0 |\psi_1\rangle = 0, \qquad \langle\psi_0| E_1 |\psi_0\rangle = 0. \tag{4.20}$$

Positivity implies $E_0 |\psi_1\rangle = 0$ and $E_1 |\psi_0\rangle = 0$. Therefore, in the qubit case, $\mathrm{supp}\, E_0 \subseteq \mathrm{span}\{\left|\psi_1^{\perp}\right\rangle\}$ and $\mathrm{supp}\, E_1 \subseteq \mathrm{span}\{\left|\psi_0^{\perp}\right\rangle\}$. A nonzero inconclusive effect is unavoidable when the states are nonorthogonal: if there were none, a perfect deterministic discriminator would contradict the Helstrom bound.

4.9 Reconstruction and statistical precision

The most direct single-qubit tomography estimates $r_x = \mathrm{Tr}(X\rho)$, $r_y = \mathrm{Tr}(Y\rho)$, $r_z = \mathrm{Tr}(Z\rho)$ by measuring independently prepared copies in the three Pauli eigenbases. The reconstructed operator is $\tilde{\rho} = (I + \hat{\boldsymbol{r}} \cdot \boldsymbol{\sigma})/2$. Finite sampling may give $\|\hat{\boldsymbol{r}}\| > 1$, which is not physical; a constrained reconstruction must project or optimize inside the Bloch ball.

Example 4.15 (Physicality constraint from incomplete data)**.** Suppose $r_x = r_y = 1/2$ are known but r_z is unmeasured. Positivity requires

$$\frac{1}{4} + \frac{1}{4} + r_z^2 \le 1, \quad \text{hence} \quad -\frac{1}{\sqrt{2}} \le r_z \le \frac{1}{\sqrt{2}}. \tag{4.21}$$

No other restriction follows from the Bloch-ball characterization, so this interval is tight.

Proposition 4.16 (Hoeffding bound for a Pauli estimate)**.** *If an observable takes only values* ± 1 *and is measured independently on* n *copies, then the empirical mean* $\bar{X}$ *satisfies*

$$\mathbb{P}(\left|\bar{X} - \mathbb{E}X\right| \ge \varepsilon) \le 2\exp(-n\varepsilon^2/2). \tag{4.22}$$

Proof. The observations lie in an interval of width two. The standard bounded-difference exponential-moment proof of Hoeffding's inequality gives $2e^{-2n\varepsilon^2/(2^2)}$. □

Consequently $n \ge (2/\varepsilon^2)\log(6/\delta)$ samples per Pauli observable suffice to estimate all three coordinates within ε simultaneously with failure probability at most δ, by a union bound. In dimension d, full tomography requires $d^2 - 1$ real parameters, so the number of settings and precision cost grow with the state-space dimension.

4.10 Unequal priors and the geometry of binary discrimination

Let $\rho_0 = |0\rangle\langle 0|$ and $\rho_1 = |+\rangle\langle +|$, with prior p of state zero. Their Helstrom operator is

$$\Delta_p = p\rho_0 - (1-p)\rho_1 = \frac{1}{2}\begin{pmatrix} 3p-1 & -(1-p) \\ -(1-p) & -(1-p) \end{pmatrix}. \tag{4.23}$$

Its trace and determinant are $2p-1$ and $-p(1-p)/2$. Thus the eigenvalues are

$$\lambda_\pm = \frac{1}{2}\left(2p - 1 \pm \sqrt{p^2 + (1-p)^2}\right). \tag{4.24}$$

For every strict interior prior $0 < p < 1$, the determinant is negative, so the eigenvalues have opposite signs. The trace norm is their difference, yielding

$$P^*_{\text{succ}} = \frac{1}{2}\left(1 + \sqrt{p^2 + (1-p)^2}\right). \tag{4.25}$$

The optimal measurement basis depends continuously on p. In Bloch-vector language, Δ_p has scalar part $(2p-1)I/2$ and vector part $[-(1-p)X + pZ]/2$, so the optimal decision axis points along

$$\boldsymbol{n}_p = \frac{(-(1-p), 0, p)}{\sqrt{p^2 + (1-p)^2}}. \tag{4.26}$$

The scalar part changes the relative strengths of the hypotheses, while the vector part determines the discriminating measurement basis. At $p = 1/2$ the axis bisects the appropriate directions and recovers the equal-prior expression.

4.10.1 Comparison with blind guessing

Without measuring the system, the optimal decision is to guess the more likely hypothesis and succeed with probability $P_{\text{blind}} = \max\{p, 1-p\}$. For the two distinct nonorthogonal pure states and any strict interior prior, the optimal Helstrom success probability is strictly larger. This follows from

$$\sqrt{p^2 + (1-p)^2} > |2p-1| \quad \text{when } 0 < p < 1, \tag{4.27}$$

which is equivalent to $2p(1-p) > 0$. At endpoint priors there is no uncertainty about the label, so both strategies succeed with probability one.

4.11 Classical Bayesian decision after a fixed measurement

Suppose Alice sends one of two diagonal qubit states

$$\rho_0 = \text{diag}(3/4, 1/4), \qquad \rho_1 = \text{diag}(1/4, 3/4) \tag{4.28}$$

with priors p and $1-p$. Bob measures in the computational basis but chooses his guess based on the prior. Upon outcome zero, the unnormalized posterior weights are $3p/4$ and $(1-p)/4$; Bob should guess state zero exactly when $3p \geq 1-p$, or $p \geq 1/4$. Upon outcome

one, the weights are $p/4$ and $3(1-p)/4$; Bob should guess state zero exactly when $p \geq 3/4$.

$$P_{\text{succ}}^{\text{fixed}} = \begin{cases} 1-p, & 0 \leq p \leq 1/4, \\ 3/4, & 1/4 \leq p \leq 3/4, \\ p, & 3/4 \leq p \leq 1. \end{cases} \tag{4.29}$$

The piecewise behavior arises because the measurement does not always change the posterior optimal decision. At $p = 0.1$, outcome zero is more likely under ρ_0 than outcome one, but the prior is so strongly biased toward ρ_1 that Bob should still always guess state one; his success is 0.9.

4.12 Two-state discrimination versus accessible information

The Helstrom objective maximizes the probability of guessing a discrete hypothesis correctly. The mutual-information objective instead maximizes $I(X:Y)$ over measurements, which is a different optimization functional. Even for binary ensembles, one should identify which operational criterion is relevant before choosing a measurement. For many signals, a POVM with more than two outcomes may be useful; a projective two-outcome measurement is not generally sufficient.

Example 4.17 (Posterior calculation)**.** Let $p = 1/3$ in the diagonal example. The probability of outcome zero is $q_0 = (3p + 1 - p)/4 = (1 + 2p)/4 = 5/12$. Its posterior probability for label zero is

$$\mathbb{P}(X = 0 \mid Y = 0) = \frac{3p/4}{(1+2p)/4} = \frac{3p}{1+2p} = \frac{3}{5}. \tag{4.30}$$

For outcome one, the posterior is $p/(3-2p) = 1/7$. The decoder therefore guesses zero on outcome zero and one on outcome one, attaining success probability $3/4$.

CHAPTER 5

Quantum Circuits and Universality

5.1 Computational model and elementary gates

Definition 5.1 (Unitary circuit)**.** An n-qubit circuit is a composition $U = U_T \cdots U_2 U_1$, where each U_j is a one- or few-qubit gate tensored with identities on untouched wires. Its size is the number of elementary gates and its depth is the number of parallel layers. Measurement and classical feedforward extend this unitary model to general quantum operations.

The Hadamard, phase and T gates are

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \mathrm{diag}(1, i), \quad T = \mathrm{diag}(1, e^{i\pi/4}). \tag{5.1}$$

For $P \in \{X, Y, Z\}$, the rotations $R_P(\theta) = e^{-i\theta P/2} = \cos(\theta/2) I - i \sin(\theta/2) P$ act on the Bloch vector by the corresponding spatial rotation. Pauli multiplication obeys $XY = iZ$, $YZ = iX$, $ZX = iY$.

Proposition 5.2 (Euler decomposition)**.** *For every $U \in U(2)$, there are real $\alpha, \beta, \gamma, \delta$ such that*

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta). \tag{5.2}$$

Definition 5.3 (Controlled unitary)**.** The controlled-U gate is $C(U) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$. The controlled-NOT gate has $U = X$; the Toffoli gate maps $|a, b, c\rangle \mapsto |a, b, c \oplus ab\rangle$.

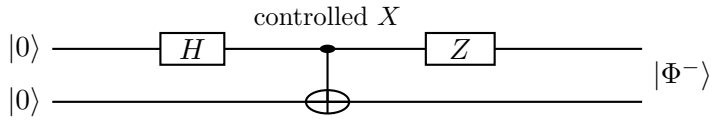


Figure 5.1: Native TikZ representation of a Hadamard followed by CNOT. The prepared joint state is $(|00\rangle + |11\rangle)/\sqrt{2}$; the Z gate shown afterward maps it to $(|00\rangle - |11\rangle)/\sqrt{2}$.

5.2 No-cloning, controlled operations, and uncomputation

Theorem 5.4 (No-cloning)**.** *There is no fixed unitary U and blank state $|0\rangle$ such that $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for every unknown pure state $|\psi\rangle$.*

Proof. For two states $|\psi\rangle, |\phi\rangle$, preservation of inner products under U would require $\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2$, which cannot hold for an arbitrary overlap strictly between zero and one. □

Proposition 5.5 (Compute–act–uncompute)**.** *Suppose a reversible subcircuit computes $g(x)$ into work registers, i.e., $C|x\rangle|0\rangle = |x\rangle|g(x)\rangle$. Then $C^\dagger V C$ implements a controlled action V depending on $g(x)$ and restores the work registers to $|0\rangle$ whenever V leaves the computed controls intact.*

Remark 5.6. Garbage removal is essential for interference: leaving distinguishable work states can destroy the coherence of the computational branches.

5.3 Universality and synthesis

Definition 5.7 (Approximate universality)**.** A gate family $\mathcal{G}$ is approximately universal if for each n, $U \in SU(2^n)$ and $\varepsilon > 0$, a circuit from $\mathcal{G}$ approximates U to operator-norm error at most ε (allowing an immaterial global phase).

Two-level unitaries can eliminate matrix entries through Givens-like transformations, and arbitrary n-qubit unitaries decompose into two-level factors. Together with CNOT and one-qubit rotations, this gives universality. The finite set $\{H, T, \mathrm{CNOT}\}$ is approximately universal.

Theorem 5.8 (Solovay–Kitaev, qualitative form)**.** *For a fixed finite inverse-closed gate set densely generating $SU(2)$, any one-qubit unitary can be approximated to accuracy ε using a circuit of length $O(\log^c(1/\varepsilon))$ for a universal constant $c > 0$ under the theorem's standard assumptions.*

The theorem addresses precision scaling, not the exponential cost of synthesizing an arbitrary unitary acting on exponentially large dimension.

5.4 Clifford circuits and stabilizers

Definition 5.9 (Clifford unitary)**.** An n-qubit unitary U is Clifford if $UPU^\dagger$ is a Pauli operator up to phase for every Pauli string P. The standard generators are H, S, and CNOT.

Theorem 5.10 (Gottesman–Knill)**.** *Stabilizer-state preparation, Clifford unitaries, Pauli-basis measurements, and classical feedforward can be simulated efficiently on a classical computer within the stabilizer formalism.*

Adding non-Clifford T gates enables universal quantum computation; entanglement or superposition alone is therefore insufficient to establish a computational speedup.

5.5 Hadamard and SWAP tests

Proposition 5.11 (Hadamard test)**.** *Given a controlled unitary U, an input state ρ, and an ancilla initialized in $|+\rangle$, the expectation of the ancilla's X outcome is $\Re \operatorname{Tr}(U\rho)$. Measuring Y with the appropriate sign convention recovers the imaginary part.*

Proposition 5.12 (SWAP test)**.** *For independent input states ρ, σ on equal-dimensional registers, the symmetric/antisymmetric outcome probabilities are*

$$p_\pm = \frac{1 \pm \operatorname{Tr}(\rho\sigma)}{2}. \tag{5.3}$$

In particular, for pure states $|\psi\rangle, |\phi\rangle$, $p_+ = \frac{1}{2}(1 + |\langle\psi|\phi\rangle|^2)$.

Proof. The controlled-SWAP circuit with Hadamards on the control measures $(I \pm S)/2$ on the two registers. The identity $\operatorname{Tr}[S(\rho \otimes \sigma)] = \operatorname{Tr}(\rho\sigma)$ completes the derivation. □

5.6 Single-qubit synthesis and controlled logic

The elementary matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{5.4}$$

provide a useful generating language. Up to global phase, any $U \in U(2)$ admits an Euler-angle decomposition

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta). \tag{5.5}$$

To see this, factor out the phase so the remaining matrix belongs to $SU(2)$ and write it as $\left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a} \end{smallmatrix}\right)$ with $|a|^2 + |b|^2 = 1$. Choose γ so that $\cos(\gamma/2) = |a|$ and $\sin(\gamma/2) = |b|$, then select β, δ to match the phases of a, b. This decomposition separates the continuous part of one-qubit unitary control from entangling operations.

Example 5.13 (CNOT creates a Bell pair)**.** Starting with $|0\rangle\,|0\rangle$, the Hadamard on the first register gives $(|0\rangle + |1\rangle)\,|0\rangle\,/\sqrt{2}$. A CNOT with that register as control maps $|10\rangle \mapsto |11\rangle$ while leaving $|00\rangle$ invariant, giving

$$\mathrm{CNOT}(H \otimes I)\,|00\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{5.6}$$

The reduced density operator of either qubit is $I/2$, not $|+\rangle\langle+|$. Local purity is lost even though the global evolution remains unitary.

5.6.1 Multi-controlled operations and reversible arithmetic

A controlled gate $\Lambda(U) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$ is unitary whenever U is. Multi-controlled NOT is a classical reversible operation on computational-basis strings, so it embeds consistently into quantum mechanics. Quantum algorithms often compute an intermediate Boolean predicate into an ancilla, use the ancilla to control a phase or bit flip, and then reverse the computation. If

$$U_f\,|x\rangle\,|0\rangle = |x\rangle\,|f(x)\rangle\,, \tag{5.7}$$

then $U_f^\dagger$ removes a work register from a coherently computed state, preventing unwanted entanglement with the output. This *uncomputation* is necessary when the intermediate register is not part of the desired answer.

5.7 Circuit identities from parity and Pauli strings

Let $P = \sigma_{a_1} \otimes \cdots \otimes \sigma_{a_n}$ be a Hermitian Pauli string. As $P^2 = I$,

$$e^{-i\theta P} = \cos\theta\, I - i \sin\theta\, P. \tag{5.8}$$

Local Clifford basis changes map each nonidentity X or Y to Z, after which a CNOT parity ladder computes the total Z eigenvalue parity onto a chosen target qubit. Apply $R_z(2\theta)$ to the target and reverse the CNOT ladder and local basis changes. On a computational basis vector of parity q, the phase is $e^{-i\theta(-1)^q}$, which equals the eigenphase of $e^{-i\theta Z^{\otimes k}}$.

Example 5.14 (Three-body interaction)**.** Let $C = \mathrm{CNOT}_{1\to 3}\mathrm{CNOT}_{2\to 3}$ (right-to-left

operator convention may be used because these two gates commute). Its action is

$$C\left|x_1x_2x_3\right\rangle = \left|x_1x_2(x_3 \oplus x_1 \oplus x_2)\right\rangle. \tag{5.9}$$

Conjugating a target rotation gives

$$C^\dagger(I \otimes I \otimes R_z(2\theta))C = e^{-i\theta Z\otimes Z\otimes Z}. \tag{5.10}$$

A common factor-of-two error is to use $R_z(\theta)$; recall its computational eigenphases are $e^{\mp i\theta/2}$, whereas the desired Pauli-string phases are $e^{\mp i\theta}$.

5.8 Information extraction through interference tests

5.8.1 The real and imaginary Hadamard tests

Prepare a control in $|0\rangle$ and the data register in $|\psi\rangle$. Apply H on the control, controlled-U, and a final H. Immediately before measurement, the state is

$$\frac{1}{2}\left|0\right\rangle(I+U)\left|\psi\right\rangle + \frac{1}{2}\left|1\right\rangle(I-U)\left|\psi\right\rangle. \tag{5.11}$$

Therefore

$$p(0) = \frac{1}{4}\left\langle\psi\right|(2I+U+U^\dagger)\left|\psi\right\rangle = \frac{1+\Re\left\langle\psi\right|U\left|\psi\right\rangle}{2}, \tag{5.12}$$

$$p(1) = \frac{1-\Re\left\langle\psi\right|U\left|\psi\right\rangle}{2}. \tag{5.13}$$

Hence $\mathbb{E}[(-1)^b] = p(0) - p(1) = \Re\left\langle\psi\right|U\left|\psi\right\rangle$. Inserting $S^\dagger$ on the ancilla before its final Hadamard yields the imaginary part $\Im\left\langle\psi\right|U\left|\psi\right\rangle$ for the usual gate convention; the sign should always be checked by propagating the i-phase explicitly.

Example 5.15 (A one-qubit diagonal test)**.** For $|\psi\rangle = |+\rangle$ and $U = R_z(\theta)$,

$$\left\langle+\right|R_z(\theta)\left|+\right\rangle = \frac{1}{2}(e^{-i\theta/2} + e^{i\theta/2}) = \cos(\theta/2). \tag{5.14}$$

The Hadamard-test success probability is $p(0) = (1+\cos(\theta/2))/2$. At $\theta = \pi$, it becomes $1/2$; at $\theta = 0$, it is one.

5.8.2 The SWAP test

Let the data registers hold density matrices ρ and σ. The controlled-SWAP and two Hadamards yield

$$p(0) = \frac{1}{2}\left(1 + \mathrm{Tr}[S(\rho\otimes\sigma)]\right) = \frac{1}{2}(1+\mathrm{Tr}\,\rho\sigma), \tag{5.15}$$

because the swap trick gives $\mathrm{Tr}[S(A\otimes B)] = \mathrm{Tr}(AB)$. For pure states this becomes

$$p(0) = \frac{1}{2}(1+|\langle\psi|\phi\rangle|^2). \tag{5.16}$$

Identical pure states give $p(0) = 1$, orthogonal states give $1/2$, and identical maximally mixed qubits give $p(0) = 3/4$. Thus $p(0) = 1$ does not necessarily mean two unknown mixed

states coincide; the test measures Hilbert–Schmidt overlap, not an equality predicate.

5.9 Universal gate approximation and the role of Clifford structure

A finite gate set is approximately universal if products of its elements are dense in the required special unitary group. Clifford gates alone map Pauli operators to Pauli operators under conjugation, and their circuits have efficient stabilizer-based classical simulation. Adding a non-Clifford gate such as $T = \mathrm{diag}(1, e^{i\pi/4})$ produces a universal generating family together with suitable Clifford gates. Universality is a statement about the closure of achievable unitaries; efficiency additionally requires the approximation length to scale reasonably with target error.

Remark 5.16 (Solovay–Kitaev hypotheses). The qualitative Solovay–Kitaev theorem concerns suitable inverse-closed finite dense gate sets in fixed dimension. Its polylogarithmic approximation length should not be confused with a uniform efficient compilation algorithm for arbitrary exponentially large n-qubit operators: the target representation and locality model matter.

5.9.1 Stabilizer generators

For a stabilizer state on n qubits, an abelian Pauli subgroup with n independent commuting generators uniquely fixes the state. The Bell state $|\Phi^+\rangle$ is stabilized by $X \otimes X$ and $Z \otimes Z$. Since these operators commute, the joint $+1$ eigenspace is one-dimensional. A Clifford unitary maps each generator to another Pauli string, reducing the evolution of the stabilizer group to discrete updates rather than amplitudes of all 2^n basis states.

Example 5.17 (Conjugating stabilizers through CNOT)**.** With control c and target t,

$$X_c \mapsto X_c X_t, \quad Z_t \mapsto Z_c Z_t, \quad Z_c \mapsto Z_c, \quad X_t \mapsto X_t. \tag{5.17}$$

Starting from $|+0\rangle$ with generators X_c and Z_t, CNOT maps them to $X_c X_t$ and $Z_c Z_t$, precisely the stabilizers of $|\Phi^+\rangle$.

5.10 Interference tests for a general input density operator

The Hadamard test does not require the data input to be pure. By linearity, for mixed ρ and unitary U the control bias is $\Re \operatorname{Tr}(U\rho)$. It can be verified by expanding a convex decomposition $\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$ and averaging pure-state test probabilities. The same experiment measures the imaginary part after a suitable control phase gate.

Example 5.18 (A coherent versus incoherent input)**.** For $U = X$, the state $\rho = |+\rangle\langle+|$ gives $\operatorname{Tr}(X\rho) = 1$ and hence $p(0) = 1$. The incoherent state $\rho = I/2$ gives $\operatorname{Tr}(X\rho) = 0$ and $p(0) = 1/2$. The circuit distinguishes the known observable expectation of the two states without reconstructing them.

5.10.1 Sampling errors in overlap estimation

A SWAP test produces a Bernoulli outcome with mean $p_0 = (1 + \operatorname{Tr} \rho\sigma)/2$. After n independent repetitions, an estimator is $\widehat{F} = 2\hat{p}_0 - 1$. Hoeffding's inequality yields

$$\mathbb{P}(\left|\widehat{F} - \operatorname{Tr} \rho\sigma\right| \geq \varepsilon) \leq 2e^{-n\varepsilon^2/2}. \tag{5.18}$$

Thus additive precision ε with failure probability δ requires $n = O(\varepsilon^{-2} \log(1/\delta))$ copies of each input state. The circuit has a small per-copy gate count only when controlled-SWAP is available at reasonable cost. The sample complexity cannot be omitted when claiming practical efficiency.

5.11 Controlled operations and no-cloning as linearity constraints

Assume a unitary U clones both $|\psi\rangle$ and $|\phi\rangle$ from a blank state:

$$U |\psi\rangle |0\rangle = |\psi\rangle |\psi\rangle , \qquad U |\phi\rangle |0\rangle = |\phi\rangle |\phi\rangle . \tag{5.19}$$

Inner-product preservation implies $\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2$. If $c = \langle\psi|\phi\rangle$, then $c = c^2$, hence $c \in \{0, 1\}$ for the phases chosen to satisfy the cloning equations. More invariantly, taking absolute values yields $|c| = |c|^2$, so the states are orthogonal or identical up to phase. Nonorthogonal unknown states cannot be cloned perfectly. This proof depends only on unitarity and the specification that both copies reproduce their respective inputs.

Remark 5.19 (Copying classical basis states). CNOT acts as $|x\rangle |0\rangle \mapsto |x\rangle |x\rangle$ for $x \in \{0, 1\}$, so there is no contradiction with copying computational-basis classical data. For a superposition $a |0\rangle + b |1\rangle$, the same operation produces $a |00\rangle + b |11\rangle$, not $(a |0\rangle + b |1\rangle)^{\otimes 2}$.

5.12 Gate synthesis with a controlled phase gadget

The controlled-Z gate is $CZ = \operatorname{diag}(1, 1, 1, -1)$ and is equivalent to a CNOT conjugated by a Hadamard on the target:

$$CZ = (I \otimes H) \operatorname{CNOT} (I \otimes H). \tag{5.20}$$

A gate $e^{-i\theta Z\otimes Z}$ is diagonal with eigenphases $e^{-i\theta}$ on even parity basis states and $e^{i\theta}$ on odd parity states. Implement a CNOT from the first qubit to the second, then $R_z(2\theta)$ on the second, then the same CNOT. The target temporarily holds $x_1 \oplus x_2$, so its phase is exactly the required parity-dependent phase.

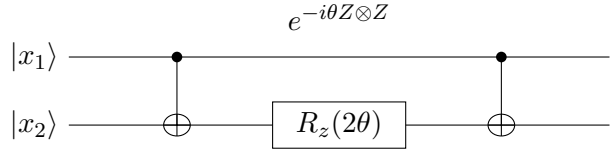


Figure 5.2: A native TikZ parity gadget implementing a two-qubit ZZ rotation using two CNOT gates and one Z rotation.

CHAPTER 6

Hamiltonian Simulation and Complexity Models

6.1 Simulation tasks

Problem 6.1 (Hamiltonian simulation)**.** Given an efficiently specified Hermitian H acting on n qubits, a time t, and an error tolerance $\varepsilon > 0$, construct a quantum circuit $\widetilde{U}$ satisfying

$$\left\| \widetilde{U} - e^{-itH} \right\|_\infty \leq \varepsilon. \tag{6.1}$$

Costs are measured in elementary gates, queries to a specified oracle, and circuit depth; the encoding of H must be part of the problem specification.

6.2 Pauli-string exponentials

For a Pauli string $P = P_1 \otimes \cdots \otimes P_k$, $P^2 = I$ and

$$e^{-it\alpha P} = \cos(\alpha t) I - i \sin(\alpha t) P. \tag{6.2}$$

Conjugate each X by H and each Y by $S^\dagger H$ (with $B_Y = HS^\dagger$ so that $B_Y Y B_Y^\dagger = Z$) to map to a Z string, compute parity onto a target using CNOT, rotate the target by $R_z(2\alpha t)$, and uncompute. The operation acts as $e^{-it\alpha P}$ on the data, up to global phase. The precise basis-change direction must be paired with the chosen conjugation convention.

6.3 Product formulas and error

Theorem 6.2 (Lie–Trotter product formula)**.** *For bounded Hermitian matrices $H_1, \ldots, H_L$,*

$$e^{-it\sum_j H_j} = \lim_{r\to\infty} \left(\prod_{j=1}^{L} e^{-itH_j/r} \right)^r. \tag{6.3}$$

For fixed bounded t, a first-order formula typically incurs global operator-norm error $O(t^2/r)$ with constants depending on commutator norms.

Proof. At step size $\delta = t/r$, $e^{-i\delta(H_1+H_2)} - e^{-i\delta H_1} e^{-i\delta H_2} = O(\delta^2)$ by Taylor expansion. A telescoping sum over r unitary steps bounds the accumulated error by $rO(\delta^2) = O(t^2/r)$. Generalization to finitely many terms is analogous. □

Proposition 6.3 (Symmetric product formula)**.** *For two terms,*

$$e^{-it(H_1+H_2)} = \left(e^{-itH_1/(2r)} e^{-itH_2/r} e^{-itH_1/(2r)} \right)^r + O(t^3/r^2), \tag{6.4}$$

with bounded fixed H_1, H_2 and suitable norm-dependent constants.

6.4 Density-matrix exponentiation

Let S be the SWAP operator on two isomorphic systems. For a stored state ρ and a target σ, applying $e^{-iS\delta}$ and tracing out the stored system yields

$$\mathrm{Tr}_A[e^{-iS\delta}(\rho_A \otimes \sigma_B)e^{iS\delta}] = \sigma - i\delta[\rho, \sigma] + O(\delta^2). \tag{6.5}$$

Iterating with fresh copies of ρ approximates the conjugation $\sigma \mapsto e^{-it\rho}\sigma e^{it\rho}$. This is a copy-access model and must not be confused with arbitrary black-box access to matrix entries.

6.5 Decision, query, and verification complexity

Definition 6.4 (Decision problem)**.** A language is a set $L \subseteq \{0,1\}^*$. Its characteristic function determines whether a bit string x belongs to L. A promise problem specifies disjoint sets of yes- and no-instances; behavior outside their union is irrelevant.

Definition 6.5 (Complexity classes)**.** P contains languages decidable by deterministic polynomial-time classical algorithms; NP contains languages with polynomial-length witnesses verifiable in deterministic polynomial time. BPP and BQP allow classical randomized and quantum algorithms, respectively, with error at most $1/3$ on every valid input. QMA is the quantum-witness analogue of MA with polynomial-time quantum verification. PSPACE and EXP bound classical space polynomially and time exponentially, respectively.

A standard inclusion chain is

$$\mathsf{P} \subseteq \mathsf{BPP} \subseteq \mathsf{BQP} \subseteq \mathsf{PP} \subseteq \mathsf{PSPACE} \subseteq \mathsf{EXP}, \tag{6.6}$$

where the containments are not all known to be strict. Also $\mathsf{NP} \subseteq \mathsf{PSPACE}$ and $\mathsf{QMA} \subseteq \mathsf{PP}$. Whether $\mathsf{P} = \mathsf{NP}$ or BQP contains NP-complete languages remains unresolved.

Definition 6.6 (Quantum query model)**.** An input function $f : \{0,1\}^n \to \{0,1\}^m$ is accessed via a unitary oracle $O_f |x\rangle |z\rangle = |x\rangle |z \oplus f(x)\rangle$. Quantum query complexity minimizes the number of calls to O_f needed to solve a stated promise problem with a specified success criterion.

6.6 Phase kickback and interference

If $f : \{0,1\}^n \to \{0,1\}$, then setting the response qubit to $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ gives

$$O_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle . \tag{6.7}$$

The function value is encoded as a relative phase, not as a directly accessible list of all 2^n evaluations. Interference under a subsequent unitary concentrates amplitudes of decision-relevant global properties.

6.7 Simulation accuracy and product formulas

6.7.1 The task and its norm dependence

Given a Hermitian H and evolution time t, Hamiltonian simulation aims to implement a circuit V such that $\left\|V - e^{-itH}\right\|_\infty \le \varepsilon$, or a comparably strong operational approximation. The statement is incomplete without an access model: explicit local terms, sparse-matrix oracles, block encodings, and copies of a density matrix have different costs. An error tolerance expressed in operator norm bounds the Euclidean error on every input vector, but sample complexity for estimating a final observable remains an additional cost.

Example 6.7 (One exactly solvable Hamiltonian)**.** For $H = \omega Z$,

$$e^{-itH} = \begin{pmatrix} e^{-i\omega t} & 0 \\ 0 & e^{i\omega t} \end{pmatrix} = R_z(2\omega t). \tag{6.8}$$

The evolution of $|+\rangle$ is $(e^{-i\omega t}|0\rangle + e^{i\omega t}|1\rangle)/\sqrt{2}$; measuring X gives expectation $\cos(2\omega t)$, which can be verified from the Bloch-sphere rotation picture.

6.7.2 First-order Lie product formula and commutators

Let $H = A + B$. For small δ, the expansions

$$e^{-i\delta(A+B)} = I - i\delta(A+B) - \frac{\delta^2}{2}(A+B)^2 + O(\delta^3), \tag{6.9}$$

$$e^{-i\delta A}e^{-i\delta B} = I - i\delta(A+B) - \frac{\delta^2}{2}(A^2+B^2) - \delta^2 AB + O(\delta^3) \tag{6.10}$$

differ at second order by $-\delta^2[A,B]/2$ (with the sign depending on the difference orientation). Consequently the local error is $O(\delta^2 \|[A,B]\|)$ and the r-slice global error is typically $O(t^2 \|[A,B]\|/r)$ at fixed bounded terms. The cumulative bound follows from a telescoping sum

$$U^r - V^r = \sum_{j=0}^{r-1} U^{r-j-1}(U-V)V^j, \tag{6.11}$$

whose norm is at most $r\|U-V\|$ for unitary U, V.

Example 6.8 (A noncommuting Pauli Hamiltonian)**.** Set $A = aX$, $B = bZ$. Since $[X,Z] = -2iY$, we obtain $[A,B] = -2iabY$ and $\|[A,B]\| = 2|ab|$. Thus one cannot use $e^{-itaX}e^{-itbZ}$ as an exact factorization unless $ab = 0$. With r steps, the leading error has order $|ab|\,t^2/r$.

6.7.3 Second-order symmetric splitting

The Strang step

$$S_2(\delta) = e^{-i\delta A/2}e^{-i\delta B}e^{-i\delta A/2} \tag{6.12}$$

is symmetric under time reversal: $S_2(-\delta) = S_2(\delta)^{-1}$. Its even-order logarithmic error terms cancel, yielding a local $O(\delta^3)$ error for bounded A, B and a global $O(t^3/r^2)$ behavior, with constants controlled by nested commutators such as $[A,[A,B]]$ and $[B,[A,B]]$. If A commutes with B, both first- and second-order formulas are exact.

6.8 Density matrix exponentiation and access limitations

For two registers with dimensions matching ρ, let S be the SWAP operator and consider a small partial-SWAP $e^{-i\delta S}$. Expanding through first order and tracing out an auxiliary register in state ρ gives

$$\mathrm{Tr}_E\left[e^{-i\delta S}(\sigma \otimes \rho)e^{i\delta S}\right] = \sigma - i\delta[\rho, \sigma] + O(\delta^2). \tag{6.13}$$

This approximates the effect of $e^{-i\delta\rho}\sigma e^{i\delta\rho}$ on the target. Repeating the operation with fresh copies of ρ simulates its exponentiation. The fresh-copy requirement and finite-step error are resource costs; this procedure does not give free random access to the entries of an unknown matrix.

6.9 Complexity-theoretic models and promises

A deterministic classical computation is a family of circuits or Turing-machine transitions acting on bit strings. A uniform polynomial-size quantum circuit family is a sequence of efficiently constructible unitary circuits, with designated computational-basis measurements on output registers. The class BQP contains promise decision problems for which acceptance probability is at least 2/3 on yes instances and at most 1/3 on no instances. The constants can be amplified exponentially by independent repetition and majority vote with polynomial overhead.

Definition 6.9 (Quantum verification)**.** A promise problem lies in QMA if a polynomial-time uniform quantum verifier receives a polynomial-qubit witness $|\psi\rangle$ and has completeness and soundness bounded away from one another: a witness gives acceptance at least 2/3 on yes instances, whereas every witness gives acceptance at most 1/3 on no instances.

The witness may be entangled across its qubits. Unlike BQP, which has no external quantum witness, QMA formulates a proof-verification task. Whether particular complexity classes coincide is generally unknown; a circuit-model exposition must not claim a proved separation that has not been established.

6.9.1 Query complexity versus overall computation

An oracle O_f represents one unit of access to an unknown function f. The query complexity of an algorithm counts applications of O_f and $O_f^\dagger$, often while treating other elementary gates as free. In the standard Boolean reversible oracle,

$$O_f\,|x\rangle\,|b\rangle = |x\rangle\,|b \oplus f(x)\rangle\,. \tag{6.14}$$

Preparing the output qubit in $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ gives

$$O_f\,|x\rangle\,|-\rangle = (-1)^{f(x)}\,|x\rangle\,|-\rangle\,, \tag{6.15}$$

which implements phase kickback. This transformation is the starting point for Deutsch–Jozsa, Bernstein–Vazirani, Fourier sampling, and Grover-type procedures.

Example 6.10 (Query access is not input loading)**.** If an instance consists of an arbitrary list of $N = 2^n$ classical numbers, constructing a coherent index oracle for that list may itself

require $\Omega(N)$ preprocessing or a specialized memory structure. A poly(n) query bound does not imply an end-to-end poly(n) algorithm unless the oracle access cost is separately justified.

6.10 An exact comparison of Trotter steps with the target evolution

Take $H = X + Z$. Since $(X + Z)^2 = 2I$, the exact unitary is

$$U(t) = e^{-it(X+Z)} = \cos(\sqrt{2}t)I - i\frac{\sin(\sqrt{2}t)}{\sqrt{2}}(X + Z). \tag{6.16}$$

By contrast a first-order split step is

$$V(t) = e^{-itX}e^{-itZ} \tag{6.17}$$

$$= \cos^2 t\, I - i \sin t \cos t\, (X + Z) + i \sin^2 t\, Y, \tag{6.18}$$

where $XZ = -iY$ has been used. The additional Y term is a direct signature of non-commutativity. Its leading magnitude t^2 is consistent with the Baker–Campbell–Hausdorff commutator correction. With r steps of duration t/r, the unwanted term per step is $O(t^2/r^2)$, so the total leading error decreases as $O(t^2/r)$.

Example 6.11 (A limiting comparison)**.** At $t = \pi/2$, the one-step product is $e^{-i\pi X/2}e^{-i\pi Z/2} = (-iX)(-iZ) = -XZ = iY$, whereas the exact evolution is $\cos(\pi/\sqrt{2})I - i\sin(\pi/\sqrt{2})(X + Z)/\sqrt{2}$. They are very different unitaries. Product formulas are controlled approximations only when their step size is sufficiently small relative to the norms and commutators.

6.11 Phase-kickback constructions from a reversible oracle

The Boolean oracle $O_f |x\rangle |b\rangle = |x\rangle |b \oplus f(x)\rangle$ is unitary because it permutes computational-basis vectors. For the target $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, $X |-\rangle = - |-\rangle$, so

$$O_f |x\rangle |-\rangle = |x\rangle\ X^{f(x)} |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle . \tag{6.19}$$

On a superposition, linearity preserves coherence of the phase labels:

$$O_f \left(\sum_x \alpha_x |x\rangle \right) |-\rangle = \left(\sum_x (-1)^{f(x)} \alpha_x |x\rangle \right) |-\rangle . \tag{6.20}$$

The target register factorizes from the input after the query and can be ignored. By contrast, starting the target in $|0\rangle$ computes $f(x)$ into a separate register and generally entangles it with x.

6.12 Promise amplification without altering the accepted language

Suppose a quantum decision procedure accepts a yes instance with probability at least $2/3$ and a no instance with probability at most $1/3$. Perform m independent runs and accept if a majority accept. For each fixed input, the number of accepted runs is a sum of independent Bernoulli variables. Hoeffding's inequality bounds the wrong-majority event by

$$\mathbb{P}(\text{error}) \leq e^{-2m(1/6)^2} = e^{-m/18}. \tag{6.21}$$

Thus error δ is achieved with $m \geq 18 \ln(1/\delta)$ independent trials. The bound is conservative but sufficient to justify replacing the constant completeness-soundness gap by exponentially small error with polynomial overhead.

6.13 Why controlled global phases can matter

An unconditional replacement $U \mapsto e^{i\phi}U$ has no effect on outcome probabilities of a circuit that applies only U to a system. However, controlled-U and controlled-$(e^{i\phi}U)$ differ by the phase applied to the control's $|1\rangle$ branch:

$$\Lambda(e^{i\phi}U) = (|0\rangle\langle 0| + e^{i\phi}\,|1\rangle\langle 1|) \otimes I\ \Lambda(U). \tag{6.22}$$

This relative phase can be detected by a subsequent Hadamard on the control. In phase estimation the eigenphase of U is exactly such a relative phase, so the global phase convention of the controlled unitary must be fixed.

CHAPTER 7

Oracle Algorithms and Amplitude Amplification

7.1 Fourier sampling over the Boolean cube

For $x, s \in \{0,1\}^n$, write $s \cdot x = \sum_i s_i x_i \pmod 2$. The Walsh–Hadamard transform acts as

$$H^{\otimes n} |x\rangle = 2^{-n/2} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle . \tag{7.1}$$

This identity underlies phase-query algorithms.

7.2 Deutsch–Jozsa

Problem 7.1 (Deutsch–Jozsa promise)**.** Given query access to $f : \{0,1\}^n \to \{0,1\}$ promised either to be constant or balanced ($|f^{-1}(0)| = |f^{-1}(1)| = 2^{n-1}$), decide which case holds.

Prepare $|0^n\rangle |-\rangle$, apply $H^{\otimes n}$ to the first register, query O_f, then apply $H^{\otimes n}$ again. The amplitude of $|0^n\rangle$ is $2^{-n} \sum_x (-1)^{f(x)}$: it has magnitude one for a constant function and zero for a balanced one. The quantum algorithm uses one oracle query and solves the promise problem exactly.

7.3 Bernstein–Vazirani

Problem 7.2 (Hidden linear function)**.** For an unknown $s \in \{0,1\}^n$, the oracle is $f_s(x) = s \cdot x \pmod 2$. Recover s.

Phase kickback creates $2^{-n/2} \sum_x (-1)^{s \cdot x} |x\rangle = H^{\otimes n} |s\rangle$. A final $H^{\otimes n}$ returns $|s\rangle$ with certainty using one quantum query.

7.4 Simon's problem

Problem 7.3 (Hidden XOR mask)**.** Given $f : \{0,1\}^n \to \{0,1\}^m$ with the promise that some nonzero $s \in \{0,1\}^n$ satisfies $f(x) = f(y)$ if and only if $y \in \{x, x \oplus s\}$, determine s.

Querying a uniform superposition and measuring the output yields $(|x\rangle + |x \oplus s\rangle)/\sqrt{2}$ in the input register. After $H^{\otimes n}$, only strings y with $y \cdot s = 0$ occur, uniformly over the orthogonal hyperplane. Collect $O(n)$ independent linear constraints with high probability and solve them by Gaussian elimination over $\mathbb{F}_2$ to recover s.

7.5 Grover search

Problem 7.4 (Unstructured search)**.** Given an oracle recognizing a marked subset $G \subseteq [N]$ of size $M \geq 1$, output a marked element with constant success probability, minimizing oracle calls.

Let $|g\rangle = M^{-1/2} \sum_{x \in G} |x\rangle$ and $|b\rangle = (N-M)^{-1/2} \sum_{x \notin G} |x\rangle$. Define $\sin^2 \theta = M/N$, $|u\rangle = \sin\theta\,|g\rangle + \cos\theta\,|b\rangle$. The phase oracle $O_G = I - 2\sum_{x \in G} |x\rangle\langle x|$ and diffusion $D = 2\,|u\rangle\langle u| - I$ induce the Grover iterate $G = DO_G$.

Theorem 7.5 (Amplitude rotation)**.** *For $r \geq 0$, $G^r\,|u\rangle = \sin((2r+1)\theta)\,|g\rangle + \cos((2r+1)\theta)\,|b\rangle$ and hence*

$$P_{\text{success}}(r) = \sin^2((2r+1)\theta). \tag{7.2}$$

When $M \ll N$, choosing r nearest to $\pi/(4\theta) - 1/2$ gives $\Theta(\sqrt{N/M})$ oracle queries for constant success probability.

Proof. Both reflections preserve the two-dimensional span of $|g\rangle, |b\rangle$. Their product is rotation by 2θ toward $|g\rangle$. Induction on the iterate count yields the formula. □

Remark 7.6. For unstructured black-box search, the $\Theta(\sqrt{N/M})$ scaling is optimal up to constants in the usual bounded-error query model. Knowledge of M enables near-optimal stopping; unknown M requires a modified schedule or amplitude-estimation strategy.

7.6 General amplitude amplification

If a state-preparation unitary A produces $A\,|0\rangle = \sqrt{a}\,|\text{good}\rangle + \sqrt{1-a}\,|\text{bad}\rangle$, reflections about the prepared state and good subspace boost the good probability from a toward one in $O(1/\sqrt{a})$ iterations, provided these reflections can be implemented. This formulation abstracts away the uniform-search assumption.

7.7 Boolean Fourier analysis and phase oracles

Functions $f : \{0,1\}^n \to \mathbb{R}$ can be expanded in the orthonormal character basis

$$\chi_s(x) = (-1)^{s\cdot x}, \qquad \widehat{f}(s) = 2^{-n} \sum_{x\in\{0,1\}^n} f(x)(-1)^{s\cdot x}, \tag{7.3}$$

where $s \cdot x = \bigoplus_{j=1}^{n} s_j x_j$. Orthogonality is the identity

$$2^{-n} \sum_{x} \chi_s(x)\chi_t(x) = \mathbf{1}\{s = t\}. \tag{7.4}$$

Indeed the sum factorizes over coordinates; for $s_j \neq t_j$ the contribution $1 + (-1)$ vanishes. Accordingly Fourier inversion and Parseval's identity follow:

$$f(x) = \sum_{s} \widehat{f}(s)\chi_s(x), \qquad \sum_{s} \widehat{f}(s)^2 = \mathbb{E}_x f(x)^2. \tag{7.5}$$

Proposition 7.7 (Fourier sampling from a phase query)**.** *For $f : \{0,1\}^n \to \{\pm 1\}$ with efficient phase access $U_f |x\rangle = f(x) |x\rangle$, the circuit $H^{\otimes n} U_f H^{\otimes n} |0^n\rangle$ produces*

$$\sum_{s\in\{0,1\}^n} \widehat{f}(s) |s\rangle . \tag{7.6}$$

Therefore measuring produces index s with probability $\widehat{f}(s)^2$.

Proof. The first Hadamards prepare $2^{-n/2} \sum_x |x\rangle$. The phase query multiplies each amplitude by $f(x)$. A final Hadamard transform maps $|x\rangle$ to $2^{-n/2} \sum_s (-1)^{s\cdot x} |s\rangle$, so the coefficient of $|s\rangle$ is $2^{-n} \sum_x f(x)(-1)^{s\cdot x} = \widehat{f}(s)$. Parseval ensures normalization. □

7.8 Promise problems and deterministic interference

7.8.1 Deutsch–Jozsa: exact cancellation

The promise is that $f : \{0,1\}^n \to \{0,1\}$ is either constant or balanced. Prepare the query state $2^{-n/2} \sum_x |x\rangle |-\rangle$, perform one standard oracle query, and apply $H^{\otimes n}$ to the input. The amplitude of $|0^n\rangle$ is

$$2^{-n} \sum_{x} (-1)^{f(x)} = \begin{cases} +1 \text{ or } -1, & f \text{ constant}, \\ 0, & f \text{ balanced}. \end{cases} \tag{7.7}$$

Therefore measurement of all zeros identifies the constant case with certainty; any other outcome identifies the balanced case. The correctness depends on the stated promise; without it, the same circuit estimates the global bias rather than solving arbitrary function classification in one query.

Example 7.8 (Four-bit balanced oracle)**.** Take $f(x) = x_1 \oplus x_3$ on $\{0,1\}^4$. Then $(-1)^{f(x)} = (-1)^{(1010)\cdot x}$ under the coordinate order (x_1, x_2, x_3, x_4). The transformed register is exactly $|1010\rangle$, and it is never $|0000\rangle$. This is simultaneously a Deutsch–Jozsa balanced instance and a Bernstein–Vazirani instance.

7.8.2 Bernstein–Vazirani: exact hidden-string recovery

Suppose $f_s(x) = s \cdot x \pmod 2$ for an unknown $s \in \{0,1\}^n$. The phase $(-1)^{f_s(x)}$ is precisely the character $\chi_s(x)$. Its Fourier spectrum has a single unit coefficient at s, because

$$2^{-n} \sum_x (-1)^{(s \oplus t)\cdot x} = \mathbf{1}\{s = t\}. \tag{7.8}$$

Thus one quantum query and one measurement reveal the full hidden string exactly. A deterministic classical query algorithm needs n linearly independent function evaluations in the worst case to determine all n bits of s. This is an oracle separation, not a claim that arbitrary length-n input strings can be learned with one black-box bit read.

7.9 Simon periodicity and linear algebra over $\mathbb{F}_2$

Let $f : \{0,1\}^n \to \mathcal{Y}$ satisfy the promise $f(x) = f(y)$ if and only if $y = x$ or $y = x \oplus s$, where $s \neq 0$ is an unknown mask. Start with a uniform superposition on the query register and compute $f(x)$ into a second register:

$$2^{-n/2} \sum_x |x\rangle\,|f(x)\rangle\,. \tag{7.9}$$

Conditioned on observing a particular function value, the first register collapses to $(|x\rangle + |x \oplus s\rangle)/\sqrt{2}$. Applying Hadamards gives an amplitude proportional to

$$(-1)^{x\cdot y}\big[1 + (-1)^{s\cdot y}\big], \tag{7.10}$$

which vanishes unless $s \cdot y = 0 \pmod 2$. Among the 2^{n-1} vectors satisfying that equation, the measurement distribution is uniform. Independent repetitions yield homogeneous linear constraints on s, which can be solved by Gaussian elimination over $\mathbb{F}_2$.

Example 7.9 (Three-bit hidden mask)**.** Suppose $s = (1,0,1)$. Every sample y obeys $y_1 \oplus y_3 = 0$, so possible outputs include $000, 010, 101, 111$. If the samples 010 and 101 are obtained, the constraints give $s_2 = 0$ and $s_1 = s_3$. With the nonzero-mask promise, the unique solution is 101.

Proposition 7.10 (Sampling complexity for Simon's problem)**.** *After $n - 1 + t$ independent uniformly random samples from the subspace $s^\perp$, the probability that the samples fail to span $s^\perp$ is at most 2^{-t}.*

Proof. The subspace has dimension $n - 1$. If the sample span is proper, there is a nonzero linear functional on $s^\perp$ that vanishes on every sample. For a fixed nonzero functional, each sample satisfies it with probability $1/2$. Taking a union bound over fewer than 2^{n-1} nonzero functionals gives failure probability less than $2^{n-1}2^{-(n-1+t)} = 2^{-t}$. □

7.10 Grover iteration as a two-dimensional rotation

Let N be the search-space size, M the number of marked elements with $1 \le M < N$. Introduce normalized equal superpositions of marked and unmarked states,

$$|w\rangle = \frac{1}{\sqrt{M}} \sum_{f(x)=1} |x\rangle , \qquad |r\rangle = \frac{1}{\sqrt{N-M}} \sum_{f(x)=0} |x\rangle . \tag{7.11}$$

The uniform starting state decomposes as

$$|s\rangle = \sqrt{\frac{M}{N}}\,|w\rangle + \sqrt{1-\frac{M}{N}}\,|r\rangle = \sin\theta\,|w\rangle + \cos\theta\,|r\rangle , \qquad \theta = \arcsin\sqrt{M/N}. \tag{7.12}$$

Define the marking reflection $O = I - 2\Pi_{\text{marked}}$ and the diffusion reflection $D = 2\,|s\rangle\langle s| - I$. In the basis $(|w\rangle , |r\rangle)$, their product acts as a planar rotation by 2θ. Hence after k Grover iterations,

$$G^k\,|s\rangle = \sin((2k+1)\theta)\,|w\rangle + \cos((2k+1)\theta)\,|r\rangle . \tag{7.13}$$

The success probability is $\sin^2((2k+1)\theta)$. For $M \ll N$ the first maximum occurs near

$$k \approx \frac{\pi}{4\theta} - \frac{1}{2} \sim \frac{\pi}{4}\sqrt{\frac{N}{M}}. \tag{7.14}$$

Example 7.11 (Eight-element search)**.** For $N = 8, M = 2$, $\sin\theta = 1/2$, so $\theta = \pi/6$. One Grover iteration gives angle $3\theta = \pi/2$ and succeeds with probability one. Two iterations instead give angle $5\pi/6$ and success probability $1/4$. Repeating beyond the optimal iteration count may reduce success, an effect hidden by a mere big-O query statement.

Example 7.12 (A large search with several solutions)**.** For $N = 1024$ and $M = 16$, $\theta = \arcsin(1/8) \approx 0.12533$. The first near-optimal count is $k \approx \pi/(4\theta) - 1/2 \approx 5.77$, so $k = 6$ iterations give success probability $\sin^2(13\theta) \approx 0.997$. The scaling is $\Theta(\sqrt{N/M})$, rather than $\Theta(\sqrt{N})$ when M grows with N.

7.10.1 General amplitude amplification

Let $A\,|0\rangle = \sqrt{a}\,|\psi_{\text{good}}\rangle + \sqrt{1-a}\,|\psi_{\text{bad}}\rangle$ with orthogonal good and bad sectors. Define reflections about the starting state and the good subspace. Their product again gives a rotation in the two-dimensional span of $|\psi_{\text{good}}\rangle$ and $|\psi_{\text{bad}}\rangle$, leading to $O(1/\sqrt{a})$ invocations of $A, A^\dagger$ for constant success when a is known. If the success probability is not known, blindly iterating the standard Grover operator can overshoot; randomized schedules or phase-adjusted procedures address this limitation.

Remark 7.13 (Optimality in the unstructured query model)*.* The quadratic gain is asymptotically tight: unstructured search with one marked item has quantum query lower bound $\Omega(\sqrt{N})$. This lower bound applies to black-box access, and makes no claim about structured search problems that admit additional exploitable algebraic information.

7.11 Explicit Boolean Fourier spectra

Example 7.14 (Parity and dictator functions)**.** For $f(x) = (-1)^{x_1}$, the only nonzero Fourier coefficient is $\widehat{f}(10\cdots0) = 1$. Fourier sampling therefore deterministically outputs the first-coordinate label. For constant $f \equiv -1$, the only coefficient is $\widehat{f}(0^n) = -1$, and measurement always outputs 0^n: the phase sign is global and unobservable. Both facts follow immediately from character orthogonality.

Example 7.15 (A function with nontrivial Fourier support)**.** Let $n = 2$ and $f(00) = f(01) = f(10) = 1$, $f(11) = -1$. Evaluating the four Fourier coefficients gives

$$\widehat{f}(00) = \frac{1}{2}, \quad \widehat{f}(01) = \frac{1}{2}, \quad \widehat{f}(10) = \frac{1}{2}, \quad \widehat{f}(11) = -\frac{1}{2}. \tag{7.15}$$

The Fourier-sampling circuit has uniform probability $1/4$ over the four labels, despite the input Boolean function being highly nonuniform in the computational truth table. The negative Fourier coefficient contributes a relative output phase but not a negative measurement probability.

7.12 The precise two-dimensional Grover matrix

In the ordered basis $(|w\rangle, |r\rangle)$, the marking oracle is $O = \mathrm{diag}(-1, 1)$. With $|s\rangle = (\sin\theta, \cos\theta)^{\mathsf{T}}$, the reflection about $|s\rangle$ is

$$D = 2|s\rangle\langle s| - I = \begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}. \tag{7.16}$$

Thus the Grover iterate is

$$G = DO = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix}. \tag{7.17}$$

Multiplication by $(\sin\theta, \cos\theta)^{\mathsf{T}}$ gives $(\sin 3\theta, \cos 3\theta)^{\mathsf{T}}$, and induction yields $(\sin(2k+1)\theta, \cos(2k+1)\theta)^{\mathsf{T}}$ after k iterations. The matrix proof makes clear that the algorithm acts entirely in a two-dimensional invariant subspace, no matter how large the original Hilbert space.

7.13 Robust handling of an unknown marked fraction

If M is not known in advance, the angle $\theta = \arcsin\sqrt{M/N}$ is unknown. A fixed iteration count might produce a high success probability for one M but a low one for another because the Grover rotation oscillates. A standard approach samples a number of iterations from a growing interval, measures, and verifies a candidate by querying f. The resulting expected query complexity remains $O(\sqrt{N/M})$ under appropriate assumptions without relying on an exact prior estimate of M. Verification matters: amplitude amplification produces a candidate, not a formal certificate that it is marked unless the predicate can be checked.

Example 7.16 (Overshooting)**.** For $N = 8$ and $M = 2$, $\theta = \pi/6$ and one iteration succeeds with probability one. After four iterations the angle is $9\theta = 3\pi/2$, also giving success

probability one; but after two iterations the success probability falls to 1/4. This oscillation demonstrates why more Grover queries do not imply monotonically better success.

7.14 Simon's linear equations as a complete algorithmic pipeline

Let $n = 4$ and suppose the unknown mask is $s = (1, 0, 1, 1)$. Each Fourier sample y satisfies

$$y_1 \oplus y_3 \oplus y_4 = 0. \tag{7.18}$$

Possible independent samples include $y^{(1)} = 0100$, $y^{(2)} = 1010$, and $y^{(3)} = 1001$. Writing the system $Ys = 0$ over $\mathbb{F}_2$ produces

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \end{pmatrix} = 0. \tag{7.19}$$

The equations force $s_2 = 0$, $s_3 = s_1$, and $s_4 = s_1$. The nonzero-promise condition selects $s = (1, 0, 1, 1)$. If the sampled rows were linearly dependent, Gaussian elimination would leave several candidate masks, and more oracle queries would be needed.

CHAPTER 8

Quantum Fourier Transform, Phase Estimation, and Factoring

8.1 Finite Fourier transforms

Definition 8.1 (Quantum Fourier transform)**.** For $N \geq 2$, $\omega_N = e^{2\pi i/N}$ and

$$F_N |x\rangle = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega_N^{xy} |y\rangle , \qquad F_N^\dagger |y\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \omega_N^{-xy} |x\rangle . \tag{8.1}$$

Orthogonality of roots of unity implies $F_N^\dagger F_N = I$. For $N = 2^n$, standard exact circuits implement F_N using $O(n^2)$ one-qubit Hadamard and controlled-phase gates, plus $O(n)$ swaps; approximate QFT admits additional gate savings.

Proposition 8.2 (Product representation)**.** *Writing $x = x_{n-1} \cdots x_0$ in binary and $0.x_j \cdots x_0$ for a binary fraction,*

$$F_{2^n} |x_{n-1} \cdots x_0\rangle = 2^{-n/2} \bigotimes_{j=0}^{n-1} (|0\rangle + e^{2\pi i\, 0.x_j \cdots x_0} |1\rangle), \tag{8.2}$$

where tensor factors appear in reversed output-bit order relative to the conventional circuit without final swaps.

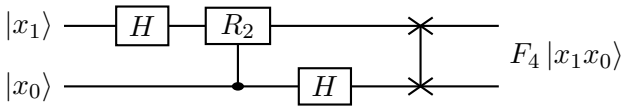


Figure 8.1: Two-qubit QFT structure, using a controlled phase $R_2 = \mathrm{diag}(1, i)$, Hadamards, and a final bit reversal. Gate-order conventions must match the sign convention of F_N.

8.2 Phase estimation

Problem 8.3 (Eigenphase estimation)**.** Given a unitary U and an eigenstate $U |\psi\rangle = e^{2\pi i\theta} |\psi\rangle$, estimate $\theta \in [0, 1)$ using controlled powers U^{2^j}, with specified error and success probability.

Starting with m control qubits in $|+\rangle^{\otimes m}$, controlled powers produce

$$\frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} e^{2\pi i k\theta} |k\rangle |\psi\rangle . \tag{8.3}$$

Applying $F_{2^m}^\dagger$ to the first register gives $|a\rangle$ with certainty if $\theta = a/2^m$. If θ is not representable

exactly, outcome a has probability

$$p(a) = \frac{1}{2^{2m}} \left| \sum_{k=0}^{2^m-1} e^{2\pi i k(\theta - a/2^m)} \right|^2, \tag{8.4}$$

which is peaked at the nearest grid points. Extra precision bits and repetition control the error probability.

Example 8.4 (Exact binary phase)**.** With $m = 3$ and $\theta = 5/8$, the control register is $|101\rangle$ after inverse QFT, so the measured integer is $a = 5$.

8.3 Order finding and Shor factoring

Problem 8.5 (Order finding)**.** For coprime integers a, N, the multiplicative order $r = \mathrm{ord}_N(a)$ is the least positive r such that $a^r \equiv 1 \pmod{N}$.

Define $U_a |y\rangle = |ay \bmod N\rangle$ on the invertible residue subspace. Its eigenphases include s/r, and QPE yields rational approximations from which continued fractions can recover r if precision is sufficient and the sampled numerator is suitable.

Theorem 8.6 (Factoring reduction)**.** *If N is odd and composite, a is coprime to N, $r = \mathrm{ord}_N(a)$ is even, and $a^{r/2} \not\equiv -1 \pmod{N}$, then at least one of*

$$\gcd(a^{r/2} - 1, N), \qquad \gcd(a^{r/2} + 1, N) \tag{8.5}$$

provides a nontrivial factor of N.

Proof. Because $(a^{r/2} - 1)(a^{r/2} + 1) \equiv 0 \pmod{N}$, factors of N are distributed among the two factors. The hypotheses exclude $a^{r/2} \equiv \pm 1 \pmod{N}$, making the gcd step nontrivial for at least one choice. □

Remark 8.7. The quantum advantage arises from efficient period finding, while modular arithmetic, continued fractions, and gcd are classical polynomial-time routines. A quantum Fourier transform does not permit full readout of all complex Fourier amplitudes by one measurement.

8.4 Quantum linear-system algorithms

Problem 8.8 (Quantum linear-system state preparation)**.** Given efficient oracle or block-encoding access to a well-conditioned suitably normalized matrix A and a prepared state $|b\rangle$, produce a state approximating $|x\rangle = A^{-1} |b\rangle / \left\| A^{-1} |b\rangle \right\|$.

Phase estimation on an operator encoding A and a controlled eigenvalue-dependent rotation implement the central inverse-eigenvalue operation. Resource bounds depend on the condition number κ, precision ε, data-access model, and output format. The output is a quantum state, not an explicit list of all entries of x, so exponential claims relative to classical solvers require careful treatment of state preparation and readout costs.

8.5 Exact Fourier identities and circuit conventions

For $N \geq 1$, define $\omega_N = e^{2\pi i/N}$ and

$$F_N |x\rangle = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega_N^{xy} |y\rangle . \tag{8.6}$$

The matrix F_N is unitary because

$$\langle x| F_N^\dagger F_N |x'\rangle = \frac{1}{N} \sum_{y=0}^{N-1} \omega_N^{(x'-x)y} = \delta_{xx'}. \tag{8.7}$$

The finite geometric sum is N if $x = x' \pmod N$ and zero otherwise. The inverse transform replaces ω_N with ω_N^{-1}. Fixing the sign of the Fourier phase is essential: phase-estimation circuits using an inverse QFT must be paired with the consistent controlled-unitary convention.

Example 8.9 (Fourier transform of $|1\rangle$ for $N = 4$)**.** With positive-phase convention, $\omega_4 = i$, so

$$F_4 |1\rangle = \frac{1}{2}(|0\rangle + i |1\rangle - |2\rangle - i |3\rangle). \tag{8.8}$$

The factor $1/2$ is mandatory for normalization, and the sign of the i term changes for the inverse transform.

8.5.1 Recursive construction with an explicit output permutation

Let $N = 2^n$, $D_N = \mathrm{diag}(1, \omega_{2N}, \ldots, \omega_{2N}^{N-1})$ and $C_N = |0\rangle\langle 0| \otimes I_N + |1\rangle\langle 1| \otimes D_N$. Define

$$U_N = (I_2 \otimes F_N) C_N (H \otimes I_N) = \frac{1}{\sqrt{2}} \begin{pmatrix} F_N & F_N \\ F_N D_N & -F_N D_N \end{pmatrix}. \tag{8.9}$$

We identify input basis $|bN + j\rangle$ with $|b\rangle |j\rangle$, and let the permutation $\Pi_N |a\rangle |k\rangle = |2k + a\rangle$. For output $y = 2k + a$ and input $x = bN + j$, matrix multiplication gives

$$\langle y| \Pi_N U_N |x\rangle = \frac{1}{\sqrt{2N}} (-1)^{ab} \omega_N^{jk} \omega_{2N}^{aj}, \tag{8.10}$$

$$\omega_{2N}^{xy} = \omega_{2N}^{(bN+j)(2k+a)} = (-1)^{ab} \omega_N^{jk} \omega_{2N}^{aj}. \tag{8.11}$$

Hence the exact recursive identity is

$$\boxed{F_{2N} = \Pi_N (I_2 \otimes F_N) C_N (H \otimes I_N).} \tag{8.12}$$

The permutation Π_N moves the leading output qubit to the least significant position; it must not be silently dropped when the computational basis ordering is fixed. Repeating the construction yields $O(n^2)$ controlled-phase gates and Hadamards, plus a possible final bit reversal.

8.6 Phase estimation as a discrete interference calculation

Let $U|u\rangle = e^{2\pi i\varphi}|u\rangle$, $0 \le \varphi < 1$. For an m-qubit control register with $M = 2^m$, controlled powers $U^{2^0}, \dots, U^{2^{m-1}}$ yield

$$\frac{1}{\sqrt{M}} \sum_{k=0}^{M-1} e^{2\pi i k\varphi} |k\rangle |u\rangle . \tag{8.13}$$

Applying $F_M^\dagger$ gives the amplitude

$$\alpha_y = \frac{1}{M} \sum_{k=0}^{M-1} e^{2\pi i k(\varphi - y/M)}. \tag{8.14}$$

If $\varphi - y/M$ is not an integer, the geometric series yields

$$|\alpha_y|^2 = \frac{1}{M^2} \frac{\sin^2(\pi M(\varphi - y/M))}{\sin^2(\pi(\varphi - y/M))}. \tag{8.15}$$

The removable singularity is interpreted by continuity, giving probability one when $\varphi = y/M$. The distribution is concentrated near integers $y \approx M\varphi$; it is not flat when the phase is not exactly represented.

Example 8.10 (Three control qubits and an exactly representable phase)**.** For $m = 3$ and $\varphi = 5/8$, the phase register before inverse QFT is $F_8|5\rangle$, so the inverse QFT returns $|5\rangle = |101\rangle$ with certainty. The output string 101 is the binary expansion of 5, not 011; its interpretation uses most significant bit first.

Example 8.11 (An inexact phase)**.** Let $m = 2$ and $\varphi = 1/3$. Then $M = 4$ and the nearest integers to $4/3$ are $y = 1$ and $y = 2$. The probability of $y = 1$ is

$$p(1) = \frac{1}{16} \frac{\sin^2(\pi/3)}{\sin^2(\pi/12)} \approx 0.69976. \tag{8.16}$$

The algorithm need not produce the nearest integer with certainty, but repetition and a larger control register can increase useful precision.

8.6.1 Eigenstate requirements and phase mixtures

If the input is $|\psi\rangle = \sum_j c_j |u_j\rangle$ with distinct eigenphases φ_j, phase estimation produces a superposition of branch states. Measurement yields a mixture of the individual phase-estimation distributions weighted by $|c_j|^2$. One cannot interpret such an output as a single well-defined phase unless the input lies in one eigenspace (or all occupied eigenvectors share the same eigenphase).

8.7 From order finding to integer factorization

For a composite odd integer N, choose a uniformly with $1 < a < N$. If $\gcd(a, N) > 1$, a nontrivial factor is already known. Otherwise find the multiplicative order $r = \mathrm{ord}_N(a)$ defined by $a^r \equiv 1 \pmod N$ and $a^j \not\equiv 1 \pmod N$ for $1 \le j < r$. The modular multiplication unitary

$$U_a |x\rangle = |ax \bmod N\rangle \tag{8.17}$$

is extended as a permutation to a convenient power-of-two register. Phase estimation on an appropriate input samples rational phases k/r, which can be recovered from a sufficiently accurate measured fraction using continued fractions.

Proposition 8.12 (Factor extraction from a suitable order)**.** *If r is even and $a^{r/2} \not\equiv -1 \pmod N$, then*

$$\gcd(a^{r/2} - 1, N), \qquad \gcd(a^{r/2} + 1, N) \tag{8.18}$$

provide nontrivial factors (at least one is nontrivial, and for the standard odd composite case both are proper divisors when the conditions hold).

Proof. Write $x = a^{r/2}$. Then $x^2 \equiv 1 \pmod N$, so N divides $(x-1)(x+1)$. Minimality of r implies $x \not\equiv 1 \pmod N$, and the stated condition excludes $x \equiv -1 \pmod N$. Therefore neither $x-1$ nor $x+1$ is a multiple of N, while their product is, allowing nontrivial common divisors with N. □

Example 8.13 (Factoring 15)**.** Choose $a = 2$, coprime to 15. The powers $2, 4, 8, 1$ modulo 15 show $r = 4$. Since $r/2 = 2$ and $2^2 = 4 \not\equiv -1 \pmod{15}$, compute

$$\gcd(4 - 1, 15) = 3, \qquad \gcd(4 + 1, 15) = 5. \tag{8.19}$$

This is the classical postprocessing stage of Shor's reduction. Quantum phase estimation supplies information about r, but the final greatest-common-divisor calculations are classical.

8.8 Quantum linear systems: state preparation is not vector readout

For Hermitian A with eigenvalues bounded away from zero and an efficiently preparable $|b\rangle = \sum_j b_j |u_j\rangle$, the target is the normalized state proportional to

$$A^{-1} |b\rangle = \sum_j \frac{b_j}{\lambda_j} |u_j\rangle . \tag{8.20}$$

Phase estimation on a unitary encoding A can correlate eigenvalue estimates with eigenvectors. Controlled transformations approximate reciprocal eigenvalue amplitudes, and conditional uncomputation yields the target state after appropriate postselection or amplitude amplification. The condition number $\kappa = \|A\|_\infty \left\|A^{-1}\right\|_\infty$ enters because small eigenvalues require large reciprocals. This task does not automatically return all N coordinates of $A^{-1}b$ as classical data: reading a general length-N vector remains output-size constrained.

Example 8.14 (A diagonal two-dimensional system)**.** Let $A = \mathrm{diag}(1, 1/2)$ and $|b\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Then $A^{-1} |b\rangle = (|0\rangle + 2 |1\rangle)/\sqrt{2}$ and its normalized state is

$$|x\rangle = \frac{|0\rangle + 2 |1\rangle}{\sqrt{5}}. \tag{8.21}$$

Computing expectation values of observables on $|x\rangle$ can be useful without reconstructing the entire solution vector.

8.9 Fourier transforms on periodic input states

Consider a positive integer N and a periodic superposition of the form

$$|\psi\rangle = \frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} |x_0 + jr \bmod N\rangle \,, \tag{8.22}$$

where the listed basis states are distinct. Applying F_N yields

$$F_N |\psi\rangle = \frac{1}{\sqrt{LN}} \sum_{y=0}^{N-1} \omega_N^{x_0 y} \left(\sum_{j=0}^{L-1} \omega_N^{jry} \right) |y\rangle \,. \tag{8.23}$$

The inner geometric sum is large when ry/N is close to an integer, explaining why period-finding measurements cluster near multiples of N/r. When r divides N and $L = N/r$, the inner sum is exactly L if y is a multiple of N/r and zero otherwise. Thus the Fourier transform maps an ideal periodic coset state to a superposition on the dual subgroup.

Example 8.15 (An exact periodic transform)**.** Take $N = 8$, $r = 2$, and $x_0 = 1$. Then

$$|\psi\rangle = \frac{1}{2}(|1\rangle + |3\rangle + |5\rangle + |7\rangle). \tag{8.24}$$

Its QFT has support only at $y = 0, 4$. The amplitudes are $1/\sqrt{2}$ at zero and $-1/\sqrt{2}$ at four, so

$$F_8 |\psi\rangle = \frac{|0\rangle - |4\rangle}{\sqrt{2}}. \tag{8.25}$$

The measurement probabilities are both $1/2$; the sign difference encodes the coset offset x_0 and is invisible in a single computational-basis measurement.

8.10 Phase-estimation precision and a two-register calculation

For $M = 2^m$, a phase $\varphi = k/M$ has exact binary representation in the m-qubit register and phase estimation returns k with certainty on an eigenstate. For a general phase, let y be the nearest integer to $M\varphi$. Then $\delta = \varphi - y/M$ obeys $|\delta| \le 1/(2M)$, and the exact probability is

$$p(y) = \frac{1}{M^2} \left(\frac{\sin(\pi M \delta)}{\sin(\pi \delta)} \right)^2 . \tag{8.26}$$

Using $|\sin x| \le |x|$ and $|\sin x| \ge 2\,|x|/\pi$ for $|x| \le \pi/2$, the nearest-integer outcome has probability at least $4/\pi^2$ for the standard phase-estimation setting away from removable singularities. This lower bound is a constant rather than a guarantee of exact output in one trial.

Example 8.16 (A binary phase check)**.** With $m = 4$ and $\varphi = 11/16$, the controlled powers imprint the phases $e^{2\pi i 2^j 11/16}$ onto control qubit j. The pre-inverse-QFT register is $F_{16}|11\rangle$, and inverse QFT gives $|11\rangle$, whose four-bit representation is 1011. This verifies directly that the measured integer equals $M\varphi$, not a reversed bit string when swaps are included.

8.11 Continued fractions and a concrete period recovery

If a reduced fraction k/r approximates a real number x with

$$|x - k/r| < \frac{1}{2r^2}, \tag{8.27}$$

then k/r is a convergent of the continued-fraction expansion of x. In Shor order finding, phase estimation produces $x = y/M$ close to k/r, and a sufficiently large control register allows continued fractions to propose the denominator r. The denominator of a reduced sampled fraction can be a proper divisor of the true order if $\gcd(k, r) > 1$, so verifying $a^r \equiv 1 \pmod{N}$ and repeating the experiment may be necessary.

Example 8.17 (Order-four ambiguity)**.** Suppose the true order is $r = 4$ but the sampled phase corresponds to $k = 2$, giving $k/r = 1/2$. Continued fractions recover denominator two, not four. Indeed $2^2 \not\equiv 1 \pmod{15}$, so the candidate order two fails verification. A phase with coprime numerator such as $1/4$ or $3/4$ reveals denominator four.

8.12 Classical postprocessing and complexity accounting

Factoring by the order-finding reduction invokes modular exponentiation and controlled modular multiplications, each realizable by reversible arithmetic circuits of size polynomial in the bit length $\log N$. The quantum subroutine generates data from which the order is inferred. Euclid's gcd algorithm and continued-fraction arithmetic are polynomial-time classical postprocessing. The role of the QFT is to convert the periodic phase structure into a measurable estimate; it does not produce the prime factors by direct measurement.

CHAPTER 9

Nonlocality and Quantum Communication

9.1 Resource models and the CHSH game

Let $[c \to c]$ denote a noiseless classical-bit channel, $[q \to q]$ a noiseless qubit channel, and $[qq]$ a shared Bell pair. Resource inequalities specify what can be achieved with local operations and the given communication or entanglement resources.

Problem 9.1 (CHSH nonlocal game)**.** A referee independently samples uniform bits $x, y \in \{0, 1\}$, gives x to Alice and y to Bob, and accepts their bits a, b if $a \oplus b = xy$. Alice and Bob may agree beforehand on a strategy but cannot communicate after receiving their questions.

Proposition 9.2 (Classical CHSH value)**.** *Any local deterministic strategy succeeds on at most three of the four input pairs. Shared randomness cannot improve the optimum, so* $p_{\text{win}}^{\text{cl}} = 3/4$.

Proof. The four equations $a_0 \oplus b_0 = 0$, $a_0 \oplus b_1 = 0$, $a_1 \oplus b_0 = 0$, $a_1 \oplus b_1 = 1$ have an inconsistent XOR sum. Their simultaneous satisfaction is impossible. A strategy satisfying three equations exists; shared randomness only averages deterministic success rates. □

Theorem 9.3 (Tsirelson bound)**.** *For local binary quantum observables* A_0, A_1, B_0, B_1 *with eigenvalues* ± 1*, the CHSH expression obeys*

$$\left|\langle A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1)\rangle\right| \leq 2\sqrt{2}. \tag{9.1}$$

Consequently $p_{\text{win}}^{\text{qm}} \leq \frac{1}{2} + \frac{1}{2\sqrt{2}} = \cos^2(\pi/8)$*, and an entangled two-qubit strategy attains the bound.*

Proof. For $\mathsf{B} = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1)$, expansion gives $\mathsf{B}^2 = 4I - [A_0, A_1] \otimes [B_0, B_1]$. Since each commutator has norm at most two, $\|\mathsf{B}\|_\infty^2 \leq 8$. The winning probability is $1/2 + \langle \mathsf{B} \rangle / 8$. □

9.2 Bell pairs and superdense coding

Define the Bell basis $|\Phi^{\pm}\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$ and $|\Psi^{\pm}\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}$.

Proposition 9.4 (Superdense coding)**.** *One shared Bell pair and transmission of one qubit suffice to send two classical bits perfectly. Alice applies one of* $\{I, X, Z, XZ\}$ *to her half of* $|\Phi^+\rangle$*; Bob performs a Bell-basis measurement on the resulting two-qubit state to decode the message.*

The corresponding exact idealized resource identity is $[q \to q] + [qq] \succeq 2[c \to c]$. Entanglement alone does not provide signalling: Bob's reduced state is $I/2$ for every encoded message before the qubit arrives.

9.3 Teleportation and preservation of references

Theorem 9.5 (Qubit teleportation)**.** *A shared Bell pair, a Bell-basis measurement by Alice on the input and her share, two classical bits from Alice to Bob, and a Pauli correction by Bob implement the identity qubit channel.*

Using Bell labels $|\beta_{ab}\rangle = (Z^a X^b \otimes I)\,|\Phi^+\rangle$, the coherent identity, with global phases absorbed in the chosen Bell vectors, is

$$|\psi\rangle_Q \left|\Phi^+\right\rangle_{AB} = \frac{1}{2} \sum_{a,b\in\{0,1\}} |\beta_{ab}\rangle_{QA} \otimes X^b Z^a\,|\psi\rangle_B\,. \tag{9.2}$$

After outcome (a, b), Bob applies $Z^a X^b$. Every Bell outcome has probability $1/4$, independently of the unknown input.

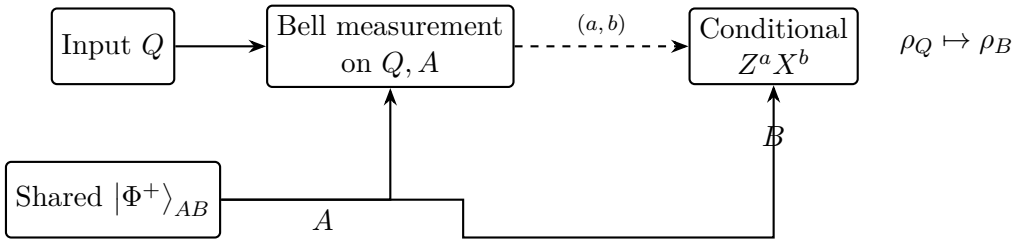


Figure 9.1: Teleportation resources and classical feedforward. Dashed arrows encode two classical bits; the solid lower path denotes Bob's retained share of entanglement.

Proposition 9.6 (Complete preservation)**.** *Let $\mathcal{T}_{Q\to B}$ be the entire teleportation protocol. For any auxiliary reference R and any joint state ρ_{RQ},*

$$(\mathrm{id}_R \otimes \mathcal{T}_{Q\to B})(\rho_{RQ}) = \rho_{RB}, \tag{9.3}$$

where B is identified with the original input space. In particular, entanglement between R and Q is preserved and transferred to $R : B$.

Proof. Apply the teleportation identity to the Q component of a general pure state $|\Psi\rangle_{RQ} = |r_0\rangle_R\,|0\rangle_Q + |r_1\rangle_R\,|1\rangle_Q$. The conditional correction restores $|\Psi\rangle_{RB}$ without acting on R. Extend by linearity to mixed inputs. □

The inverse idealized resource relation is $[qq] + 2[c \to c] \succeq [q \to q]$. Teleportation never permits faster-than-light signalling because Bob requires Alice's classical message.

9.4 Noisy resources and generalized qudits

For a d-level system, generalized shift and clock operators are $X_d\,|j\rangle = |j + 1 \bmod d\rangle$ and $Z_d\,|j\rangle = \omega_d^j\,|j\rangle$. They generate d^2 distinct generalized Pauli labels, supporting d-level Bell measurements, dense coding, and teleportation. Under noisy shared entanglement or channels, protocol performance depends on the induced effective quantum channel and can be quantified by entanglement fidelity.

9.5 Gate teleportation

Replacing Bob's share by a resource state containing an intended gate U can implement $U|\psi\rangle$ by teleportation, provided the measurement outcome-dependent byproduct operators can be corrected or propagated through later operations. Clifford gates send Pauli byproducts to Pauli byproducts under conjugation, simplifying the classical correction rule.

9.6 Correlations that do not admit a local hidden-variable model

9.6.1 The classical CHSH inequality

Alice and Bob receive uniformly random bits $x, y \in \{0,1\}$ and produce outputs $a, b \in \{0,1\}$. The winning condition is $a \oplus b = xy$. A deterministic local strategy chooses four bits a_0, a_1, b_0, b_1. Satisfying all four input conditions would require

$$a_0 \oplus b_0 = 0, \quad a_0 \oplus b_1 = 0, \quad a_1 \oplus b_0 = 0, \quad a_1 \oplus b_1 = 1. \tag{9.4}$$

XOR of the left sides is zero, because each output bit appears twice, whereas XOR of the right sides is one. Thus no deterministic local strategy wins all four cases. At most three can be won, and a fixed strategy such as always outputting zero wins exactly three. Shared randomness merely forms a convex combination of deterministic strategies, so

$$\omega_{\text{classical}} = \frac{3}{4}. \tag{9.5}$$

This conclusion assumes the inputs are independent of hidden shared randomness and there is no communication after inputs are revealed.

9.6.2 The quantum operator proof

Encode binary outputs by observables with eigenvalues ± 1. Define

$$\mathsf{C} = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1), \tag{9.6}$$

where $A_x^2 = B_y^2 = I$ and operators on distinct parties commute. Expanding gives

$$\mathsf{C}^2 = 4I - [A_0, A_1] \otimes [B_0, B_1]. \tag{9.7}$$

Because $\|[A_0, A_1]\|_\infty \leq 2$ and likewise for Bob, $\|\mathsf{C}^2\|_\infty \leq 8$, yielding $\|\mathsf{C}\|_\infty \leq 2\sqrt{2}$. The winning probability is

$$p_{\text{win}} = \frac{1}{2} + \frac{1}{8}\langle \mathsf{C} \rangle \leq \frac{1}{2} + \frac{1}{2\sqrt{2}} = \cos^2(\pi/8). \tag{9.8}$$

The bound is attained with $|\Phi^+\rangle$, $A_0 = Z$, $A_1 = X$, $B_0 = (Z + X)/\sqrt{2}$, $B_1 = (Z - X)/\sqrt{2}$. Each relevant Bell correlator can be evaluated using $\langle Z \otimes Z \rangle = \langle X \otimes X \rangle = 1$ and vanishing mixed correlators.

Example 9.7 (Computing one CHSH correlator)**.** For the specified settings,

$$\langle A_0 \otimes B_0 \rangle = \frac{1}{\sqrt{2}} \left(\langle Z \otimes Z \rangle + \langle Z \otimes X \rangle \right) = \frac{1}{\sqrt{2}}. \tag{9.9}$$

The four signed terms sum to $2\sqrt{2}$. This establishes an explicit violation of the classical ceiling $\langle \mathsf{C} \rangle \leq 2$ while respecting Tsirelson's quantum bound.

9.7 Dense coding as a basis transformation

Suppose Alice and Bob share $\left|\Phi^+\right\rangle_{AB}$. Alice receives two classical bits (a, b) and applies $Z^a X^b$ to her half, producing one of the four orthogonal Bell states. She sends her single qubit to Bob. Bob applies $\mathrm{CNOT}_{A\to B}$ followed by H_A and measures both qubits in the computational basis, recovering the two bits exactly. Explicitly,

$$(I \otimes I)\left|\Phi^+\right\rangle = \left|\Phi^+\right\rangle, \qquad (Z \otimes I)\left|\Phi^+\right\rangle = \left|\Phi^-\right\rangle,$$
$$(X \otimes I)\left|\Phi^+\right\rangle = \left|\Psi^+\right\rangle, \qquad (ZX \otimes I)\left|\Phi^+\right\rangle = \pm\left|\Psi^-\right\rangle.$$

The global sign in the last expression has no operational significance. One shared ebit plus one transmitted qubit supports two bits of noiseless classical communication. Without the pre-shared entanglement, a single qubit cannot transmit two reliably distinguishable classical bits in the one-shot unconstrained setting.

9.8 Teleportation with explicit Bell measurement branches

Let an unknown input $|\psi\rangle_Q = \alpha|0\rangle + \beta|1\rangle$ and shared Bell pair $\left|\Phi^+\right\rangle_{AB}$ be given. In the order QAB,

$$|\psi\rangle_Q \left|\Phi^+\right\rangle_{AB} = \frac{1}{2}\big[\left|\Phi^+\right\rangle_{QA}(\alpha|0\rangle + \beta|1\rangle)_B \tag{9.10}$$
$$+ \left|\Phi^-\right\rangle_{QA}(\alpha|0\rangle - \beta|1\rangle)_B \tag{9.11}$$
$$+ \left|\Psi^+\right\rangle_{QA}(\beta|0\rangle + \alpha|1\rangle)_B \tag{9.12}$$
$$+ \left|\Psi^-\right\rangle_{QA}(-\beta|0\rangle + \alpha|1\rangle)_B\big]. \tag{9.13}$$

Every Bell outcome has probability $1/4$, independent of α, β. The corresponding uncorrected Bob states are $I|\psi\rangle$, $Z|\psi\rangle$, $X|\psi\rangle$, and $XZ|\psi\rangle$ (up to sign). Alice communicates the two-bit outcome, and Bob applies the inverse Pauli correction. The protocol consumes one Bell pair and two classical bits, transferring an arbitrary qubit state without physically sending that qubit through the Alice-to-Bob channel.

9.8.1 Why teleportation preserves entanglement with a reference

Let R be inaccessible and consider the general pure reference-input state

$$|\Psi\rangle_{RQ} = |r_0\rangle_R|0\rangle_Q + |r_1\rangle_R|1\rangle_Q, \quad \|r_0\|^2 + \|r_1\|^2 = 1. \tag{9.14}$$

The reference vectors need not be orthogonal or normalized individually. Apply the Bell expansion linearly to each $|j\rangle_Q$. Conditional on any Bell result, Bob has $(I_R \otimes P_{ab})|\Psi\rangle_{RB}/2$ with the same reference register. The Pauli correction gives exactly $|\Psi\rangle_{RB}$. Thus the implemented channel is the identity even when the input is part of an entangled state; merely checking its action on isolated pure states would hide this important requirement.

9.9 Noisy communication and resource accounting

9.9.1 Classical communication through noisy channels

For a classical channel $W(y|x)$, an (n, M) code consists of an encoder mapping one of M messages to an input sequence of length n, and a decoder mapping output sequences to message estimates. Its rate is $(\log M)/n$ bits per channel use. The Shannon coding theorem identifies the supremum of asymptotically reliable rates as $C(W) = \max_{p_X} I(X : Y)$ for a discrete memoryless channel.

9.9.2 Erasure effects on teleportation

If the two classical correction bits are erased or corrupted, Bob may fail to apply the intended Pauli. Suppose the correction message is replaced by an independent uniformly random two-bit message; then averaging over the uncorrected Pauli gives the completely depolarizing channel

$$\frac{1}{4}\sum_{P\in\{I,X,Y,Z\}} P\rho P = \frac{I}{2}. \tag{9.15}$$

Thus noisy classical correction cannot be ignored: entanglement alone is insufficient for faithful teleportation. Conversely, if the classical channel is noiseless but the entangled pair is noisy, the induced channel is a Pauli channel determined by the Bell-basis populations of the shared resource under the standard teleportation protocol.

Example 9.8 (Qudit resource accounting)**.** For dimension d, the maximally entangled state $|\Phi_d\rangle = d^{-1/2}\sum_{j=0}^{d-1}|jj\rangle$ enables teleportation by a generalized Bell measurement with d^2 outcomes. Consequently the correction label carries $\log(d^2) = 2\log d$ classical bits. Dense coding uses the same d^2 orthogonal generalized Bell states to transmit $2\log d$ classical bits with one qudit transmission and one shared maximally entangled pair.

9.10 Gate teleportation and one-way computation

If the Bell-pair resource is modified by a unitary U on Bob's half, then teleportation produces $UP_{ab}|\psi\rangle$ before correction. The desired state is $U|\psi\rangle$, so the conditional correction is $UP_{ab}^{\dagger}U^{\dagger}$, not necessarily a Pauli operator. For Clifford U, conjugation preserves Pauli operators, and corrections remain easy to track. For non-Clifford gates, conditional corrections may leave the Pauli group and require additional resources or adaptive processing. Measurement-based quantum computation formalizes this feed-forward behavior using measurements on graph states.

9.11 CHSH values expressed as measurable correlators

Suppose dichotomic outputs are represented by signs $(-1)^a$ and $(-1)^b$. The probability that $a \oplus b = xy$ for given inputs is

$$p_{xy} = \frac{1}{2}\left(1 + (-1)^{xy}\langle A_x \otimes B_y\rangle\right). \tag{9.16}$$

Uniformly averaging the four input pairs gives

$$p_{\text{win}} = \frac{1}{4}\sum_{x,y} p_{xy} \tag{9.17}$$

$$= \frac{1}{2} + \frac{1}{8}\big(\langle A_0B_0\rangle + \langle A_0B_1\rangle + \langle A_1B_0\rangle - \langle A_1B_1\rangle\big). \tag{9.18}$$

The CHSH quantity S is the parenthesized combination. Classical local models obey $|S| \le 2$, while quantum strategies obey $|S| \le 2\sqrt{2}$. Therefore the corresponding game-winning ceilings are $3/4$ and $(2+\sqrt{2})/4 \approx 0.85355$, respectively. The separation is operational even though neither party can signal through the entangled state alone.

9.12 Teleportation written as a channel

Denote the completely positive map associated with Bell outcome (a, b) by $\mathcal{I}_{ab}$. In ideal teleportation using $|\Phi^+\rangle$, the unnormalized Bob output is

$$\mathcal{I}_{ab}(\rho) = \frac{1}{4} P_{ab}\rho P_{ab}^\dagger, \tag{9.19}$$

with a suitable Pauli family P_{ab}. Its trace is $1/4$, independent of input. The conditional recovery map applies $P_{ab}^\dagger$. Summing corrected branches yields

$$\sum_{a,b} P_{ab}^\dagger \mathcal{I}_{ab}(\rho) P_{ab} = \frac{1}{4}\sum_{a,b} \rho = \rho. \tag{9.20}$$

This formula demonstrates that the overall protocol implements the identity CPTP map, not merely a correct transformation on a selected input state.

Example 9.9 (Teleporting a reference-entangled state)**.** Prepare $|\Phi^+\rangle_{RQ}$, where R is not touched by the protocol. Ideal teleportation sends Q to Bob's output B and must produce $|\Phi^+\rangle_{RB}$. The reduced output on B remains $I/2$ both before and after a completely depolarizing channel, so checking only $\rho_B = I/2$ would not distinguish faithful teleportation from total information loss. Preserving the joint RB state is the correct test.

9.13 Classical messages and the no-signalling constraint

Before Bob receives Alice's correction label, his averaged conditional state is

$$\sum_{a,b} \mathcal{I}_{ab}(\rho) = \frac{1}{4}\sum_{a,b} P_{ab}\rho P_{ab}^\dagger = I/2. \tag{9.21}$$

The right-hand side is independent of ρ. Thus Bob cannot identify the unknown input from his local qubit before the classical message arrives. This explicitly realizes no-signalling within the teleportation protocol: entanglement and local measurement alone do not transfer a readable message faster than classical communication.

9.13.1 The teleportation Pauli twirl

Write a qubit density operator as $\rho = (I + r_x X + r_y Y + r_z Z)/2$. Conjugation by the four Pauli operators flips different Bloch components. Averaging over I, X, Y, Z cancels each nonidentity Pauli coefficient, leaving

$$\frac{1}{4} \sum_{P \in \{I,X,Y,Z\}} P \rho P = \frac{I}{2}. \tag{9.22}$$

This identity also explains the completely mixed output when the classical correction is lost and no outcome information is available. Pauli twirling more generally converts certain noise descriptions into Pauli-diagonal forms after averaging suitable conjugations, though the exact implemented channel depends on which random operations and records are retained.

CHAPTER 10

Quantum Channels: Structure and Examples

10.1 Completely positive trace-preserving maps

Definition 10.1 (Quantum channel)**.** A quantum channel $\mathcal{N} : \mathcal{L}(A) \to \mathcal{L}(B)$ is a linear, trace-preserving, completely positive (CPTP) map. Complete positivity means that $\mathrm{id}_R \otimes \mathcal{N}$ is positive on every $\mathcal{L}(R \otimes A)$ for every finite reference R.

Ordinary positivity is insufficient: the matrix-transpose map is positive on isolated density matrices but not completely positive, as can be verified by acting on one half of a maximally entangled state.

Theorem 10.2 (Kraus representation)**.** *A linear map $\mathcal{N}$ is CPTP if and only if there are operators $K_j : A \to B$ such that*

$$\mathcal{N}(\rho) = \sum_j K_j \rho K_j^\dagger, \qquad \sum_j K_j^\dagger K_j = I_A. \tag{10.1}$$

Theorem 10.3 (Stinespring dilation)**.** *Every channel $\mathcal{N} : A \to B$ admits an isometry $V : A \to B \otimes E$ such that*

$$\mathcal{N}(\rho) = \mathrm{Tr}_E(V \rho V^\dagger). \tag{10.2}$$

An isometry extends to a unitary on a sufficiently large space with a fixed environmental input state.

Proof. Given Kraus operators, choose orthonormal $|j\rangle_E$ and set $V|\psi\rangle = \sum_j K_j|\psi\rangle \otimes |j\rangle$. The completeness condition gives $V^\dagger V = I$ and partial tracing yields the channel. Conversely, expanding V in an environment basis produces Kraus operators $K_j = (I_B \otimes \langle j|)V$. □

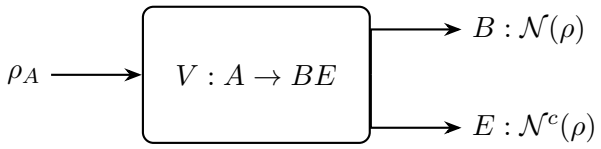


Figure 10.1: Stinespring isometry and its two reduced channels. The complementary output is obtained by tracing out B rather than E.

10.2 Choi–Jamiołkowski correspondence

Definition 10.4 (Normalized Choi operator)**.** Let $d_A = \dim A$. Fix matching computational bases on R and A, and set

$$|\Phi_d\rangle = \frac{1}{\sqrt{d_A}} \sum_{j=0}^{d_A-1} |j\rangle_R \otimes |j\rangle_A \,. \tag{10.3}$$

Define

$$\begin{aligned} J_{RB}(\mathcal{N}) &= (\mathrm{id}_R \otimes \mathcal{N}_{A\to B})(|\Phi_d\rangle\langle\Phi_d|) \\ &= \frac{1}{d_A} \sum_{i,j} |i\rangle\langle j|_R \otimes \mathcal{N}(|i\rangle\langle j|_A). \end{aligned} \tag{10.4}$$

Theorem 10.5 (Choi characterization)**.** *A linear $\mathcal{N} : \mathcal{L}(A) \to \mathcal{L}(B)$ is CPTP if and only if*

$$J_{RB}(\mathcal{N}) \succeq 0, \qquad \mathrm{Tr}_B\, J_{RB}(\mathcal{N}) = I_R/d_A. \tag{10.5}$$

The action on $X \in \mathcal{L}(A)$ is reconstructed by

$$\mathcal{N}(X) = d_A \,\mathrm{Tr}_R\Big[(X^T \otimes I_B) J_{RB}(\mathcal{N})\Big] \,. \tag{10.6}$$

Proof. The reconstruction follows by expanding X in matrix units. A Kraus map yields a sum of positive outer products $J = \sum_j |(I \otimes K_j)\Phi_d\rangle\langle(I \otimes K_j)\Phi_d|$. Conversely, spectrally decompose a positive J and reshape each eigenvector into a Kraus operator; the partial-trace condition supplies $\sum_j K_j^\dagger K_j = I$. □

The minimal number of nonzero Kraus operators equals $\mathrm{rank}\, J(\mathcal{N})$. The normalized and unnormalized Choi conventions differ by the factor d_A; formulas must not mix them.

10.3 Noise channels

Definition 10.6 (Common single-qubit channels)**.** For $0 \le p, q, \gamma \le 1$, define

$$\mathcal{B}_p(\rho) = (1-p)\rho + pX\rho X, \tag{10.7}$$

$$\mathcal{D}_q(\rho) = (1-q)\rho + q\frac{I}{2}, \tag{10.8}$$

$$\mathcal{Z}_p(\rho) = (1-p)\rho + pZ\rho Z, \tag{10.9}$$

$$\mathcal{A}_\gamma(\rho) = K_0\rho K_0^\dagger + K_1\rho K_1^\dagger, \tag{10.10}$$

$$K_0 = |0\rangle\langle 0| + \sqrt{1-\gamma}\,|1\rangle\langle 1|\,, \quad K_1 = \sqrt{\gamma}\,|0\rangle\langle 1|\,. \tag{10.11}$$

A bit-flip channel has Kraus operators $\sqrt{1-p}I, \sqrt{p}X$ and a dilation

$$V|\psi\rangle = \sqrt{1-p}\,|\psi\rangle\,|0\rangle + \sqrt{p}\,X\,|\psi\rangle\,|1\rangle\,. \tag{10.12}$$

The depolarizing channel shrinks the Bloch vector by $1-q$. Amplitude damping maps excited population to ground population and introduces irreversible energy loss. An erasure channel flags the erased subspace orthogonally to valid outputs.

10.4 Entanglement-breaking channels

Definition 10.7 (Entanglement breaking)**.** A channel $\mathcal{N}$ is entanglement breaking if $(\mathrm{id}_R \otimes \mathcal{N})(\rho_{RA})$ is separable for every reference R and every input ρ_{RA}. This holds if and only if the Choi operator $J(\mathcal{N})$ is separable across $R : B$.

Proposition 10.8 (Qubit noise thresholds)**.** *For $\mathcal{B}_p$ with $0 \leq p \leq 1$, the channel is entanglement breaking precisely at $p = 1/2$. For $\mathcal{D}_q$ with $0 \leq q \leq 1$, it is entanglement breaking precisely when $q \geq 2/3$.*

Proof. The Choi state $J(\mathcal{B}_p)$ has partial-transpose eigenvalues $1/2, 1/2, (1-2p)/2, (2p-1)/2$. Hence PPT holds only at $p = 1/2$. Also $J(\mathcal{D}_q) = (1-q)\left|\Phi^+\right\rangle\!\left\langle\Phi^+\right| + qI/4$ has partial-transpose eigenvalues $(2-q)/4$ three times and $(3q-2)/4$ once. The two-qubit PPT criterion establishes the thresholds. □

10.5 Complementary channels

Given a Stinespring isometry $V : A \to BE$, the complementary channel is $\mathcal{N}^c(\rho) = \mathrm{Tr}_B(V\rho V^\dagger)$. The joint state of output B and environment E remains pure if the input was pure and V is applied without an additional mixed ancillary preparation. This purification relation is central to quantum capacity and recoverability.

10.6 Quantum operations from an environment model

10.6.1 Kraus operators obtained from an isometry

Let $V : A \to B \otimes E$ be an isometry satisfying $V^\dagger V = I_A$. Choose an orthonormal environment basis $\{|j\rangle_E\}$ and define $K_j = (I_B \otimes \langle j|_E)V$. Then

$$\mathrm{Tr}_E(V\rho V^\dagger) = \sum_j K_j \rho K_j^\dagger, \qquad \sum_j K_j^\dagger K_j = V^\dagger V = I_A. \tag{10.13}$$

This proves the Kraus form of every channel represented by such an isometry. Conversely, given Kraus operators obeying $\sum_j K_j^\dagger K_j = I$, the map $V|\psi\rangle = \sum_j K_j|\psi\rangle|j\rangle_E$ is an isometry and can be extended to a unitary on a sufficiently large system-environment space.

Example 10.9 (Bit-flip noise with an explicit dilation)**.** Define $\mathcal{N}_p(\rho) = (1-p)\rho + pX\rho X$, $0 \le p \le 1$. A Kraus representation is

$$K_0 = \sqrt{1-p}\, I, \qquad K_1 = \sqrt{p}\, X. \tag{10.14}$$

An isometry is $V|\psi\rangle = \sqrt{1-p}\,|\psi\rangle|0\rangle + \sqrt{p}X|\psi\rangle|1\rangle$. The environment vectors are orthogonal, so cross terms vanish upon partial trace. To complete V to a unitary for a single-qubit environment, select an orthonormal completion of its two-dimensional image in the four-dimensional joint space. The physical channel does not depend on the choice of completion.

10.6.2 Trace preservation and complete positivity

A linear map $\mathcal{N}$ is trace preserving if $\mathrm{Tr}\,\mathcal{N}(X) = \mathrm{Tr}\,X$ for all operators X. A positive map preserves positive operators; complete positivity additionally requires $\mathrm{id}_R \otimes \mathcal{N}$ to preserve positivity for every auxiliary dimension. The transpose map $T(\rho) = \rho^{\mathsf{T}}$ is positive and trace preserving, but not completely positive. Indeed applying $\mathrm{id} \otimes T$ to the Bell projector yields a partial transpose with a negative eigenvalue $-1/2$. This counterexample establishes the operational need for complete positivity when a system may be entangled with an untouched reference.

10.7 The Choi operator and equivalent channel representations

Let A have dimension d and define $|\Phi_d\rangle = d^{-1/2}\sum_{i=0}^{d-1}|ii\rangle$. The normalized Choi operator of $\mathcal{N} : A \to B$ is

$$J_{RB}(\mathcal{N}) = (\mathrm{id}_R \otimes \mathcal{N}_A)(|\Phi_d\rangle\langle\Phi_d|), \quad R \simeq A. \tag{10.15}$$

Expanding its matrix units gives

$$J(\mathcal{N}) = \frac{1}{d}\sum_{i,j} |i\rangle\langle j|_R \otimes \mathcal{N}(|i\rangle\langle j|_A). \tag{10.16}$$

The action of the channel can be reconstructed as

$$\mathcal{N}(X) = d\ \mathrm{Tr}_R[(X^{\mathsf{T}} \otimes I_B)J(\mathcal{N})]. \tag{10.17}$$

The transpose is with respect to the computational basis used for $|\Phi_d\rangle$. The conditions for a valid CPTP map are

$$J(\mathcal{N}) \succeq 0, \qquad \mathrm{Tr}_B\, J(\mathcal{N}) = I_R/d. \tag{10.18}$$

Thus channel identification may be cast as positive semidefinite matrix reconstruction under affine marginal constraints.

Example 10.10 (Choi state of the bit-flip channel)**.** Because $(I \otimes X)\left|\Phi^+\right\rangle = \left|\Psi^+\right\rangle$,

$$J(\mathcal{N}_p) = (1-p)\left|\Phi^+\right\rangle\!\left\langle\Phi^+\right| + p\left|\Psi^+\right\rangle\!\left\langle\Psi^+\right|. \tag{10.19}$$

The Bell vectors are orthonormal. For $0 < p < 1$, the Choi rank is two, so the minimal Kraus representation needs two linearly independent operators. At $p = 0$ or $p = 1$ the channel is unitary and has Choi rank one.

10.7.1 Why Choi rank equals minimal Kraus rank

Given a Kraus family $\{K_j\}$, vectorize each K_j to $|K_j\rangle\!\rangle = (I \otimes K_j)\sum_i |ii\rangle$. Then $dJ(\mathcal{N}) = \sum_j |K_j\rangle\!\rangle\langle\!\langle K_j|$, implying $\mathrm{rank}\, J \le \#\{K_j\}$. Conversely, diagonalize the positive Choi matrix and reshape each eigenvector with nonzero eigenvalue into an operator; the resulting spectral decomposition supplies exactly $\mathrm{rank}\, J$ Kraus operators. Hence the minimal environment dimension in an isometric Stinespring representation equals $\mathrm{rank}\, J$.

10.8 Four noise models in detail

10.8.1 Amplitude damping

A zero-temperature decay channel with parameter γ has Kraus operators

$$K_0 = |0\rangle\langle 0| + \sqrt{1-\gamma}\,|1\rangle\langle 1|, \quad K_1 = \sqrt{\gamma}\,|0\rangle\langle 1|. \tag{10.20}$$

For $\rho = \left(\begin{smallmatrix} a & c \\ \overline{c} & b \end{smallmatrix}\right)$,

$$\mathcal{A}_\gamma(\rho) = \begin{pmatrix} a + \gamma b & \sqrt{1-\gamma}\, c \\ \sqrt{1-\gamma}\,\overline{c} & (1-\gamma)b \end{pmatrix}. \tag{10.21}$$

This channel is nonunital: $\mathcal{A}_\gamma(I/2) \neq I/2$ for $\gamma > 0$. The excited-state population decays while coherence is attenuated by $\sqrt{1-\gamma}$.

10.8.2 Dephasing

The phase-flip channel $\mathcal{N}_p^{(Z)}(\rho) = (1-p)\rho + pZ\rho Z$ preserves diagonal entries and multiplies the off-diagonal entries by $(1-2p)$. For $p = 1/2$ it becomes complete dephasing in the Z basis, a measure-and-prepare channel in that basis.

10.8.3 Depolarizing noise

For $0 \le q \le 1$, $\mathcal{D}_q(\rho) = (1-q)\rho + q\,\mathrm{Tr}(\rho)I/2$. On Bloch vectors it acts by $\boldsymbol{r} \mapsto (1-q)\boldsymbol{r}$. Its normalized Choi state is isotropic,

$$J(\mathcal{D}_q) = (1-q)\left|\Phi^+\right\rangle\!\left\langle\Phi^+\right| + \frac{q}{4} I_4. \tag{10.22}$$

Its eigenvalues are $1 - 3q/4$ and $q/4$ with multiplicity three; the state has rank four for $q > 0$ and rank one at $q = 0$.

10.8.4 Erasure

A d-dimensional erasure channel with probability ϵ has output dimension $d + 1$ and acts as

$$\mathcal{E}_\epsilon(\rho) = (1 - \epsilon)\rho \oplus \epsilon \operatorname{Tr}\rho\, |\perp\rangle\langle\perp| . \tag{10.23}$$

The orthogonal erasure flag is available to the receiver. This differs qualitatively from depolarizing noise, where the receiver is not told which uses suffered errors.

10.9 Entanglement-breaking criteria by Choi separability

A channel $\mathcal{N}$ is entanglement breaking if $\mathrm{id}_R \otimes \mathcal{N}$ produces a separable state for every reference-input joint state. In finite dimension this is equivalent to $J(\mathcal{N})$ being separable across $R : B$, and also to a measure-and-prepare representation

$$\mathcal{N}(\rho) = \sum_y \operatorname{Tr}(M_y \rho)\sigma_y, \tag{10.24}$$

where $\{M_y\}$ is a POVM and the σ_y are output states.

Example 10.11 (Depolarizing threshold on qubits)**.** The qubit depolarizing Choi state is an isotropic two-qubit state with fidelity $F = \langle\Phi^+|J|\Phi^+\rangle = 1 - 3q/4$. In dimension two, this family is separable exactly when $F \le 1/2$, i.e.

$$q \ge \frac{2}{3}. \tag{10.25}$$

Thus $\mathcal{D}_q$ is entanglement breaking for $q \ge 2/3$. At $q = 2/3$ its Choi state lies on the PPT boundary. The bit-flip channel $(1 - p)\rho + pX\rho X$ is a complete dephasing channel in the X basis at $p = 1/2$ and is entanglement breaking precisely at that central value within $0 \le p \le 1$.

10.10 Choi matrices calculated element by element

10.10.1 Bit-flip channel in the computational basis

For $\mathcal{N}_p(\rho) = (1 - p)\rho + pX\rho X$, the normalized Choi state is

$$J_p = (1 - p)\left|\Phi^+\right\rangle\!\left\langle\Phi^+\right| + p\left|\Psi^+\right\rangle\!\left\langle\Psi^+\right|. \tag{10.26}$$

In the ordered basis $00, 01, 10, 11$ its matrix is

$$J_p = \frac{1}{2}\begin{pmatrix} 1-p & 0 & 0 & 1-p \\ 0 & p & p & 0 \\ 0 & p & p & 0 \\ 1-p & 0 & 0 & 1-p \end{pmatrix}. \tag{10.27}$$

This is positive because it is a convex mixture of orthogonal rank-one projectors. Its spectrum is $\{1-p, p, 0, 0\}$. Taking the partial transpose on the output register gives

$$J_p^{T_B} = \frac{1}{2}\begin{pmatrix} 1-p & 0 & 0 & p \\ 0 & p & 1-p & 0 \\ 0 & 1-p & p & 0 \\ p & 0 & 0 & 1-p \end{pmatrix}. \tag{10.28}$$

The two 2×2 corner and middle blocks have eigenvalues $1/2$, $(1-2p)/2$, $1/2$, and $(2p-1)/2$, respectively. One eigenvalue is negative unless $p = 1/2$. Since this is a 2×2 bipartite state, PPT is equivalent to separability. Thus $\mathcal{N}_p$ is entanglement breaking exactly when $p = 1/2$.

10.10.2 Depolarizing Choi state

For $\mathcal{D}_q(\rho) = (1-q)\rho + q\,\mathrm{Tr}(\rho)I/2$,

$$J_q = (1-q)\left|\Phi^+\right\rangle\!\left\langle\Phi^+\right| + qI_4/4. \tag{10.29}$$

Its spectrum is $1-3q/4, q/4, q/4, q/4$. Partial transposition of the Bell projector equals half the SWAP operator, so

$$J_q^{T_B} = \frac{1-q}{2}S + \frac{q}{4}I_4. \tag{10.30}$$

SWAP has eigenvalue $+1$ on the three-dimensional symmetric subspace and -1 on the one-dimensional antisymmetric subspace. Hence the eigenvalues of the partial transpose are $(2-q)/4$ (multiplicity three) and $(3q-2)/4$ (multiplicity one). PPT holds exactly when $q \geq 2/3$, and in 2×2 this is precisely the entanglement-breaking threshold.

10.11 Constructing a unitary dilation explicitly

Let $a = \sqrt{1-p}$ and $b = \sqrt{p}$ for the bit-flip channel, with system and environment both qubits. Define the isometry by

$$V\left|0\right\rangle = a\left|00\right\rangle + b\left|11\right\rangle, \tag{10.31}$$
$$V\left|1\right\rangle = a\left|10\right\rangle + b\left|01\right\rangle. \tag{10.32}$$

These images are orthonormal because they lie in orthogonal parity subspaces and each has squared norm $a^2 + b^2 = 1$. One possible unitary completion on the four-dimensional joint system is specified by

$$U\left|00\right\rangle = a\left|00\right\rangle + b\left|11\right\rangle, \tag{10.33}$$
$$U\left|10\right\rangle = a\left|10\right\rangle + b\left|01\right\rangle, \tag{10.34}$$
$$U\left|01\right\rangle = -b\left|00\right\rangle + a\left|11\right\rangle, \tag{10.35}$$
$$U\left|11\right\rangle = -b\left|10\right\rangle + a\left|01\right\rangle. \tag{10.36}$$

The four output columns are orthonormal, so U is unitary. With the environment initialized in $\left|0\right\rangle$, tracing it out gives precisely $(1-p)\rho + pX\rho X$. Alternative orthonormal completions lead to the same system channel on the specified environment input.

10.12 Nonunitality and the distinction between damping and Pauli noise

For amplitude damping,

$$\mathcal{A}_\gamma(I/2) = \frac{1}{2}\begin{pmatrix} 1+\gamma & 0 \\ 0 & 1-\gamma \end{pmatrix} \neq I/2 \quad (\gamma > 0). \tag{10.37}$$

For a Pauli channel $\mathcal{P}(\rho) = \sum_P p_P P\rho P^\dagger$, one has $\mathcal{P}(I/2) = I/2$. Thus an affine shift of the Bloch center witnesses nonunital noise. The amplitude-damping channel maps the Bloch vector by

$$(r_x, r_y, r_z) \mapsto (\sqrt{1-\gamma}\, r_x, \sqrt{1-\gamma}\, r_y, \gamma + (1-\gamma) r_z). \tag{10.38}$$

The positive translation in r_z represents relaxation toward $|0\rangle$; no mixture of unital Pauli conjugations can implement that translation.

CHAPTER 11

Quantum Entropies, Data Processing, and Recovery

11.1 Quantum information measures

Definition 11.1 (Quantum entropies)**.** For $\rho_{ABC} \in \mathcal{D}(ABC)$, define

$$S(A)_\rho = -\operatorname{Tr} \rho_A \log \rho_A, \tag{11.1}$$

$$S(A \mid B)_\rho = S(AB)_\rho - S(B)_\rho, \tag{11.2}$$

$$I(A:B)_\rho = S(A)_\rho + S(B)_\rho - S(AB)_\rho, \tag{11.3}$$

$$I(A:C \mid B)_\rho = S(AB)_\rho + S(BC)_\rho - S(B)_\rho - S(ABC)_\rho. \tag{11.4}$$

Unlike classical conditional entropy, quantum $S(A \mid B)$ can be negative. A maximally entangled state of Schmidt rank d has $S(A \mid B) = -\log d$, $S(A) = S(B) = \log d$, and $S(AB) = 0$.

Theorem 11.2 (Strong subadditivity)**.** *For every tripartite quantum state, $I(A:C \mid B)_\rho \geq 0$.*

Corollary 11.3 (Araki–Lieb inequality)**.** *For every bipartite state, $|S(A) - S(B)| \leq S(AB) \leq S(A) + S(B)$.*

11.2 Quantum relative entropy and monotonicity

Definition 11.4 (Umegaki relative entropy)**.** For density operators ρ, σ with $\operatorname{supp} \rho \subseteq \operatorname{supp} \sigma$,

$$D(\rho \| \sigma) = \operatorname{Tr} \rho (\log \rho - \log \sigma); \tag{11.5}$$

otherwise set $D(\rho\|\sigma) = +\infty$.

Theorem 11.5 (Quantum data-processing inequality)**.** *For every channel $\mathcal{N}$,*

$$D(\rho\|\sigma) \geq D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)). \tag{11.6}$$

In particular, applying a local channel to B cannot increase $I(A:B)$.

Derivation of the mutual-information consequence. Since $I(A:B)_\rho = D(\rho_{AB}\|\rho_A \otimes \rho_B)$, apply the relative-entropy inequality to $\mathrm{id}_A \otimes \mathcal{N}_{B\to B'}$, and observe that product marginals transform to $\rho_A \otimes \mathcal{N}(\rho_B)$. □

11.3 Fidelity, recovery, and the Petz construction

Definition 11.6 (Fidelity)**.** For density operators ρ, σ, the squared Uhlmann fidelity is

$$F(\rho, \sigma) = \left(\operatorname{Tr} \sqrt{\sqrt{\rho}\, \sigma \sqrt{\rho}} \right)^2 . \tag{11.7}$$

It lies in $[0, 1]$ and equals one precisely when $\rho = \sigma$.

Definition 11.7 (Petz recovery map)**.** For $\mathcal{N} : A \to B$ and a full-rank reference state σ on its support, define the transpose/recovery map

$$\mathcal{R}_{\sigma,\mathcal{N}}(X) = \sigma^{1/2} \mathcal{N}^\dagger \left(\mathcal{N}(\sigma)^{-1/2} X \mathcal{N}(\sigma)^{-1/2} \right) \sigma^{1/2}, \tag{11.8}$$

using generalized inverses on the corresponding supports and a trace-preserving completion outside them if needed.

Theorem 11.8 (Equality in data processing)**.** *Under standard finite-dimensional support assumptions, equality $D(\rho\|\sigma) = D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ is equivalent to exact sufficiency for the pair: a recovery channel recovers both ρ and σ. The Petz map recovers them on the supported subspace.*

Theorem 11.9 (Approximate recovery, schematic quantitative form)**.** *Under finite relative-entropic support conditions, there exists a recovery channel $\mathcal{R}$ depending on σ and $\mathcal{N}$, with $\mathcal{R}(\mathcal{N}(\sigma)) = \sigma$, such that*

$$D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)) \geq -\log F(\rho, \mathcal{R}(\mathcal{N}(\rho))). \tag{11.9}$$

A suitable averaged rotated Petz recovery map can satisfy this fidelity guarantee; the unrotated Petz map should not automatically be substituted into this quantitative claim.

Example 11.10 (Bayesian reverse channel)**.** For a classical prior $q(x)$ and stochastic channel $W(y \mid x)$, the Petz formula reduces to the Bayesian reverse kernel

$$R_q(x \mid y) = \frac{q(x) W(y \mid x)}{\sum_{x'} q(x') W(y \mid x')}, \tag{11.10}$$

for outputs with nonzero denominator. It recovers the prior q exactly. Perfect recovery of another distribution p occurs when the channel loses no relative-entropy distinguishability for the pair (p, q).

11.4 Guessing probability and min-entropy

Definition 11.11 (Classical conditional min-entropy)**.** For a classical joint distribution p_{XY},

$$p_{\text{guess}}(X \mid Y) = \sum_y \max_x p(x, y), \qquad H_{\min}(X \mid Y) = -\log p_{\text{guess}}(X \mid Y). \tag{11.11}$$

Definition 11.12 (Classical–quantum min-entropy)**.** For $\rho_{XB} = \sum_x p_x \, |x\rangle\langle x| \otimes \rho_B^x$, define

$$p_{\mathrm{guess}}(X \mid B) = \max_{\{M_x\}\,\mathrm{POVM}} \sum_x p_x \operatorname{Tr}(M_x \rho_B^x), \qquad H_{\min}(X \mid B) = -\log p_{\mathrm{guess}}(X \mid B). \tag{11.12}$$

For a uniformly distributed independent n-bit secret X, $H_{\min}(X) = n$. If X is perfectly determined by the conditioning register, its conditional min-entropy is zero in the classical–quantum setting.

11.5 Quantum relative entropy and operational monotonicity

For states ρ, σ on the same finite-dimensional space, the relative entropy is $D(\rho\|\sigma) = \operatorname{Tr}\rho(\log\rho - \log\sigma)$ when $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$, and $+\infty$ otherwise. It is nonnegative and vanishes exactly at $\rho = \sigma$. The data-processing inequality states

$$D(\rho\|\sigma) \geq D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)) \tag{11.13}$$

for any CPTP map $\mathcal{N}$. The inequality is not simply a classical log-sum argument when the states do not commute; its full proof uses operator inequalities, such as operator convexity or integral representations of the logarithm.

11.5.1 A direct proof for a complete dephasing channel

Let $\Delta(\rho) = \sum_i |i\rangle\langle i|\,\rho\,|i\rangle\langle i|$. It discards off-diagonal entries in the chosen basis. If σ is diagonal in this basis, then

$$D(\rho\|\sigma) - D(\Delta\rho\|\sigma) = -S(\rho) + S(\Delta\rho) \tag{11.14}$$

$$= D(\rho\|\Delta\rho) \geq 0. \tag{11.15}$$

The last equality follows because $\log\Delta\rho$ is diagonal and hence $\operatorname{Tr}\rho\log\Delta\rho = \operatorname{Tr}\Delta\rho\log\Delta\rho$. This gives a self-contained special case of information loss under decoherence.

11.6 Strong subadditivity and its recovery meaning

Strong subadditivity is

$$S(AB) + S(BC) \geq S(B) + S(ABC). \tag{11.16}$$

Equivalently, the conditional mutual information $I(A:C|B)$ is nonnegative. To see its relation to relative entropy, note that

$$I(A:C|B) = D(\rho_{ABC}\|I_A \otimes \rho_{BC}) - D(\rho_{AB}\|I_A \otimes \rho_B)\,, \tag{11.17}$$

where the second arguments are positive operators and the logarithmic expression is interpreted with matching support. Applying data processing under Tr_C proves the inequality. The state-dependent equality condition can be characterized using a recovery channel that reconstructs the correlations with C from system B alone.

Example 11.13 (A classical Markov state embedded diagonally)**.** Let A and C be conditionally independent given B, with $p(a,b,c) = p(b)p(a|b)p(c|b)$. Embed the distribution as the diagonal density operator

$$\rho_{ABC} = \sum_{a,b,c} p(a,b,c)\,|abc\rangle\langle abc|\,. \tag{11.18}$$

Its quantum conditional mutual information equals the classical $I(A:C|B) = 0$. A recovery map that reads the classical b and prepares the conditional C distribution exactly reconstructs the original state. The quantum equality condition extends this Markov

intuition to noncommuting systems.

11.7 The Petz map and exact data-processing equality

Let $\mathcal{N} : A \to B$ be a CPTP map and let σ_A be a positive reference state. The Petz map is

$$\mathcal{R}_{\sigma,\mathcal{N}}(X) = \sigma^{1/2}\mathcal{N}^\dagger\left(\mathcal{N}(\sigma)^{-1/2}X\mathcal{N}(\sigma)^{-1/2}\right)\sigma^{1/2}, \tag{11.19}$$

where inverses are restricted to supports. The Hilbert–Schmidt adjoint satisfies $\mathrm{Tr}[Y\mathcal{N}(X)] = \mathrm{Tr}[\mathcal{N}^\dagger(Y)X]$.

Proposition 11.14 (Exact reference-state recovery)**.** *With support conventions and normalized σ, $\mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\sigma)) = \sigma$.*

Proof. Inside the support of $\mathcal{N}(\sigma)$, the sandwiched expression equals the support projector $\Pi_{\mathcal{N}(\sigma)}$. The adjoint sends the identity to the identity because $\mathcal{N}$ is trace preserving. On the support of σ, one obtains $\sigma^{1/2}I\sigma^{1/2} = \sigma$, with any necessary restriction or trace-preserving extension outside the support. □

Remark 11.15 (Equality versus approximate recovery)*.* Exact equality in the relative-entropy data-processing inequality for the pair (ρ, σ) is equivalent, under the appropriate finite-dimensional support conditions, to the existence of a recovery map reconstructing both ρ and σ. The Petz map gives the canonical exact recovery expression. Quantitative approximate-recovery theorems use suitably rotated or averaged Petz maps; one must not assume the unrotated Petz map always attains the strongest available fidelity bound.

11.7.1 Bayes' rule as a commuting Petz map

Consider a classical channel $W(y|x)$ and reference input distribution $q(x) > 0$. Its reference output is $q_Y(y) = \sum_x W(y|x)q(x)$. The transpose recovery kernel is

$$R_q(x|y) = \frac{q(x)W(y|x)}{q_Y(y)} \quad \text{for } q_Y(y) > 0. \tag{11.20}$$

This is exactly Bayesian inversion. It is stochastic because $\sum_x R_q(x|y) = 1$. Applying it to the reference output gives $\sum_y R_q(x|y)q_Y(y) = q(x)$, matching exact recovery of the reference state. The quantum Petz formula is a noncommutative extension of this weighted reverse-channel construction.

11.8 Fidelity and one-shot min-entropy

Define the root fidelity by $F_{\mathrm{root}}(\rho, \sigma) = \left\|\sqrt{\rho}\sqrt{\sigma}\right\|_1$ and its square by $F(\rho, \sigma) = F_{\mathrm{root}}(\rho, \sigma)^2$. The distinction between these two conventions matters when comparing textbook formulas. For pure states, $F(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = |\langle\psi|\phi\rangle|^2$. The trace distance $T(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ obeys the Fuchs–van de Graaf bounds

$$1 - \sqrt{F(\rho, \sigma)} \le T(\rho, \sigma) \le \sqrt{1 - F(\rho, \sigma)}. \tag{11.21}$$

These inequalities connect a recovery theorem phrased in fidelity to a bound on observable outcome distributions.

11.8.1 Classical–quantum guessing probability

For a cq state $\rho_{XB} = \sum_x p_x \, |x\rangle\langle x| \otimes \rho_B^x$, the optimal guessing probability is

$$p_{\text{guess}}(X|B) = \max_{\{M_x\}} \sum_x p_x \, \text{Tr}(M_x \rho_B^x), \tag{11.22}$$

where the maximization ranges over POVMs on B. The conditional min-entropy is

$$H_{\min}(X|B) = -\log p_{\text{guess}}(X|B). \tag{11.23}$$

When all ρ_B^x coincide, Bob's register contains no information about X and $p_{\text{guess}} = \max_x p_x$. When the conditional states have orthogonal supports, a projective measurement distinguishes them exactly and $H_{\min}(X|B) = 0$.

Example 11.16 (Binary pure-state guessing)**.** For two equiprobable pure states $|0\rangle$ and $|+\rangle$, the optimal POVM is the Helstrom measurement and

$$p_{\text{guess}}(X|B) = \frac{1}{2}\left(1 + \frac{1}{\sqrt{2}}\right). \tag{11.24}$$

Consequently the conditional min-entropy is $-\log_2[(1+1/\sqrt{2})/2]$, strictly between zero and one bit.

11.9 Entropy inequalities compared on explicit tripartite states

11.9.1 A classically correlated Markov chain

Let B be an unbiased bit, and conditioned on $B = b$, set $A = C = b$ deterministically. The density operator is

$$\rho_{ABC} = \frac{1}{2} \, |000\rangle\langle 000| + \frac{1}{2} \, |111\rangle\langle 111| \, . \tag{11.25}$$

Its nontrivial marginals A, B, C, AB, and BC each have entropy one; the full state also has entropy one. Therefore

$$I(A:C|B) = S(AB) + S(BC) - S(B) - S(ABC) = 1 + 1 - 1 - 1 = 0. \tag{11.26}$$

Despite perfect marginal correlations between A and C, they are conditionally independent given B. Such an example shows that vanishing conditional mutual information does not mean absence of unconditional correlation.

11.9.2 A genuinely quantum tripartite example

For the pure GHZ vector $|\text{GHZ}\rangle_{ABC}$, $S(ABC) = 0$, $S(B) = 1$, and $S(AB) = S(BC) = 1$. Thus

$$I(A:C|B) = 1 + 1 - 1 - 0 = 1. \tag{11.27}$$

The GHZ state is not a quantum Markov chain conditioned on B even though each two-qubit marginal is separable. Quantum conditional mutual information detects a constraint on how global correlations are assembled, not simply the presence or absence of bipartite entanglement in a particular reduction.

11.10 Recovery maps for discarding an uncorrelated subsystem

Let $\rho_{AB} = \rho_A \otimes \tau_B$ and consider $\mathcal{N} = \mathrm{Tr}_B$. For the reference state $\sigma_{AB} = \sigma_A \otimes \tau_B$ with faithful factors, the Petz map for the partial trace is

$$\mathcal{R}_{\sigma,\mathrm{Tr}_B}(X_A) = \sigma_{AB}^{1/2}\left[\sigma_A^{-1/2} X_A \sigma_A^{-1/2} \otimes I_B\right]\sigma_{AB}^{1/2} \tag{11.28}$$

$$= X_A \otimes \tau_B. \tag{11.29}$$

The last equality uses the commuting product tensor factors and $\tau_B^{1/2} I \tau_B^{1/2} = \tau_B$. Therefore the recovered state is exactly $\rho_A \otimes \tau_B = \rho_{AB}$. The equality

$$D(\rho_A \otimes \tau_B \| \sigma_A \otimes \tau_B) = D(\rho_A \| \sigma_A) \tag{11.30}$$

shows explicitly why there is no relative-entropy loss when the discarded system carries no distinguishing information about the input hypothesis.

11.11 From fidelity to expectation-value error

For any two states ρ, σ and any Hermitian observable M with $\|M\|_\infty \leq 1$, the trace-norm duality gives

$$|\mathrm{Tr}[M(\rho - \sigma)]| \leq \|\rho - \sigma\|_1 = 2T(\rho, \sigma). \tag{11.31}$$

If a recovery theorem guarantees squared fidelity $F(\rho, \sigma) \geq 1 - \varepsilon$, the Fuchs–van de Graaf upper bound gives

$$T(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)} \leq \sqrt{\varepsilon}, \tag{11.32}$$

so every bounded observable expectation changes by at most $2\sqrt{\varepsilon}$. This operational conversion is useful when a theorem reports recovery fidelity but the experimental criterion is an expectation value or success probability.

11.12 Conditional min-entropy and optimal discrimination

For binary equiprobable cq states, the min-entropy is determined by the Helstrom trace norm:

$$H_{\min}(X|B) = -\log\left[\frac{1}{2} + \frac{1}{4}\|\rho_0 - \rho_1\|_1\right]. \tag{11.33}$$

For $\rho_0 = |0\rangle\langle 0|$ and $\rho_1 = |+\rangle\langle +|$, the trace norm is $\sqrt{2}$ and $H_{\min}(X|B) = -\log[(2+\sqrt{2})/4]$. Orthogonal states have trace norm two and zero conditional min-entropy; identical states have trace norm zero and one bit of conditional min-entropy. Both cases agree with $H_{\min}(X|B) = -\log p_{\mathrm{guess}}(X|B)$.

CHAPTER 12

Classical and Quantum Capacities of Noisy Quantum Channels

12.1 Classical messages transmitted through quantum outputs

Definition 12.1 (Classical–quantum state and Holevo information)**.** For an ensemble $\mathcal{E} = \{p_x, \rho_x\}$ and quantum register B, define

$$\rho_{XB} = \sum_x p_x \, |x\rangle\langle x| \otimes \rho_x, \quad \bar{\rho} = \sum_x p_x \rho_x, \quad \chi(\mathcal{E}) = S(\bar{\rho}) - \sum_x p_x S(\rho_x). \tag{12.1}$$

Then $\chi(\mathcal{E}) = I(X:B)_\rho$.

Entropy identity. The eigenvalues of ρ_{XB} are $p_x \lambda_{xj}$, where λ_{xj} are the eigenvalues of ρ_x. Hence $S(XB) = H(p) + \sum_x p_x S(\rho_x)$, $S(X) = H(p)$, and $S(B) = S(\bar{\rho})$. □

Theorem 12.2 (Holevo bound)**.** *For any measurement on B producing classical outcome Y,*

$$I(X:Y) \le I(X:B) = \chi(\mathcal{E}) \le \log d_B. \tag{12.2}$$

Proof. Measurement is a channel $B \to Y$, so data processing gives the first inequality. The second follows from $\chi = S(\bar{\rho}) - \sum_x p_x S(\rho_x) \le S(\bar{\rho}) \le \log d_B$. □

Definition 12.3 (One-use Holevo optimization)**.** For a channel $\mathcal{N} : A \to B$,

$$\chi^*(\mathcal{N}) = \max_{\{p_x, \rho_x\}} \left[S\left(\sum_x p_x \mathcal{N}(\rho_x) \right) - \sum_x p_x S(\mathcal{N}(\rho_x)) \right]. \tag{12.3}$$

Theorem 12.4 (Unassisted classical capacity)**.** *For a finite-dimensional memoryless channel, with arbitrary block encodings and collective decoding,*

$$C(\mathcal{N}) = \lim_{n\to\infty} \frac{1}{n} \chi^*(\mathcal{N}^{\otimes n}) = \sup_{n \ge 1} \frac{1}{n} \chi^*(\mathcal{N}^{\otimes n}). \tag{12.4}$$

If χ^* is additive for the channel, the regularization reduces to $C(\mathcal{N}) = \chi^*(\mathcal{N})$.

Example 12.5 (Depolarizing channel)**.** For $\mathcal{D}_q(\rho) = (1-q)\rho + qI/2$, a pure input has output spectrum $\{1 - q/2, q/2\}$. Every ensemble therefore satisfies $\chi \le 1 - h_2(q/2)$; two equiprobable orthogonal pure signals attain equality. Additivity for this qubit depolarizing channel gives

$$C(\mathcal{D}_q) = 1 - h_2(q/2). \tag{12.5}$$

12.2 Entanglement-assisted classical communication

Definition 12.6 (Channel mutual information)**.** Let $|\phi\rangle_{RA}$ purify the input ρ_A, and set $\omega_{RB} = (\mathrm{id}_R \otimes \mathcal{N})(|\phi\rangle\langle\phi|)$. Define

$$I(\mathcal{N}) = \max_{\rho_A} I(R:B)_\omega. \tag{12.6}$$

Theorem 12.7 (Entanglement-assisted classical capacity)**.** *For a finite-dimensional memoryless channel, with unlimited preshared entanglement,*

$$C_E(\mathcal{N}) = I(\mathcal{N}). \tag{12.7}$$

Moreover, $I(\mathcal{N} \otimes \mathcal{M}) = I(\mathcal{N}) + I(\mathcal{M})$, so no regularization is needed.

For the identity qubit channel, $C_E(\mathrm{id}_2) = 2$ bits/use, consistent with superdense coding.

12.3 Quantum communication and coherent information

Definition 12.8 (Quantum transmission code)**.** An (n, K, ε) quantum code consists of an encoder $\mathcal{E} : \mathcal{L}(\mathbb{C}^K) \to \mathcal{L}(A^{\otimes n})$ and decoder $\mathcal{D} : \mathcal{L}(B^{\otimes n}) \to \mathcal{L}(\mathbb{C}^K)$ such that for every pure $|\phi\rangle_{R\mathbb{C}^K}$ (with $\dim R = K$), the final state is within trace distance ε of $|\phi\rangle\langle\phi|$ under the canonical identification of the logical input and output.

A rate Q is achievable when $\varepsilon_n \to 0$ and $\liminf_n (\log K_n)/n \geq Q$. The quantum capacity $Q(\mathcal{N})$ is the supremum of achievable rates.

Definition 12.9 (Coherent information)**.** For $\omega_{RB} = (\mathrm{id} \otimes \mathcal{N})(\phi_{RA})$ with ϕ pure, let

$$I(R\rangle B)_\omega = S(B)_\omega - S(RB)_\omega = -S(R \mid B)_\omega, \quad Q^{(1)}(\mathcal{N}) = \max_{\phi_{RA} \text{ pure}} I(R\rangle B)_\omega. \tag{12.8}$$

Theorem 12.10 (Quantum capacity formula)**.** *For a finite-dimensional memoryless quantum channel,*

$$Q(\mathcal{N}) = \lim_{n\to\infty} \frac{1}{n} Q^{(1)}(\mathcal{N}^{\otimes n}). \tag{12.9}$$

Unlike entanglement-assisted mutual information, coherent information is not additive for every channel.

12.4 Degradable channels

Definition 12.11 (Degradable channel)**.** A channel $\mathcal{N} : A \to B$ with complementary channel $\mathcal{N}^c : A \to E$ is degradable if there exists a CPTP map $\mathcal{T} : B \to E$ satisfying $\mathcal{N}^c = \mathcal{T} \circ \mathcal{N}$.

For degradable channels, coherent information is additive and $Q(\mathcal{N}) = Q^{(1)}(\mathcal{N})$. For antidegradable channels, the environment can simulate Bob's output, and the no-cloning argument entails zero quantum capacity.

12.5 Capacity comparison

Task	Optimized quantity	Regularization
Classical, unassisted	Holevo information $\chi^*(\mathcal{N}^{\otimes n})$	In general required
Classical, entanglement assisted	Quantum mutual information $I(\mathcal{N})$	Not required
Quantum-state transmission	Coherent information $Q^{(1)}(\mathcal{N}^{\otimes n})$	In general required

These capacities are operationally distinct; a measured one-use accessible-information quantity must not be equated with the full asymptotic classical capacity.

12.6 Three distinct communication tasks and their code models

A channel may carry classical labels encoded in quantum states, unknown quantum states, or classical labels assisted by pre-shared entanglement. These are different tasks. In a classical communication code for a quantum channel $\mathcal{N} : A \to B$, the encoder chooses an input state ρ_m for message m, the decoder performs a POVM $\{\Lambda_m\}$, and the average success probability is $M^{-1} \sum_m \mathrm{Tr}[\Lambda_m \mathcal{N}(\rho_m)]$. In a quantum transmission code, the encoder and decoder are CPTP maps and fidelity to the identity channel on a reference-entangled input is the natural performance requirement.

12.7 Holevo information and its operational origin

For an ensemble $\mathcal{E} = \{p_x, \rho_x\}$, introduce the cq state

$$\rho_{XB} = \sum_x p_x \, |x\rangle\langle x| \otimes \rho_x. \tag{12.10}$$

It has marginals $\rho_X = \sum_x p_x \, |x\rangle\langle x|$ and $\bar{\rho} = \sum_x p_x \rho_x$. Its joint entropy is

$$S(XB) = H(p) + \sum_x p_x S(\rho_x), \tag{12.11}$$

because the density matrix is block diagonal, with blocks $p_x \rho_x$. Therefore

$$I(X : B) = S(\bar{\rho}) - \sum_x p_x S(\rho_x) =: \chi(\mathcal{E}). \tag{12.12}$$

If Bob measures a POVM, the measurement is a channel $\mathcal{M} : B \to Y$ into a classical register. Quantum data processing gives

$$I(X : Y) \leq I(X : B) = \chi(\mathcal{E}), \tag{12.13}$$

which is the Holevo bound. It constrains accessible mutual information, not the cardinality of distinct preparations.

Example 12.12 (A nonorthogonal binary ensemble)**.** Let $p_0 = p_1 = 1/2$, $\rho_0 = |0\rangle\langle 0|$, $\rho_1 = |+\rangle\langle +|$. Each signal is pure. The average state is

$$\bar{\rho} = \frac{1}{2} \, |0\rangle\langle 0| + \frac{1}{2} \, |+\rangle\langle +| = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}. \tag{12.14}$$

Its eigenvalues are $\lambda_\pm = (2 \pm \sqrt{2})/4$, so

$$\chi = h_2\left(\frac{2+\sqrt{2}}{4}\right) \simeq 0.60088 \text{ bits}. \tag{12.15}$$

The Helstrom measurement minimizes classification error for these two hypotheses; it does not automatically maximize mutual information for every general ensemble. Even though two states can be distinguished with error below $1/2$, the ensemble cannot convey one full

accessible bit per use.

12.7.1 Product-state coding and regularization

For a single channel $\mathcal{N}$, define

$$\chi^*(\mathcal{N}) = \sup_{\{p_x,\rho_x\}} \left[S\left(\sum_x p_x \mathcal{N}(\rho_x)\right) - \sum_x p_x S(\mathcal{N}(\rho_x)) \right]. \tag{12.16}$$

Product-state encodings across independent channel uses achieve rates up to $\chi^*(\mathcal{N})$ with sufficiently long block codes and collective output measurement. If the sender may employ entangled inputs across uses, the unrestricted classical capacity is

$$C(\mathcal{N}) = \lim_{n\to\infty} \frac{1}{n} \chi^*(\mathcal{N}^{\otimes n}). \tag{12.17}$$

The regularization is essential in general, because additivity of the one-use Holevo optimization fails for some channels. It must not be replaced silently by $\chi^*(\mathcal{N})$ as a universal identity.

12.8 Entanglement-assisted classical capacity

Suppose Alice and Bob may consume arbitrarily many pre-shared entangled states and use n copies of $\mathcal{N}$ for classical communication. The entanglement-assisted capacity is single-letter:

$$C_E(\mathcal{N}) = \max_{\rho_A} I(R:B)_{(\mathrm{id}_R \otimes \mathcal{N})(|\phi\rangle\langle\phi|)}, \tag{12.18}$$

where $|\phi\rangle_{RA}$ purifies ρ_A. The reference system appears because mutual information must quantify correlations between the input information and the channel output. Different purifications produce locally isometric references and do not change the entropy expression.

Example 12.13 (The noiseless qudit channel)**.** For $\mathcal{N} = \mathrm{id}_d$, choose $\rho_A = I/d$ and a maximally entangled purification. Then $S(R) = S(B) = \log d$ and $S(RB) = 0$, giving

$$C_E(\mathrm{id}_d) = 2\log d. \tag{12.19}$$

This is precisely the superdense-coding rate; it does not contradict the unassisted classical capacity $C(\mathrm{id}_d) = \log d$.

12.9 Quantum capacity and coherent information

An (n, k) quantum code maps a k-qubit logical system into n channel inputs and then decodes the n outputs. Its rate is k/n transmitted qubits per use, with entanglement fidelity approaching one for the identity action on the logical system and an untouched reference. Given a state ρ_A with purification $|\phi\rangle_{RA}$,

$$I_c(\rho, \mathcal{N}) = S(B) - S(RB) = -S(R|B) \tag{12.20}$$

is the coherent information. Unlike mutual information, coherent information need not be nonnegative. The general quantum capacity formula involves regularization:

$$Q(\mathcal{N}) = \lim_{n\to\infty} \frac{1}{n} \max_{\rho_{A^n}} I_c(\rho_{A^n}, \mathcal{N}^{\otimes n}). \tag{12.21}$$

It captures the possibility that entangled input blocks improve coding rates beyond any independent single-use optimization. For degradable channels the coherent information is additive under the appropriate conditions and a one-use formula applies.

Example 12.14 (The qubit erasure channel)**.** For $\mathcal{E}_\epsilon(\rho) = (1-\epsilon)\rho \oplus \epsilon\,|\perp\rangle\langle\perp|$, the receiver gets the input intact with probability $1-\epsilon$, while the complementary channel gets it with probability ϵ. On a maximally mixed input the coherent information is

$$I_c = (1-2\epsilon)\log 2 = 1-2\epsilon. \tag{12.22}$$

The known quantum capacity of the qubit erasure channel is

$$Q(\mathcal{E}_\epsilon) = \max\{0, 1-2\epsilon\}. \tag{12.23}$$

For $\epsilon \geq 1/2$, the receiver is at least as erased as the environment and unassisted quantum transmission has zero asymptotic rate.

12.10 Capacity calculations for symmetric qubit noise

The qubit depolarizing channel $\mathcal{D}_q(\rho) = (1-q)\rho + qI/2$ is unitarily covariant. The single-use Holevo optimization is attained by a uniform ensemble of orthogonal pure states. The average output is maximally mixed, with entropy one, while each pure input produces output eigenvalues $1-q/2$ and $q/2$. Therefore

$$\chi^*(\mathcal{D}_q) = 1 - h_2(q/2). \tag{12.24}$$

For this covariant unital qubit channel the unassisted classical capacity equals this expression, but the conclusion relies on the relevant additivity property; covariance alone should not be promoted to an additivity theorem for arbitrary high-dimensional channels.

Example 12.15 (Numerical depolarization benchmark)**.** For $q = 1/2$, the two output eigenvalues from a pure input are $3/4, 1/4$. Thus the classical capacity is

$$C(\mathcal{D}_{1/2}) = 1 - h_2(1/4) \simeq 0.18872 \text{ bit/use}. \tag{12.25}$$

The physical channel still outputs a qubit on every use, but distinguishability of classical signals is degraded.

12.11 Single-use Holevo optimization for Pauli-covariant noise

Let a qubit depolarizing channel act as $\mathcal{D}_q(\rho) = (1-q)\rho + qI/2$. For a pure input, the output eigenvalues are $1-q/2$ and $q/2$, so every pure input has output entropy $h_2(q/2)$. For any mixed input, concavity of entropy and the shrinking Bloch vector imply that the output

entropy cannot be smaller than the pure-input value. For an arbitrary ensemble $\{p_x, \rho_x\}$,

$$\chi = S\left(\sum_x p_x \mathcal{D}_q(\rho_x)\right) - \sum_x p_x S(\mathcal{D}_q(\rho_x)) \tag{12.26}$$

$$\leq 1 - h_2(q/2). \tag{12.27}$$

Choose equiprobable computational-basis inputs $|0\rangle, |1\rangle$. The average output is $I/2$ and each output has entropy $h_2(q/2)$, attaining the bound. Hence

$$\chi^*(\mathcal{D}_q) = 1 - h_2(q/2). \tag{12.28}$$

The equality is a single-use optimization. Equating it with the asymptotic classical capacity additionally uses known additivity in this special qubit channel class.

12.12 A direct comparison of three benchmark channels

Channel	Capacity or information identity
Noiseless qubit	$C = 1$, $C_E = 2$, $Q = 1$ bit/qubit per use
Completely depolarizing qubit	$C = 0$, $C_E = 0$, $Q = 0$
Qubit erasure with ϵ	$C = 1 - \epsilon$, $Q = \max\{0, 1 - 2\epsilon\}$

The first row reflects the difference between an unassisted classical code, an entanglement-assisted code, and a quantum transmission code. The second row is zero because every input produces the same output density matrix. The erasure row illustrates that reliable classical communication may persist after quantum capacity has fallen to zero, because classically identifiable unerased positions can still transmit data.

12.13 Classical capacity of an erasure quantum channel

For $\mathcal{E}_\epsilon$ with input dimension d, consider an orthogonal computational-basis ensemble with uniform prior. Upon non-erasure, the label is recovered perfectly; upon erasure, no input information survives. Therefore

$$I(X : Y) = (1 - \epsilon) \log d. \tag{12.29}$$

The upper bound follows from the erasure flag: conditioned on a non-erasure, at most $\log d$ bits can be conveyed by a d-dimensional output; conditioned on erasure, the receiver sees a fixed orthogonal symbol. Thus

$$C(\mathcal{E}_\epsilon) = (1 - \epsilon) \log d. \tag{12.30}$$

For $d = 2$ this becomes $1-\epsilon$ classical bits per use, while the quantum capacity is $\max\{0, 1-2\epsilon\}$ qubits per use. The extra factor of ϵ in quantum capacity reflects the stricter requirement of preserving coherent entanglement with a reference.

12.14 Coherent information on a pure input versus a mixed input

If the channel input is pure and unentangled with a reference, a purifying reference can be taken one-dimensional. The channel output B has entropy $S(B)$ and the joint RB state has the same entropy; hence $I_c = 0$. Thus maximizing coherent information for a channel generally requires searching over mixed input density matrices, even though ordinary classical ensembles are often optimized using pure signal states. This distinction is important when deriving the erasure-channel rate from a maximally mixed input rather than a pure computational-basis state.

CHAPTER 13

Continuous-Variable Quantum Information

13.1 Canonical quadratures and covariance matrices

Definition 13.1 (Bosonic mode and quadratures)**.** A single bosonic mode obeys $[a, a^\dagger] = I$. In the normalization used throughout this chapter,

$$\hat{x} = a + a^\dagger, \qquad \hat{p} = -i(a - a^\dagger), \qquad [\hat{x}, \hat{p}] = 2iI. \tag{13.1}$$

Vacuum has $\langle \hat{x} \rangle = \langle \hat{p} \rangle = 0$ and $\mathrm{Var}(\hat{x}) = \mathrm{Var}(\hat{p}) = 1$ (shot-noise units).

For m modes, write $\hat{\boldsymbol{r}} = (\hat{x}_1, \hat{p}_1, \ldots, \hat{x}_m, \hat{p}_m)^T$, and $\Omega = \bigoplus_{j=1}^{m} \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, so that $[\hat{r}_j, \hat{r}_k] = 2i\Omega_{jk} I$.

Definition 13.2 (First and second moments)**.** The displacement vector is $d_j = \mathrm{Tr}(\rho \hat{r}_j)$ and the real symmetric covariance matrix is

$$V_{jk} = \frac{1}{2} \mathrm{Tr}\, \rho \{\hat{r}_j - d_j, \hat{r}_k - d_k\}. \tag{13.2}$$

A physical Gaussian covariance matrix satisfies $V + i\Omega \succeq 0$.

Theorem 13.3 (Canonical uncertainty)**.** *For one mode,* $\mathrm{Var}(\hat{x}) \mathrm{Var}(\hat{p}) - \mathrm{Cov}(\hat{x}, \hat{p})^2 \geq 1$. *Equivalently* $\det V \geq 1$.

13.2 Gaussian states

Definition 13.4 (Gaussian state)**.** A state is Gaussian when its Weyl characteristic function is Gaussian. Such a state is determined completely by its mean vector $\boldsymbol{d}$ and covariance matrix V; higher cumulants vanish. A pure one-mode Gaussian state has $\det V = 1$.

A coherent state $|\alpha\rangle = D(\alpha)|0\rangle$ with $D(\alpha) = e^{\alpha a^\dagger - \alpha^* a}$ satisfies $a|\alpha\rangle = \alpha|\alpha\rangle$. Its moments are $\boldsymbol{d} = (2\Re\alpha, 2\Im\alpha)$, $V = I_2$, and its photon-number distribution is $e^{-|\alpha|^2} |\alpha|^{2n}/n!$. A thermal state with mean photon number $\bar{n}$ has zero displacement and covariance $V = (2\bar{n} + 1) I_2$.

Definition 13.5 (Single-mode squeezing)**.** Let $S(r) = \exp\left[\frac{r}{2}(a^2 - a^{\dagger 2})\right]$, $r \in \mathbb{R}$. The squeezed vacuum $S(r)|0\rangle$ has

$$V_r = \mathrm{diag}(e^{-2r}, e^{2r}). \tag{13.3}$$

The squeezing level relative to vacuum is $s_{\mathrm{dB}} = -10 \log_{10} V_x$ when the x-variance is squeezed.

Example 13.6 (Six-decibel squeezing)**.** A reported 6 dB reduction below vacuum corresponds to $V_x = 10^{-6/10} \approx 0.251$ and, for a pure squeezed state, $V_p = 1/V_x \approx 3.981$.

13.3 Symplectic Gaussian transformations

Definition 13.7 (Gaussian unitary)**.** A unitary generated by a Hamiltonian at most quadratic in quadratures induces an affine transformation $\hat{\boldsymbol{r}} \mapsto S\hat{\boldsymbol{r}} + \boldsymbol{s}$ with $S\Omega S^T = \Omega$. Accordingly,

$$\boldsymbol{d} \mapsto S\boldsymbol{d} + \boldsymbol{s}, \qquad V \mapsto SVS^T. \tag{13.4}$$

The phase shift $e^{-i\theta a^\dagger a}$ rotates phase space. A beam splitter with intensity transmissivity η mixes two modes through an orthogonal symplectic matrix. Two-mode squeezing produces an entangled Gaussian state with EPR-like quadrature correlations $\mathrm{Var}(x_A - x_B)$ and $\mathrm{Var}(p_A + p_B)$ decreasing as the squeezing increases.

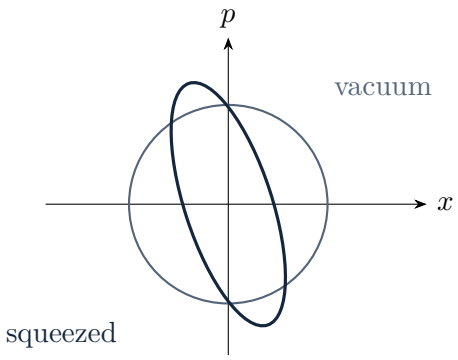


Figure 13.1: Quadrature covariance contours in shot-noise units. A symplectic squeeze contracts one quadrature while expanding its conjugate.

13.4 Loss and open Gaussian channels

A pure-loss channel with transmissivity $0 \le \eta \le 1$ is modeled by mixing the signal with an environmental vacuum mode. For a single-mode input $(\boldsymbol{d}, V)$,

$$\boldsymbol{d}_{\mathrm{out}} = \sqrt{\eta}\,\boldsymbol{d}_{\mathrm{in}}, \qquad V_{\mathrm{out}} = \eta V_{\mathrm{in}} + (1-\eta) I_2. \tag{13.5}$$

For squeezed input $V_x = e^{-2r}$ at $\eta = 1/2$, the output variance is $V_x^{\mathrm{out}} = (1 + e^{-2r})/2$: loss injects vacuum noise and prevents arbitrarily strong observable squeezing at fixed nonunit transmissivity.

13.5 Homodyne and heterodyne detection

Definition 13.8 (Homodyne observable)**.** A balanced homodyne receiver with local-oscillator phase θ measures $x_\theta = x\cos\theta + p\sin\theta$. With a sufficiently strong coherent local oscillator and ideal mode matching, the difference photocurrent is proportional to this quadrature.

Heterodyne detection jointly estimates both conjugate quadratures by splitting the signal and homodyning the outputs at orthogonal phases. The unavoidable vacuum entering the unused port adds one shot-noise unit per quadrature in the standard rescaled heterodyne-outcome convention. Finite efficiency η_d similarly mixes in vacuum and detector noise.

13.6 Continuous-variable teleportation

In a Gaussian teleportation protocol, Alice combines the unknown signal with her half of a two-mode squeezed state on a beam splitter, measures conjugate quadratures, sends classical real outcomes, and Bob displaces his mode. For ideal unity-gain conventions,

$$x_{\text{out}} = x_{\text{in}} + (x_B - x_A), \qquad p_{\text{out}} = p_{\text{in}} + (p_B + p_A). \tag{13.6}$$

Finite EPR squeezing therefore induces additive noise set by the residual correlations; infinite squeezing is an unphysical idealization.

13.7 Gaussian-modulated coherent-state key distribution

Alice samples classical Gaussian variables, prepares coherent states with corresponding displacements, and transmits them through an optical channel. In a simplified phase-aligned homodyne model,

$$X_B = \sqrt{\eta}\, X_A + Z, \tag{13.7}$$

where Z collects vacuum and excess noise in the adopted shot-noise convention. The classical data determine channel parameter estimates and an asymptotic collective-attack key-rate expression of the form

$$K \geq \beta I(X_A : X_B) - \chi(B : E), \tag{13.8}$$

under an explicitly selected reconciliation direction, security model, trusted-device assumptions, and finite-size treatment. This expression is not a universal security guarantee without those assumptions. Calibration of local-oscillator power, detector efficiency, mode overlap, electronic noise, and parameter-estimation confidence intervals is part of the protocol model.

13.8 Quadrature conventions and the Fock representation

For each bosonic mode define $a, a^\dagger$ with $[a, a^\dagger] = I$ and quadratures

$$\hat{x} = a + a^\dagger, \qquad \hat{p} = \frac{a - a^\dagger}{i}, \qquad [\hat{x}, \hat{p}] = 2iI. \tag{13.9}$$

This convention assigns the vacuum state $|0\rangle$ unit quadrature variances:

$$\langle 0|\hat{x}|0\rangle = \langle 0|\hat{p}|0\rangle = 0, \qquad \mathrm{Var}_0(\hat{x}) = \mathrm{Var}_0(\hat{p}) = 1. \tag{13.10}$$

The number operator is $n = a^\dagger a = (\hat{x}^2 + \hat{p}^2 - 2I)/4$, so $\hat{x}^2 + \hat{p}^2 = 4n + 2I$. In the Fock basis, $a\,|n\rangle = \sqrt{n}\,|n-1\rangle$ and $a^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$. The Fock space is infinite dimensional and finite-dimensional covariance formulas require finite second moments.

Example 13.9 (Quadrature variances of a Fock state)**.** For a number state $|n\rangle$, the first moments vanish. Since $\langle n|a^2|n\rangle = 0$ and $\langle n|aa^\dagger + a^\dagger a|n\rangle = 2n + 1$,

$$\mathrm{Var}_{|n\rangle}(\hat{x}) = \mathrm{Var}_{|n\rangle}(\hat{p}) = 2n + 1. \tag{13.11}$$

The product of variances equals $(2n+1)^2$, saturating the canonical lower bound only for $n = 0$.

13.8.1 Coherent and thermal states

The coherent state $|\alpha\rangle$ satisfies $a\,|\alpha\rangle = \alpha\,|\alpha\rangle$ and has Fock expansion

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,. \tag{13.12}$$

Its quadrature means are $\langle\hat{x}\rangle = 2\Re\alpha$, $\langle\hat{p}\rangle = 2\Im\alpha$, and its covariance matrix equals I_2. The photon-number distribution is Poisson with mean and variance $|\alpha|^2$. For a thermal state of mean occupation $\bar{n}$, the covariance matrix is $(2\bar{n}+1)I_2$ and the von Neumann entropy is

$$g(\bar{n}) = (\bar{n}+1)\log(\bar{n}+1) - \bar{n}\log\bar{n}. \tag{13.13}$$

13.9 Covariance matrices and canonical uncertainty

For an m-mode system let $\boldsymbol{R} = (\hat{x}_1, \hat{p}_1, \ldots, \hat{x}_m, \hat{p}_m)^{\mathsf{T}}$, with symplectic form

$$\Omega = \bigoplus_{j=1}^{m} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad [R_j, R_k] = 2i\Omega_{jk} I. \tag{13.14}$$

Define means $d_j = \mathrm{Tr}(\rho R_j)$ and covariance

$$V_{jk} = \frac{1}{2}\,\mathrm{Tr}\,\rho\{R_j - d_j, R_k - d_k\}. \tag{13.15}$$

Physical states obey the matrix uncertainty inequality

$$V + i\Omega \succeq 0. \tag{13.16}$$

For a one-mode diagonal covariance $V = \mathrm{diag}(v_x, v_p)$, the 2×2 principal-minor condition gives $v_x v_p \geq 1$. A covariance that violates this product constraint cannot correspond to any physical state, even if V itself is positive definite.

Example 13.10 (A covariance check)**.** Take $V = \mathrm{diag}(1/4, 4)$. It saturates $v_x v_p = 1$ and is physically realizable as a pure squeezed vacuum. In contrast $V = \mathrm{diag}(1/4, 3)$ has determinant $3/4 < 1$ and violates the uncertainty principle. No choice of displacement vector can repair an unphysical covariance matrix.

13.10 Gaussian states and their transformations

A Gaussian state is one whose characteristic function or Wigner quasiprobability is Gaussian in phase space. It is fully described by the first-moment vector $\boldsymbol{d}$ and covariance V. A Gaussian unitary implements a symplectic affine transformation

$$\boldsymbol{d} \mapsto S\boldsymbol{d} + \boldsymbol{s}, \qquad V \mapsto SVS^{\mathsf{T}}, \qquad S\Omega S^{\mathsf{T}} = \Omega. \tag{13.17}$$

Phase-space displacement changes means but not covariance; phase rotation changes the orientation of a covariance ellipse; squeezing changes its principal widths while preserving phase-space area for a single-mode pure Gaussian state.

13.10.1 Single-mode squeezing

For a real squeezing parameter r, the symplectic map is $S(r) = \mathrm{diag}(e^{-r}, e^{r})$. Acting on vacuum gives

$$V_r = S(r)S(r)^{\mathsf{T}} = \mathrm{diag}(e^{-2r}, e^{2r}). \tag{13.18}$$

The product remains one. A commonly used squeezing level expressed relative to vacuum is $10 \log_{10} V_x$ dB. Six decibels of squeezing below vacuum correspond approximately to $V_x = 10^{-6/10} \approx 0.2512$ and $V_p \approx 3.9811$ for an ideal minimum-uncertainty state. This convention uses variance, not amplitude, as the dB reference.

Example 13.11 (Photon number of a squeezed vacuum)**.** For $S(r)\,|0\rangle$, the mean photon number is $\sinh^2 r$. It follows from $n = (x^2 + p^2 - 2)/4$ that

$$\bar{n} = \frac{e^{-2r} + e^{2r} - 2}{4} = \sinh^2 r. \tag{13.19}$$

This links the covariance description to the Fock-space energy cost of squeezing.

13.10.2 Beam splitters

A passive beam splitter of transmissivity η transforms annihilation operators by

$$\begin{pmatrix} a_1' \\ a_2' \end{pmatrix} = \begin{pmatrix} \sqrt{\eta} & \sqrt{1-\eta} \\ -\sqrt{1-\eta} & \sqrt{\eta} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}. \tag{13.20}$$

The associated 4×4 real matrix acts identically on the x and p quadrature components, with the same real mixing coefficients. For two independent input states, output covariance is SVS^{T}. At $\eta = 1/2$ a balanced splitter generates two correlated outputs from suitable squeezed inputs, although classical coherent-state inputs remain a product of coherent states.

13.11 Loss and Gaussian noise in a covariance model

A pure-loss channel of transmissivity η can be realized by a beam splitter between the signal and a vacuum environment. Tracing out the environment yields

$$\boldsymbol{d} \mapsto \sqrt{\eta}\,\boldsymbol{d}, \qquad V \mapsto \eta V + (1-\eta) I_2. \tag{13.21}$$

For a coherent-state input $V = I_2$, the output is again a coherent state with attenuated displacement. For a squeezed input, loss pulls both quadrature variances toward the vacuum level one.

Example 13.12 (Loss of squeezed light)**.** Let $V_{\mathrm{in}} = \mathrm{diag}(1/4, 4)$ and $\eta = 1/2$. Then

$$V_{\mathrm{out}} = \frac{1}{2}\,\mathrm{diag}(1/4, 4) + \frac{1}{2} I_2 = \mathrm{diag}(5/8, 5/2). \tag{13.22}$$

Its determinant is $25/16 > 1$, showing that the originally pure Gaussian state has become mixed after coupling to and tracing over the environment.

13.12 Quadrature detection and Gaussian entanglement

An ideal homodyne detector measures a rotated quadrature $\hat{x}_\theta = \hat{x}\cos\theta + \hat{p}\sin\theta$. On a Gaussian state the outcome distribution is Gaussian with mean $\boldsymbol{u}_\theta^{\mathsf{T}} \boldsymbol{d}$ and variance $\boldsymbol{u}_\theta^{\mathsf{T}} V \boldsymbol{u}_\theta$, where $\boldsymbol{u}_\theta = (\cos\theta, \sin\theta)^{\mathsf{T}}$. Heterodyne detection jointly estimates both quadratures using a vacuum ancilla and therefore includes additional vacuum noise. It cannot give two simultaneous perfect projective measurements of noncommuting observables.

13.12.1 Two-mode squeezed vacuum

For squeezing parameter $r \geq 0$, the two-mode squeezed vacuum has covariance

$$V_{AB} = \begin{pmatrix} \cosh(2r) I_2 & \sinh(2r) Z \\ \sinh(2r) Z & \cosh(2r) I_2 \end{pmatrix}, \qquad Z = \mathrm{diag}(1, -1). \tag{13.23}$$

It yields $\mathrm{Var}(x_A - x_B) = 2e^{-2r}$ and $\mathrm{Var}(p_A + p_B) = 2e^{-2r}$ with the convention $\mathrm{Var}(x_A) = \mathrm{Var}(x_B) = \cosh(2r)$. The two commuting collective quadratures approach perfect correlations as $r \to \infty$, although the required energy diverges. At $r = 0$ the covariance is I_4, representing two independent vacuum modes.

13.13 Continuous-variable communication and security variables

Gaussian-modulated coherent-state communication represents a real-valued displacement signal by quantum quadrature means. In prepare-and-measure CV-QKD, Alice's modulation variance, line transmissivity, receiver detection efficiency, electronic noise, and excess noise determine the observed covariance data. The security rate against a specified class of attacks is not determined merely by a squeezing figure or a raw signal-to-noise ratio: it also depends on reconciliation efficiency, finite-size confidence intervals, trusted detector models, and the measured or bounded information available to an adversary. A standard asymptotic

reverse-reconciliation bookkeeping expression has the form

$$K \geq \beta I(A:B) - \chi(B:E), \tag{13.24}$$

where β is reconciliation efficiency and $\chi(B:E)$ is Eve's Holevo information conditioned on the protocol's specified quantum state and measurement model. This is a protocol-dependent achievable-rate bound, not a universal identity for all CV-QKD schemes.

13.14 Wigner phase-space descriptions and physical covariance

For a one-mode Gaussian state with displacement $\boldsymbol{d}$ and covariance V in the convention $[x, p] = 2i$, its Wigner function may be written

$$W(\boldsymbol{\xi}) = \frac{1}{2\pi\sqrt{\det V}} \exp\left[-\frac{1}{2}(\boldsymbol{\xi} - \boldsymbol{d})^{\mathsf{T}} V^{-1} (\boldsymbol{\xi} - \boldsymbol{d})\right]. \tag{13.25}$$

This normalization makes $\int_{\mathbb{R}^2} W(\boldsymbol{\xi})\, \mathrm{d}^2\xi = 1$ and gives the vacuum distribution $W_0(x, p) = (2\pi)^{-1} \exp[-(x^2 + p^2)/2]$. Gaussian states have nonnegative Gaussian Wigner functions, although generic non-Gaussian quantum states can have Wigner negativity. A positive Wigner function by itself is not a claim that all quantum phenomena admit a noncontextual classical probability model.

13.15 Purity from a one-mode covariance matrix

For a one-mode Gaussian state with covariance V, the symplectic eigenvalue is $\nu = \sqrt{\det V} \geq 1$. Its purity is

$$\operatorname{Tr} \rho^2 = \frac{1}{\nu} = \frac{1}{\sqrt{\det V}}. \tag{13.26}$$

A pure Gaussian state has $\det V = 1$, independently of displacement. If $\nu > 1$, the state is mixed and has the same spectrum as a thermal state of mean photon number $\bar{n} = (\nu - 1)/2$, with entropy $g((\nu - 1)/2)$.

Example 13.13 (A squeezed thermal state)**.** Let $V = \operatorname{diag}(1/2, 8)$. Then $\nu = \sqrt{4} = 2$, so the Gaussian state has purity $1/2$ and effective thermal photon number $\bar{n} = 1/2$. The quadrature variances are strongly asymmetric, but the mixedness is determined by the symplectic area $\sqrt{\det V} = 2$.

13.16 Homodyne outcomes and conditioning

For $\boldsymbol{d} = (d_x, d_p)$ and covariance V, measuring x gives a normal distribution with mean d_x and variance V_{xx}. If $V = \operatorname{diag}(e^{-2r}, e^{2r})$, homodyne detection of x yields variance e^{-2r} while detection of p yields e^{2r}. The two outcome distributions are not simultaneous marginals of two commuting projective measurements on the same mode; each is obtained in a separate measurement setting.

Example 13.14 (A displaced coherent state)**.** Take $\alpha = 1 + i/2$. Then $d_x = 2$, $d_p = 1$, and $V = I_2$. Ideal x-homodyne outcomes have distribution $\mathcal{N}(2, 1)$, while ideal p-homodyne

outcomes have distribution $\mathcal{N}(1,1)$. Increasing coherent amplitude shifts the means but does not reduce the vacuum quantum noise.

13.17 Vacuum loss as a complete positive Gaussian map

The pure-loss channel with transmissivity η acts on a one-mode covariance as $V \mapsto \eta V + (1-\eta)I$. The added matrix $(1-\eta)I$ is not an arbitrary classical adjustment: it is required by preservation of the canonical commutation structure after discarding the environment. If one naively used $V \mapsto \eta V$ without added noise, a vacuum input would yield $V' = \eta I$ and $\det V' = \eta^2 < 1$ whenever $\eta < 1$, violating the uncertainty inequality. The environment contribution restores the physical lower bound.

Example 13.15 (Attenuation at three dB)**.** At $\eta = 1/2$ a coherent amplitude α becomes $\sqrt{1/2}\,\alpha$ while the vacuum variance remains one. A strongly squeezed input with $V_x \to 0$ cannot retain arbitrarily small output variance after this loss: $V_x' \to 1/2$. Hence passive loss sets a noise floor even for ideally prepared squeezing.

CHAPTER 14

Uncertainty Relations and Quantum Processes

14.1 Variance-based uncertainty

Definition 14.1 (Covariance of observables)**.** For Hermitian operators A, B in state ρ, write

$$\mathrm{Cov}_\rho(A,B) = \frac{1}{2}\operatorname{Tr}\rho(AB+BA) - \operatorname{Tr}(\rho A)\operatorname{Tr}(\rho B). \tag{14.1}$$

Set $\Delta_\rho A = \sqrt{\mathrm{Cov}_\rho(A,A)}$.

Theorem 14.2 (Robertson–Schrödinger)**.** *For Hermitian A, B,*

$$(\Delta A)^2(\Delta B)^2 \geq \mathrm{Cov}(A,B)^2 + \frac{1}{4}\left|\operatorname{Tr}\rho[A,B]\right|^2. \tag{14.2}$$

Proof. Apply Cauchy–Schwarz to $\sqrt{\rho}(A-\langle A\rangle)$ and $\sqrt{\rho}(B-\langle B\rangle)$ in Hilbert–Schmidt space, and separate the real covariance and imaginary commutator parts. □

For $[x,p] = 2iI$, the lower bound on $\Delta x\Delta p$ is one. A variance-product lower bound can be trivial when the commutator expectation vanishes; this does not imply that the two output distributions are simultaneously sharp.

14.2 Lie-algebraic covariance constraints

Suppose Hermitian generators $X_1, \ldots, X_m$ span a representation of a compact Lie algebra, and write their covariance matrix $G_{jk} = \mathrm{Cov}_\rho(X_j, X_k)$. Under a real invertible change of generator basis $Y = TX$, the covariance transforms as $G_Y = TG_XT^T$. Consequently a metric-adjusted scalar such as $\operatorname{Tr}(MG)$ is representation-coordinate invariant when M is transformed contragrediently.

Example 14.3 (Angular momentum and Casimir)**.** For a spin-j irrep, $J_x^2+J_y^2+J_z^2 = j(j+1)I$. Thus

$$\sum_{k=x,y,z}(\Delta J_k)^2 = j(j+1) - \|\langle \boldsymbol{J}\rangle\|^2 \geq j. \tag{14.3}$$

The last inequality follows from $\|\langle \boldsymbol{J}\rangle\| \leq j$. In the convention $C_2 = 2\sum_k J_k^2$, a natural scalar uncertainty is $\mathcal{U} = 2\sum_k(\Delta J_k)^2 \geq 2j$.

Highest weights, Casimir eigenvalues, and positive roots can parameterize lower bounds beyond $\mathfrak{su}(2)$. Their numerical coefficients depend on the chosen Lie-algebra normalization and representation; a bound in one convention must not be transplanted unchanged into another.

14.3 Entropic uncertainty without memory

Definition 14.4 (Measurement overlap)**.** For two orthonormal bases $X = \{|x_i\rangle\}$ and $Z = \{|z_j\rangle\}$, set $c = \max_{i,j} |\langle x_i|z_j\rangle|^2$.

Theorem 14.5 (Maassen–Uffink)**.** *If P_X, P_Z are the outcome distributions of the two measurements on any state ρ,*

$$H(P_X) + H(P_Z) \geq -\log c. \tag{14.4}$$

The equivalent overlap-amplitude convention $c_{\text{amp}} = \max |\langle x_i|z_j\rangle|$ gives $-2\log c_{\text{amp}}$. For the qubit X and Z bases, $c = 1/2$ and the bound is one bit. More generally, Rényi entropies $H_\alpha(P) = (1-\alpha)^{-1} \log \sum_i p_i^\alpha$ admit uncertainty inequalities for conjugate orders under appropriate conditions.

14.4 Quantum side information

Theorem 14.6 (Entropic uncertainty with memory)**.** *For projective measurements X, Z on A and a bipartite state ρ_{AB},*

$$H(X \mid B) + H(Z \mid B) \geq -\log c + S(A \mid B)_\rho, \tag{14.5}$$

where $H(X \mid B)$ denotes the conditional von Neumann entropy of the postmeasurement classical–quantum state. For a tripartite pure state ρ_{ABC}, a corresponding complementary-memory form is $H(X \mid B) + H(Z \mid C) \geq -\log c$.

Remark 14.7. A negative quantum conditional entropy can relax a two-measurement uncertainty bound when the memory is entangled with the measured system. It does not contradict the nonnegativity of the postmeasurement classical–quantum conditional entropy.

For one-shot cryptographic settings, smooth conditional min- and max-entropies replace average von Neumann entropy. A representative dual-memory relation takes the form $H_{\min}^\varepsilon(X \mid B) + H_{\max}^\varepsilon(Z \mid C) \geq -\log c$ with matching smoothing and state conventions.

14.5 Majorization uncertainty

Definition 14.8 (Majorization)**.** For vectors p, q of equal dimension, write $p \prec q$ if their entries sorted in nonincreasing order obey $\sum_{i=1}^k p_i^\downarrow \leq \sum_{i=1}^k q_i^\downarrow$ for $k < d$, with equality for $k = d$.

Every Schur-concave entropy respects this order: $p \prec q \Rightarrow H(p) \geq H(q)$. State-independent majorization bounds on the direct sum or tensor product of distributions from incompatible measurements imply whole families of entropic uncertainty relations. The direct-sum and tensor-product constructions have different normalization and bounding-vector conventions.

14.6 Measurement error and disturbance

Definition 14.9 (Noise-operator formulation)**.** For an indirect measurement modeled by a unitary U coupling system and probe, an initial probe state $|\xi\rangle$, and a pointer M, define noise $N(A) = U^\dagger(I \otimes M)U - A \otimes I$ and disturbance $D(B) = U^\dagger(B \otimes I)U - B \otimes I$. Their root-mean-square values in the joint input define $\varepsilon(A)$ and $\eta(B)$.

Theorem 14.10 (Ozawa error–disturbance relation)**.** *For the usual noise-operator RMS definitions,*

$$\varepsilon(A)\eta(B) + \varepsilon(A)\Delta B + \Delta A\,\eta(B) \geq \frac{1}{2}\left|\mathrm{Tr}\,\rho[A,B]\right|. \tag{14.6}$$

Operational calibration and worst-case (state-independent) error definitions lead to distinct formulations and must not be conflated with the above state-dependent RMS inequality.

14.7 Time, energy, and dynamical bounds

Theorem 14.11 (Mandelstam–Tamm orthogonalization)**.** *For a closed pure state evolving under a time-independent Hamiltonian with constant energy dispersion ΔE, the time $\tau_\perp$ to evolve to an orthogonal state satisfies*

$$\tau_\perp \geq \frac{\pi\hbar}{2\Delta E}. \tag{14.7}$$

Time–energy bounds arise from dynamics rather than a canonical self-adjoint time operator for every Hamiltonian. Thermodynamic uncertainty relations link current fluctuations to entropy production under specific stochastic-dynamical assumptions; for example, classical stationary Markov jump processes have long-time bounds schematically of the form $\mathrm{Var}(J)/\langle J\rangle^2 \gtrsim 2/\Sigma$, with Σ the total entropy production in compatible units.

14.8 Process tensors and causal uncertainty

Definition 14.12 (Finite-step quantum process tensor)**.** A process tensor maps a sequence of intervention instruments $\mathcal{M}_0, \ldots, \mathcal{M}_{k-1}$ to the probability of an observed outcome. In its Choi representation it is a positive multilinear functional obeying normalization constraints that forbid signalling from future operations to past statistics.

Ordinary channels describe a single input–output transformation; process tensors retain temporal correlations between successive interventions, allowing non-Markovian dynamics. Passive observation interrogates a fixed evolution, whereas interactive protocols choose instruments and condition later operations on earlier results. Causal discrimination can separate direct-cause, common-cause, and mixtures when the allowed interventions are sufficiently informative. An uncertainty bound for such process measurements is model dependent and cannot be inferred by inserting ordinary state POVM overlaps without specifying the process tester and its normalization.

14.9 Variance uncertainty derived from Hilbert-space geometry

Let A, B be Hermitian operators and let ρ be a state with finite second moments. Define centered operators $\widetilde{A} = A - \mathrm{Tr}(\rho A)I$ and $\widetilde{B} = B - \mathrm{Tr}(\rho B)I$. For a pure state, apply Cauchy–Schwarz to the vectors $\widetilde{A}\,|\psi\rangle$ and $\widetilde{B}\,|\psi\rangle$; for a mixed state, apply it to $\widetilde{A}\sqrt{\rho}$ and $\widetilde{B}\sqrt{\rho}$ with the Hilbert–Schmidt inner product. In both cases,

$$\mathrm{Var}_\rho(A)\,\mathrm{Var}_\rho(B) \geq \left|\mathrm{Tr}(\rho\widetilde{A}\widetilde{B})\right|^2. \tag{14.8}$$

Decompose the expectation into real and imaginary components:

$$\mathrm{Tr}(\rho\widetilde{A}\widetilde{B}) = \underbrace{\frac{1}{2}\,\mathrm{Tr}\,\rho\{\widetilde{A},\widetilde{B}\}}_{\mathrm{Cov}_\rho(A,B)} + \frac{1}{2}\,\mathrm{Tr}\,\rho[A,B]. \tag{14.9}$$

The covariance term is real, while the commutator expectation is purely imaginary. Therefore

$$\boxed{\mathrm{Var}(A)\,\mathrm{Var}(B) \geq \mathrm{Cov}(A,B)^2 + \frac{1}{4}\,|\langle[A,B]\rangle|^2\,.} \tag{14.10}$$

Dropping the nonnegative covariance term yields the Robertson inequality, $\Delta A\Delta B \geq |\langle[A,B]\rangle|\,/2$.

Example 14.13 (Canonical quadratures)**.** With $[x,p] = 2iI$ the Robertson bound reads $\Delta x\Delta p \geq 1$. Vacuum has $\Delta x = \Delta p = 1$ and saturates the inequality. A squeezed vacuum with $\Delta x = e^{-r}$ and $\Delta p = e^{r}$ also saturates it, despite a large imbalance between the two uncertainties.

14.9.1 Why the Robertson bound can vanish for incompatible observables

For a qubit, choose $A = X$, $B = Z$. Their commutator is $[X,Z] = -2iY$. In the state $|0\rangle$, $\langle Y\rangle = 0$, so the Robertson lower bound is zero, even though measuring X is maximally uncertain. Indeed $\mathrm{Var}(X) = 1$ and $\mathrm{Var}(Z) = 0$. For a different state with Bloch vector in the xz plane, the commutator expectation still vanishes while both X and Z may have nonzero variance. Thus an expectation-value commutator inequality need not supply a state-independent positive lower bound for two incompatible measurements.

14.10 Lie-algebraic uncertainty and representation parameters

Let Hermitian generators J_x, J_y, J_z satisfy $[J_x, J_y] = iJ_z$ and cyclic permutations, using $\hbar = 1$. In the spin-j irreducible representation,

$$J_x^2 + J_y^2 + J_z^2 = j(j+1)I. \tag{14.11}$$

Taking expectations and subtracting squared means yields an exact identity for total uncertainty:

$$\sum_{k=x,y,z} \mathrm{Var}(J_k) = j(j+1) - \|\langle \boldsymbol{J}\rangle\|_2^2\,. \tag{14.12}$$

Because the mean spin length is at most j, one obtains the state-independent lower bound

$$\sum_k \mathrm{Var}(J_k) \geq j. \tag{14.13}$$

Proof. For any unit vector $\boldsymbol{n}$, $\boldsymbol{n}\cdot\boldsymbol{J}$ has eigenvalues between $-j$ and j, so $\boldsymbol{n}\cdot\langle\boldsymbol{J}\rangle \leq j$. Taking $\boldsymbol{n}$ parallel to the mean-spin vector gives $\|\langle\boldsymbol{J}\rangle\| \leq j$. Substitute into the Casimir identity. □

Example 14.14 (Spin one half)**.** Take $J_k = \sigma_k/2$. Then $\sum_k J_k^2 = 3I/4$. In a pure qubit state with unit Bloch vector, $\sum_k \langle J_k\rangle^2 = 1/4$, giving $\sum_k \mathrm{Var}(J_k) = 1/2$. A maximally mixed qubit has zero mean spin and total variance $3/4$. Both obey the universal lower bound $j = 1/2$.

14.10.1 Highest weights and root operators

Define $J_\pm = J_x \pm iJ_y$. The standard actions are

$$J_\pm \left|j,m\right\rangle = \sqrt{j(j+1) - m(m\pm 1)}\left|j,m\pm 1\right\rangle. \tag{14.14}$$

The highest-weight vector $|j,j\rangle$ is annihilated by J_+, and the allowed weights are $m = -j, -j+1, \ldots, j$. The highest weight specifies the irreducible representation, while the weight m labels a basis vector inside it. For $\mathfrak{su}(2)$ the rank is one, and the Dynkin label of the spin-j representation is $2j$, an integer. These structural relations permit uncertainty statements involving the full family of generators, rather than only a single commutator expectation.

14.11 Entropic uncertainty for two incompatible measurements

Let $\{|x_j\rangle\}$ and $\{|z_k\rangle\}$ be orthonormal bases of a d-dimensional space, representing rank-one projective measurements X and Z. Their squared overlap constant is

$$c = \max_{j,k} \left|\langle x_j|z_k\rangle\right|^2. \tag{14.15}$$

For a state ρ, define outcome distributions $p_j = \langle x_j|\,\rho\,|x_j\rangle$ and $q_k = \langle z_k|\,\rho\,|z_k\rangle$. The Maassen–Uffink relation is

$$H(X)_\rho + H(Z)_\rho \geq -\log c. \tag{14.16}$$

When the bases are mutually unbiased, $c = 1/d$ and the lower bound is $\log d$. For identical bases, $c = 1$ and the resulting lower bound is zero; it cannot certify incompatibility that is absent.

Example 14.15 (Pauli X and Z)**.** Their eigenbases are mutually unbiased, with $c = 1/2$. Thus $H(X)+H(Z) \geq 1$ for every qubit state. At $\rho = |0\rangle\langle 0|$, the Z outcomes are deterministic and the X outcomes equiprobable, so $H(Z) = 0$ and $H(X) = 1$; the inequality is tight. At $\rho = I/2$, both entropies equal one, so the inequality is not tight.

14.11.1 A proof outline and its domain of validity

For a pure state, the vector of amplitudes in the second basis is obtained from the first by a unitary whose matrix entries have maximum modulus $\sqrt{c}$. Interpolation between the $\ell_2 \to \ell_2$ norm one and the $\ell_1 \to \ell_\infty$ norm at most $\sqrt{c}$ yields a family of Rényi-entropic inequalities. Taking the Shannon limit establishes $H(X) + H(Z) \geq -\log c$. For mixed states the bound follows by concavity of measurement entropy combined with pure-state decomposition. The interpolation theorem is a substantive analytic ingredient; the inequality should not be presented as following directly from the Robertson commutator bound.

14.12 Quantum side information in uncertainty games

Suppose A is measured in basis X or Z, while a party retains a quantum memory B. The postmeasurement states are classical–quantum, and their conditional entropies quantify residual uncertainty given optimal access to the memory. For rank-one projective bases the memory-assisted uncertainty relation is

$$H(X|B) + H(Z|B) \geq -\log c + S(A|B). \tag{14.17}$$

Unlike the memory-free inequality, the lower bound depends on the input correlations. If AB is maximally entangled, then $S(A|B) = -\log d$ and the two conditional outcome entropies may both vanish: Bob can predict the outcome of either chosen basis after learning which basis Alice used.

Example 14.16 (Entanglement eliminates chosen-basis uncertainty)**.** Let $|\Phi^+\rangle_{AB} = (|00\rangle + |11\rangle)/\sqrt{2}$. If Alice measures Z, Bob's conditional state is $|0\rangle$ or $|1\rangle$ and is perfectly distinguishable. If she measures X, the conditional states are $|+\rangle$ or $|-\rangle$, again perfectly distinguishable. Thus $H(Z|B) = H(X|B) = 0$. Since $c = 1/2$ and $S(A|B) = -1$, the lower bound is $1 - 1 = 0$, attained exactly.

14.13 Majorization beyond a single entropy function

For probability vectors p, q sorted decreasingly, $p \prec q$ means $\sum_{i=1}^{k} p_i^{\downarrow} \leq \sum_{i=1}^{k} q_i^{\downarrow}$ for all k before the final equality of total sums. If $p \prec q$, then every Schur-concave entropy obeys $H(p) \geq H(q)$. Therefore a majorization statement implies a family of scalar entropic inequalities, whereas a single Shannon-entropy comparison generally does not imply majorization.

Example 14.17 (Incomparable distributions)**.** The vectors $p = (1/2, 1/4, 1/4, 0)$ and $q = (2/5, 2/5, 1/10, 1/10)$ are incomparable in majorization order: $p_1 = 1/2 > 2/5 = q_1$, whereas $p_1 + p_2 = 3/4 < 4/5 = q_1 + q_2$. A scalar entropy value cannot recover which of the two partial-sum inequalities is more restrictive. Such incomparability motivates uncertainty relations formulated as ordered probability-vector constraints.

14.14 Measurement error, disturbance, and dynamical speed

For a measurement apparatus with pointer observable M^{out} and system observable A, an rms noise parameter may be defined by $\varepsilon(A)^2 = \langle (M^{\text{out}} - A^{\text{in}})^2 \rangle$. For a second observable B, define disturbance $\eta(B)^2 = \langle (B^{\text{out}} - B^{\text{in}})^2 \rangle$. Under the standard measurement-model definitions, Ozawa's universally valid inequality is

$$\varepsilon(A)\eta(B) + \varepsilon(A)\Delta B + \Delta A \eta(B) \geq \frac{1}{2} \left| \langle [A, B] \rangle \right| . \tag{14.18}$$

The additional terms distinguish it from the generally invalid unrestricted assertion $\varepsilon(A)\eta(B) \geq |\langle [A, B] \rangle| /2$. Error–disturbance uncertainty concerns an apparatus and its action, not merely the intrinsic fluctuations of a fixed state.

14.14.1 Mandelstam–Tamm orthogonalization time

For a pure state evolving under a time-independent Hamiltonian H, the rate of motion in projective Hilbert space is controlled by energy standard deviation ΔH. The Mandelstam–Tamm inequality for evolution from an initial state to an orthogonal state is

$$t_{\perp} \geq \frac{\pi}{2\Delta H} \qquad (\hbar = 1). \tag{14.19}$$

Example 14.18 (A saturating two-level system)**.** Take $H = \omega Z$ and initial $|+\rangle$. Then $\Delta H = |\omega|$ and

$$\left\langle + \middle| e^{-it\omega Z} \middle| + \right\rangle = \cos(\omega t). \tag{14.20}$$

The first orthogonalization occurs at $t = \pi/(2\,|\omega|)$, exactly saturating the bound.

14.15 From temporal correlations to process tensors

A quantum process with interventions at ordered times cannot always be represented by independent applications of a single quantum channel. A process tensor is a multilinear map accepting a sequence of completely positive intervention maps $\mathcal{A}_0, \ldots, \mathcal{A}_{n-1}$ and returning the unnormalized final state or outcome probability. Its Choi representation is a positive operator with linear causality constraints enforcing that future choices cannot change earlier outcome probabilities.

Definition 14.19 (Operational Markov property)**.** A multitime process is operationally Markovian, relative to the allowed interventions, when the conditional future process depends only on the current system state after an intervention and not on the earlier intervention history. A general environment with memory can create temporal correlations that violate this factorization.

Classical observation of correlations at different times does not by itself identify direct causation: common causes and direct influence can be observationally indistinguishable. Quantum combs and process tensors incorporate interventions that distinguish these cases. Any proposed *causal uncertainty relation* must specify the compatible intervention instruments, the process normalization, and the particular operational optimization. A generic commutator bound between two static observables is not sufficient to characterize uncertainty in an interactive temporal experiment.

14.16 A state-independent uncertainty bound versus a state-dependent one

For $A = X$ and $B = Z$ on a qubit, the Robertson bound is

$$\Delta X \Delta Z \geq |\langle Y \rangle| \,. \tag{14.21}$$

It is state dependent and can vanish. The entropic relation $H(X) + H(Z) \geq 1$ instead holds independently of the input density matrix. Consider $|\psi\rangle = \cos(\theta/2)\,|0\rangle + \sin(\theta/2)\,|1\rangle$ with real coefficients. Here $\langle Y \rangle = 0$ for all θ, so the Robertson right-hand side is always zero, while the measurement distributions are

$$p_Z(+) = \frac{1 + \cos\theta}{2}, \qquad p_X(+) = \frac{1 + \sin\theta}{2}. \tag{14.22}$$

Both outcome entropies are generally positive. At $\theta = \pi/4$, each equals $h_2((1 + 1/\sqrt{2})/2)$, and their sum is about 1.20175 bits, visibly above the entropic lower bound one.

14.17 Three Pauli measurements on a qubit

For any qubit state $\rho = (I + \boldsymbol{r} \cdot \boldsymbol{\sigma})/2$, the entropy of the jth Pauli measurement is

$$H(\sigma_j) = h_2\left(\frac{1 + r_j}{2}\right). \tag{14.23}$$

Because $\|\boldsymbol{r}\|^2 \leq 1$, the three coordinates cannot simultaneously have magnitude one. A simple state-independent three-basis bound is

$$H(X) + H(Y) + H(Z) \geq 2 \text{ bits}. \tag{14.24}$$

It is saturated by any Pauli eigenstate: one measurement is deterministic, the other two are uniform. The inequality is stronger than merely applying the two-basis $H(X) + H(Z) \geq 1$ once. It expresses simultaneous incompatibility among three mutually unbiased qubit measurements.

14.18 An explicit majorization comparison

Let $p = (0.6, 0.2, 0.2)$ and $q = (0.5, 0.4, 0.1)$. The sorted first partial sums satisfy $0.6 > 0.5$, whereas the first two satisfy $0.8 < 0.9$. Hence neither $p \prec q$ nor $q \prec p$. A Shannon-entropy comparison between them still yields a scalar ordering, but it does not imply ordering for all Rényi entropies or all Schur-concave functionals. Majorization uncertainty relations retain more information by constraining the entire sorted outcome vector.

14.19 Temporal correlations in a collision model

Consider a system S that interacts sequentially with environmental ancillas E_1, E_2. If the ancillas are initially independent and never interact again, and each is discarded immediately, the reduced evolution may be described by a composition of CPTP maps. If E_1 later interacts

with E_2 or retains correlations with S that re-enter the dynamics, the process may exhibit memory. Measuring S at an intermediate time alters the future evolution because it changes joint system-environment correlations, even when it leaves the observed local density matrix apparently unchanged. A process-tensor description treats the intervening operation as an input to the multitime process and therefore captures such back-action.

Remark 14.20 (Causal interpretations). Observational two-time correlators alone do not identify a unique causal network. An operational claim of direct influence requires specifying which interventions are possible and which alternative common-cause explanations are excluded. This logical issue is distinct from uncertainty about the outcomes of simultaneous noncommuting observables.

CHAPTER 15

Mathematical Derivations and Case Studies

15.1 Recursive Fourier structure

Take $N = 2^n$, $D_N = \mathrm{diag}(1, \omega_{2N}, \ldots, \omega_{2N}^{N-1})$ and $C_N = |0\rangle\langle 0| \otimes I_N + |1\rangle\langle 1| \otimes D_N$. Define

$$U_N = (I_2 \otimes F_N) C_N (H \otimes I_N) = \frac{1}{\sqrt{2}} \begin{pmatrix} F_N & F_N \\ F_N D_N & -F_N D_N \end{pmatrix}. \tag{15.1}$$

Let Π_N be the output permutation $\Pi_N(|a\rangle \otimes |k\rangle) = |2k + a\rangle$. For input $x = bN + j$ and output $y = 2k + a$,

$$\langle y| \Pi_N U_N |x\rangle = \frac{1}{\sqrt{2N}} (-1)^{ab} \omega_N^{jk} \omega_{2N}^{aj} = \frac{1}{\sqrt{2N}} \omega_{2N}^{xy}. \tag{15.2}$$

The last equality follows by expanding $(bN + j)(2k + a)$ and using $\omega_{2N}^N = -1$. Thus

$$\boxed{F_{2N} = \Pi_N (I_2 \otimes F_N) C_N (H \otimes I_N).} \tag{15.3}$$

This recursion makes the bit-reversal permutation explicit rather than silently changing computational-basis ordering.

15.2 Min-entropy versus average entropy

For a finite distribution p, $p(x) \le p_{\max}$ implies $-\log p(x) \ge -\log p_{\max}$. Averaging shows $H(p) \ge H_{\min}(p)$. A sharp separation occurs for $p_0 = 1/2$ and $p_i = 1/(2N)$ for $1 \le i \le N$:

$$p_{\mathrm{guess}} = \frac{1}{2}, \qquad H_{\min} = 1, \qquad H(p) = 1 + \frac{1}{2} \log N. \tag{15.4}$$

The maximum single-shot guessing probability is fixed while mean Shannon uncertainty diverges logarithmically with alphabet size.

15.3 A qubit Helstrom calculation with unequal priors

Let $\rho_0 = |0\rangle\langle 0|$ and $\rho_1 = |+\rangle\langle +|$. For prior $p \in [0, 1]$ of ρ_0,

$$\Delta = p\rho_0 - (1 - p)\rho_1 = \begin{pmatrix} \frac{3p-1}{2} & -\frac{1-p}{2} \\ -\frac{1-p}{2} & -\frac{1-p}{2} \end{pmatrix}. \tag{15.5}$$

Its trace is $2p-1$ and determinant $-p(1-p)/2$. Hence its eigenvalues are

$$\lambda_{\pm} = \frac{2p-1 \pm \sqrt{p^2+(1-p)^2}}{2}, \tag{15.6}$$

so for $0 < p < 1$, the signs are opposite and

$$P^*_{\text{succ}} = \frac{1+\sqrt{p^2+(1-p)^2}}{2}. \tag{15.7}$$

At $p = 0, 1$, always guessing the certain state is optimal. The formula gives unity at both endpoints.

15.4 Pauli-string and test-circuit identities

The parity network $\text{CNOT}_{1\to3}\text{CNOT}_{2\to3}$ maps $|x_1, x_2, z\rangle \mapsto |x_1, x_2, z \oplus x_1 \oplus x_2\rangle$. Conjugating a single Z rotation on the last wire by this network implements a phase dependent on the parity of three bits. For $P = X \otimes Y \otimes Z$, local basis changes reduce the exponential to this parity circuit by computing the relevant Pauli eigenvalue parity. The Hadamard test and SWAP test yield, respectively, $\Re\,\text{Tr}(U\rho)$ and $\text{Tr}(\rho\sigma)$ without full tomography.

15.5 Channel spectra and thresholds

For the bit-flip Choi state $J(\mathcal{B}_p)$, the nonzero eigenvalues are $1-p, p$, so $\operatorname{rank} J = 2$ for $0 < p < 1$ and one at the endpoints. For the depolarizing Choi state $J(\mathcal{D}_q)$, the eigenvalues are $1 - 3q/4$ and $q/4$ (threefold), giving rank four for $0 < q \leq 1$ and rank one at $q = 0$. The Choi rank equals the minimal Kraus rank; these spectral computations therefore give a lower bound on the environment dimension of a minimal Stinespring representation.

15.6 Pure-state reference robustness of teleportation

Take arbitrary (possibly nonorthogonal, unnormalized) reference vectors $|r_0\rangle, |r_1\rangle$ with $\|r_0\|^2 + \|r_1\|^2 = 1$ and $|\Psi\rangle_{RQ} = |r_0\rangle|0\rangle + |r_1\rangle|1\rangle$. The Bell identity gives

$$|\Psi\rangle_{RQ}\left|\Phi^+\right\rangle_{AB} = \frac{1}{2}\sum_{a,b}|\beta_{ab}\rangle_{QA}\,(I_R \otimes X^b Z^a)\,|\Psi\rangle_{RB}\,. \tag{15.8}$$

The probability of every Bell outcome is $1/4$. The corrected reference–output state equals the original $|\Psi\rangle$, not merely the same reduced state on the teleported qubit. Linearity establishes the statement for any mixed reference–input density operator.

15.7 Single-qubit information benchmark

For an ensemble of signals on a qubit, the accessible classical information is bounded by one bit, regardless of ensemble cardinality. If a classical label is encoded in many nonorthogonal states, the bound remains $I(X:Y) \leq \chi \leq 1$; it is an information bound, not a restriction on the number of distinct quantum states that may be prepared.

15.8 Measurement-based computation on graph states

A graph state $|G\rangle$ on a simple graph $G = (V, E)$ is obtained by preparing $|+\rangle^{\otimes|V|}$ and applying CZ on every edge. It is stabilized by $K_v = X_v \prod_{u\in N(v)} Z_u$. In the one-way computational model, local measurements in equatorial bases $|\pm_\alpha\rangle = (|0\rangle \pm e^{i\alpha}|1\rangle)/\sqrt{2}$ implement logical transformations whose Pauli byproducts are determined by previous measurement outcomes.

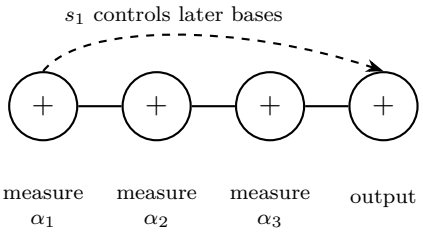


Figure 15.1: Four-site linear cluster in an abstract one-way computation. Entangling edges are CZ links, not physical wires; basis adaptations and output Pauli corrections depend on classical outcomes.

Proposition 15.1 (One-bit teleportation identity)**.** *Let $|\psi\rangle_A = a|0\rangle + b|1\rangle$, initialize B in $|+\rangle$, apply CZ_{AB}, and measure A in $\{|\pm_\alpha\rangle\}$. Conditioned on outcome $s \in \{0, 1\}$, the normalized output is*

$$|\psi_{\text{out}}\rangle = X^s H R_z(-\alpha)|\psi\rangle, \tag{15.9}$$

up to an outcome-independent global phase.

Proof. After CZ, the unnormalized state is $a|0\rangle|+\rangle + b|1\rangle|-\rangle$. Projection onto $\langle\pm_\alpha| = (\langle 0| \pm e^{-i\alpha}\langle 1|)/\sqrt{2}$ yields $(a|+\rangle \pm be^{-i\alpha}|-\rangle)/\sqrt{2}$, equal to $X^s H \operatorname{diag}(1, e^{-i\alpha})|\psi\rangle/\sqrt{2}$. Since $\operatorname{diag}(1, e^{-i\alpha})$ equals $R_z(-\alpha)$ up to phase, the result follows. □

15.9 Other algorithmic frameworks

Variational quantum algorithms optimize a parameterized state $U(\boldsymbol{\theta})|0\rangle$ against an observable objective using classical feedback; for some ansätze and random initialization regimes, gradients exhibit barren plateaus with exponentially small variance. Quantum signal processing and block-encoding methods transform polynomial functions of singular values or eigenvalues under boundedness and access assumptions. Neither framework has a universal speedup independent of input encoding, conditioning, and measurement cost.

15.10 Graph-state algebra and stabilizer identities

Let $G = (V, E)$ be a finite simple undirected graph. Associate a qubit with each vertex, initially in $|+\rangle_v$. Define its graph state by

$$|G\rangle = \prod_{\{u,v\}\in E} CZ_{uv} \, |+\rangle^{\otimes |V|} . \tag{15.10}$$

All the CZ operators commute because they are diagonal in the computational basis, so the product order is irrelevant. For every vertex v, introduce

$$K_v = X_v \prod_{u\in N(v)} Z_u. \tag{15.11}$$

Conjugating the initial stabilizer X_v through each adjacent CZ yields one factor Z_u, because $CZ_{uv} X_v CZ_{uv} = X_v Z_u$. Therefore $K_v \, |G\rangle = |G\rangle$. The generators commute, are independent, and define a one-dimensional common $+1$ eigenspace.

Example 15.2 (A two-vertex graph state)**.** For a single edge $1 - 2$,

$$|G\rangle = CZ \, |{+}{+}\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle). \tag{15.12}$$

The stabilizers are $K_1 = X_1 Z_2$ and $K_2 = Z_1 X_2$. Note that this vector is not the standard Bell state in the computational basis, but applying H to the second qubit transforms it into $|\Phi^+\rangle$.

15.11 Single-site measurement as a gate-teleportation primitive

Take an arbitrary input $|\psi\rangle = a \, |0\rangle + b \, |1\rangle$ on qubit A, prepare $|+\rangle$ on qubit B, and apply CZ_{AB}:

$$|\Psi\rangle_{AB} = a \, |0\rangle_A \, |+\rangle_B + b \, |1\rangle_A \, |-\rangle_B . \tag{15.13}$$

Measure A in the equatorial basis

$$|\pm_\alpha\rangle = \frac{|0\rangle \pm e^{i\alpha} \, |1\rangle}{\sqrt{2}}, \tag{15.14}$$

with outcome bit s (zero for plus, one for minus). Projection gives the unnormalized state

$$\langle (-1)^s_\alpha | \, \Psi\rangle = \frac{1}{\sqrt{2}} \left(a \, |+\rangle + (-1)^s b e^{-i\alpha} \, |-\rangle \right). \tag{15.15}$$

Since $H \, |0\rangle = |+\rangle$, $H \, |1\rangle = |-\rangle$, and $X \, |\pm\rangle = \pm \, |\pm\rangle$, the normalized state on B is

$$\boxed{|\psi_{\text{out}}\rangle = X^s H R_z(-\alpha) \, |\psi\rangle} \tag{15.16}$$

up to a physically irrelevant global phase. The measurement outcome probabilities are both $1/2$, independently of the input amplitudes. This is a one-bit teleportation step implementing a unitary followed by a known Pauli byproduct.

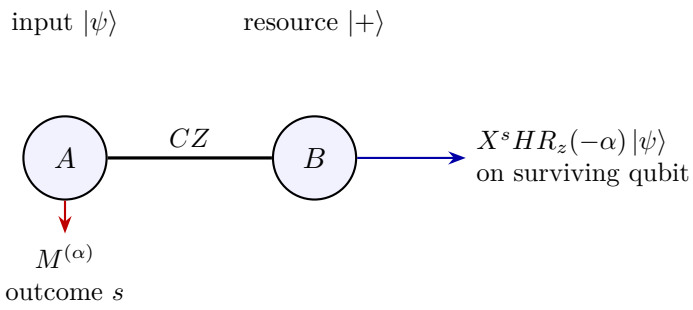


Figure 15.2: One-bit teleportation: the input is measured and the logical state propagates to the adjacent graph vertex. The Pauli byproduct is classically recorded.

15.11.1 Two sequential measurements and adaptive angles

Consider three qubits in a linear cluster. Teleport the input from vertex one to two using angle α and outcome s_1, then from two to three using angle β and outcome s_2. Ignoring global phases, the logical operator is

$$X^{s_2} H R_z(-\beta) X^{s_1} H R_z(-\alpha). \tag{15.17}$$

Use the identities $R_z(\theta)X = XR_z(-\theta)$ and $HX = ZH$ to commute the byproduct through the next operation. One finds

$$X^{s_2} H R_z(-\beta) X^{s_1} = X^{s_2} Z^{s_1} H R_z(-(-1)^{s_1}\beta). \tag{15.18}$$

Therefore, if the intended second logical rotation uses angle β_0, select the physical measurement angle $\beta = (-1)^{s_1}\beta_0$ (for the stated angle convention). The logical gates become independent of s_1 while a final known Pauli $X^{s_2}Z^{s_1}$ remains.

Example 15.3 (A concrete correction sequence)**.** Take $\alpha = 0$, $\beta_0 = \pi/4$, first outcome $s_1 = 1$, second outcome $s_2 = 0$. The second measurement is performed at angle $\beta = -\pi/4$. The output contains byproduct Z and the intended nontrivial rotation; applying Z (or recording it in a Pauli frame) restores the target logical state. If the physical second angle were kept fixed at $+\pi/4$, the implemented rotation would have the opposite sign in that branch.

15.12 Measurement calculus and the Pauli frame

A one-way pattern consists of graph-state entangling operations, local measurements, classical processing of outcomes, and final corrections. Measurement dependence is not optional bookkeeping: the choice of measurement basis may depend on earlier outcomes in order to make the logical gate independent of those random results. It is frequently cheaper to update a *Pauli frame* classically rather than physically applying every byproduct immediately. The frame stores exponents $(x, z) \in \{0, 1\}^2$ representing a pending correction $X^x Z^z$.

15.12.1 Pauli propagation rules

Under Hadamard, $HXH = Z$ and $HZH = X$, so frame bits exchange. Under a phase gate $S = \mathrm{diag}(1, i)$, $SXS^\dagger = iXZ$ (global phase irrelevant) and $SZS^\dagger = Z$, so the X frame bit

also contributes to the Z bit. Under CNOT, an X on control propagates to target and a Z on target propagates to control. These updates allow Clifford segments of a measurement-based computation to proceed without active quantum corrections.

Example 15.4 (Frame propagation through a Hadamard)**.** Suppose an intermediate state is $X^x Z^z |\phi\rangle$. Apply the intended logical H. Then

$$HX^x Z^z |\phi\rangle = Z^x X^z H |\phi\rangle = \pm X^z Z^x H |\phi\rangle . \tag{15.19}$$

Thus the frame update is $(x, z) \mapsto (z, x)$. The possible sign is a global phase and does not affect output probabilities.

15.13 A fully specified one-way single-qubit rotation

A generic single-qubit unitary up to global phase can be written as $R_z(\gamma)R_x(\beta)R_z(\alpha)$. Using $HR_z(\theta)H = R_x(\theta)$, one can realize such gates as products of alternating H and R_z rotations. A line of sufficiently many graph-state vertices, together with adaptive equatorial measurements, implements these products through repeated one-bit teleportation. The graph itself specifies the entangling resource; the computation is specified by the ordered measurement bases and the classical dependence rules. In particular, an unlabeled chain of graph vertices is not yet a complete executable measurement-based algorithm.

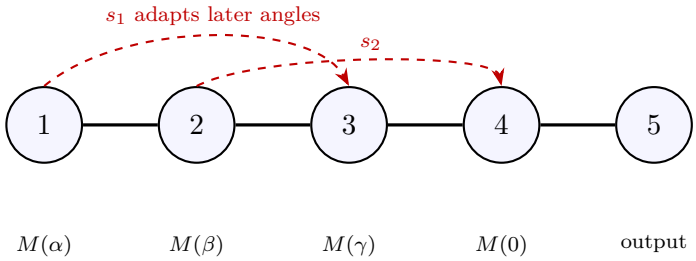


Figure 15.3: Linear graph resource with ordered equatorial measurements and classical feed-forward. The output correction is determined by the observed measurement bits.

15.14 A distinction between graph-state resources and general cluster resources

Every finite graph state is a stabilizer state, but a graph state by itself is not a universal quantum computer. Universality arises from the allowed graph family, non-Pauli measurement bases, adaptive classical feed-forward, and an output convention. Measuring only Pauli observables on graph states remains within the stabilizer formalism and is classically simulable under standard assumptions. Introducing suitably chosen non-Clifford measurement angles enables transformations outside the Clifford group.

Remark 15.5 (Normalization and byproduct convention)*.* A formula involving $HR_z(\alpha)$ versus $HR_z(-\alpha)$ depends on whether measurement basis kets use $e^{+i\alpha}$ or $e^{-i\alpha}$. The present text fixes $|\pm_\alpha\rangle = (|0\rangle \pm e^{i\alpha} |1\rangle)/\sqrt{2}$, leading to $HR_z(-\alpha)$ in the one-bit identity. Changing the basis convention changes corresponding signs; it does not alter the physical operation when all feed-forward rules are changed consistently.

15.15 Projective Pauli measurements as graph transformations

In a graph-state computation, a Pauli-Z measurement on vertex v removes that vertex from the remaining graph, up to known local Pauli byproducts on its neighbors. This can be seen directly from the preparation circuit: measuring Z_v fixes the eigenvalue $(-1)^s$ of qubit v, and each incident CZ_{uv} acts as Z_u^s on the neighbor. Removing the measured vertex therefore leaves the graph state of the induced subgraph multiplied by known local Z operators.

Example 15.6 (Three-node chain with a middle Z measurement)**.** Let the graph be the path $1-2-3$, so the resource is $CZ_{12}CZ_{23}\,|+++\rangle$. Measure vertex two in the Z basis with outcome s. If $s=0$, both incident controlled phases act trivially on the remaining qubits and the state becomes $|+\rangle_1\,|+\rangle_3$. If $s=1$, each neighbor receives Z, leaving $|-\rangle_1\,|-\rangle_3$. Up to normalization, the remaining state is

$$(Z_1Z_3)^s\,|+\rangle_1\,|+\rangle_3\,. \tag{15.20}$$

In either branch the surviving endpoint qubits are not entangled. Measuring a different basis can instead induce entanglement swapping or a logical teleportation operation.

15.16 Preparing the graph resource is not performing the computation

The graph-state preparation gates commute, but equatorial measurements generally cannot be freely reordered when later bases depend on earlier outcomes. A *measurement pattern* must specify (i) which vertices are inputs, (ii) which are outputs, (iii) the graph edges, (iv) a partial order of measurements, (v) outcome-dependent measurement angles, and (vi) final Pauli-frame corrections. These data determine whether the pattern realizes a deterministic logical unitary or a merely probabilistic branch operation.

15.16.1 Pauli corrections versus Clifford corrections

The one-bit teleportation step produces $X^sHR_z(-\alpha)$. When α is a Clifford angle, such as an integer multiple of $\pi/2$, the logical gate is Clifford and Pauli byproducts remain Pauli under conjugation. For non-Clifford angles such as $\pi/4$, commuting an unknown X correction through $R_z(-\alpha)$ changes the sign of the rotation. Hence measurement angles must be adapted or non-Pauli corrections introduced. The necessity of feed-forward is mathematical rather than an engineering convention.

15.17 A calculation of output distribution under missing feed-forward

Suppose a two-step pattern is intended to implement a fixed $HR_z(-\beta_0)HR_z(-\alpha)$. The first measurement outcome s_1 is uniform. If the second physical angle is always set equal to β_0, rather than $(-1)^{s_1}\beta_0$, the effective second rotation is $R_z(-\beta_0)$ in one branch and $R_z(+\beta_0)$ in the other, after Pauli-frame normalizations. The two branches generally produce

different pure outputs from the same input. Averaging over the unknown bit s_1 then yields a mixed output channel, and no single final Pauli correction can generally restore the intended unitary. For $\beta_0 = 0$ or a special Clifford symmetry the difference may disappear, but generic non-Clifford rotations need adaptation.

15.18 Classical control and the universality mechanism

A complete measurement-based scheme combines a suitable entangled graph resource with single-qubit measurements and classical processing. At the logical level, graph entanglement permits nonlocal gate teleportation, while arbitrary equatorial measurements provide continuous one-qubit rotations. The output need not be physically corrected after every step: maintaining a Pauli frame is sufficient when future measurement bases are updated consistently. This architecture separates the entangling resource preparation from the adaptive measurement process, while preserving exactly the same abstract unitary computational model.

CHAPTER 16

One-Shot Entropies and Information-Theoretic Security

16.1 Operational security quantities

Definition 16.1 (Classical–quantum information source)**.** A finite cq source is a state

$$\rho_{XE} = \sum_{x \in \mathcal{X}} p_x \, |x\rangle\langle x|_X \otimes \rho_E^x, \tag{16.1}$$

where the register X is classical, the states ρ_E^x may be noncommuting, and an adversary holds E. A measurement strategy is a POVM $\{M_x\}_{x \in \mathcal{X}}$, and its average guessing success probability is $\sum_x p_x \operatorname{Tr}(M_x \rho_E^x)$.

The optimal probability and min-entropy obey

$$p_{\text{guess}}(X|E) = \max_{\{M_x\}} \sum_x p_x \operatorname{Tr}(M_x \rho_E^x), \qquad H_{\min}(X|E) = -\log p_{\text{guess}}(X|E). \tag{16.2}$$

The logarithm is base two, so $H_{\min}$ is measured in bits. This single-guess task differs from the asymptotic compression task governed by Shannon entropy. A distribution may have high Shannon entropy and yet contain a highly likely individual outcome, leading to low min-entropy and a weak one-shot secrecy guarantee.

Example 16.2 (An adversary learns one of two nonorthogonal states)**.** Let $X \in \{0, 1\}$ be uniform, with Eve's states $\rho_E^0 = |0\rangle\langle 0|$ and $\rho_E^1 = |+\rangle\langle +|$. The Helstrom bound gives

$$p_{\text{guess}}(X|E) = \frac{1}{2}\left(1 + \frac{1}{\sqrt{2}}\right) \approx 0.85355, \tag{16.3}$$

so $H_{\min}(X|E) \approx 0.22845$ bits. Although $H(X) = 1$ bit, quantum side information sharply reduces uncertainty relevant to Eve's optimal prediction.

16.2 Conditional min-entropy as a semidefinite program

Theorem 16.3 (Dual characterization of cq guessing)**.** *For a cq state, the optimal guessing probability equals*

$$\min_{Y_E \succeq 0} \{\operatorname{Tr} Y_E : \; Y_E \succeq p_x \rho_E^x \text{ for all } x\}. \tag{16.4}$$

Proof. The primal optimization is

$$\max\left\{\sum_x p_x \operatorname{Tr}(M_x \rho_E^x) : M_x \succeq 0, \; \sum_x M_x = I_E\right\}. \tag{16.5}$$

Introduce the Hermitian Lagrange multiplier Y_E for $\sum_x M_x = I$. The Lagrangian becomes

$$\operatorname{Tr} Y_E + \sum_x \operatorname{Tr}[M_x(p_x\rho_E^x - Y_E)]. \tag{16.6}$$

For the supremum over $M_x \succeq 0$ to be finite, one needs $Y_E \succeq p_x\rho_E^x$ for every x. Under this condition the supremum equals $\operatorname{Tr} Y_E$. Finite-dimensional strong duality holds by strict feasibility, for example with a sufficiently large positive multiple of the identity as a dual feasible point. □

16.2.1 The general quantum conditional min-entropy

For a bipartite density operator ρ_{AE}, define

$$H_{\min}(A|E)_\rho = -\log\inf\{\lambda :\ \rho_{AE} \preceq \lambda I_A \otimes \sigma_E,\ \sigma_E \in \mathcal{D}(E)\}. \tag{16.7}$$

For cq states this reduces to the guessing-probability expression. It is possible for the quantum conditional min-entropy to be negative when A and E are entangled. For a maximally entangled state of local dimension d, one has $H_{\min}(A|E) = -\log d$ under the standard normalization.

16.3 Smoothing and finite-error operational statements

A smoothed min-entropy optimizes $H_{\min}$ over states within a specified distance ε of the actual state, typically purified distance. The smoothing definition must specify whether subnormalized states are allowed; modern one-shot entropy formulations commonly allow them. The parameter ε represents an explicit security or approximation budget and cannot be omitted when comparing different finite-block formulas.

Example 16.4 (Why smoothing matters)**.** Consider a classical distribution with a rare outcome that carries a disproportionate amount of probability relative to a desired flat support. Discarding an event of sufficiently small probability can increase the worst-case uncertainty of the remaining distribution. A smooth min-entropy can formalize this relaxation, but the increase is paid for by a nonzero distance ε from the original state.

16.4 Privacy amplification from high min-entropy

Let X be a raw key and E adversarial quantum side information. Let F be a uniformly chosen two-universal hash function from $\mathcal{X}$ to $\{0,1\}^\ell$, with the seed independent of XE. The extracted key is $K = F(X)$. A standard quantum leftover-hash bound, in one common trace-distance convention, is

$$\frac{1}{2}\left\|\rho_{KFE} - \tau_K \otimes \rho_{FE}\right\|_1 \le \frac{1}{2}\sqrt{2^{\ell - H_{\min}(X|E)}}. \tag{16.8}$$

Here $\tau_K = I_K/2^\ell$ is the ideal uniform key. The inequality exhibits the central tradeoff: extracting more bits increases the deviation from ideal uniformity, while larger conditional min-entropy reduces it.

Example 16.5 (A numerical privacy-amplification budget)**.** Suppose the certified min-

entropy is at least $m = 1000$ bits and one wishes to extract $\ell = 800$ bits. The unsmoothed leftover-hash upper bound gives

$$\frac{1}{2}\left\|\rho_{KFE} - \tau_K \otimes \rho_{FE}\right\|_1 \leq \frac{1}{2}\sqrt{2^{-200}} = 2^{-101}. \tag{16.9}$$

This estimate ignores error-correction leakage and any smoothing correction; a full protocol must subtract or bound those additional terms before fixing an output-key length.

16.5 Entropic uncertainty as a secrecy certificate

Consider a measurement in one of two incompatible orthonormal bases X and Z on a system A. If an adversary holds quantum memory E, an entropic uncertainty relation constrains the simultaneous predictability of outcomes from these bases. In an entanglement-based key-distribution analysis, a complementary-basis error estimate bounds the uncertainty of hypothetical outcomes, which can then bound an adversary's information about actual key-basis outcomes. For mutually unbiased qubit bases, the overlap constant is $c = 1/2$, giving a one-bit state-independent uncertainty scale. For the same quantum conditioning register E, the memory-assisted uncertainty relation states

$$H(X|E) + H(Z|E) \geq 1 + S(A|E) \tag{16.10}$$

under the corresponding tripartite entropic-uncertainty formulation and the specified state partition. Different tripartite variants have different right-hand sides; the conditioning systems must be stated and cannot be swapped without an accompanying theorem.

16.6 A classical binary reconciliation calculation

Suppose Alice and Bob hold correlated unbiased bits X, Y connected by a binary symmetric channel of error probability q. Their conditional Shannon entropy is

$$H(X|Y) = h_2(q). \tag{16.11}$$

Asymptotically, Slepian–Wolf-type coding permits Alice to reveal a syndrome approaching $nh_2(q)$ bits for a length-n block while Bob reconstructs her sequence using his correlated observation. This public leakage must be accounted for in a secrecy calculation; it is not free classical communication.

Example 16.6 (Reconciliation at a five-percent error rate)**.** At $q = 0.05$, $h_2(q) \approx 0.28640$ bits per raw bit. A block of 10^5 raw bits therefore has an ideal asymptotic conditional-information requirement of about 2.864×10^4 bits of reconciliation information. Practical finite-length codes typically require more leakage than this Shannon limit.

16.7 Classical, quantum, and composable security criteria

A correctness parameter quantifies the probability that Alice's and Bob's final keys disagree. A secrecy parameter quantifies the trace distance of the actual key-and-adversary state from an ideal uniform independent key. A composable security statement accounts for both and

includes all public transcripts, protocol aborts, and the authenticated-channel model. The per-signal mutual-information advantage alone is not a complete finite-size security proof.

Remark 16.7 (Postprocessing order). Parameter estimation, error correction, verification, and privacy amplification serve distinct purposes. Estimating an error rate constrains the admissible channel or adversarial state; reconciliation aligns Alice's and Bob's raw strings; verification controls residual disagreement; privacy amplification removes adversarial information. The length of the final key must be based on the entropy remaining after publicly revealed information is included.

CHAPTER 17

Gaussian Optical Protocols and Their Calculations

17.1 Balanced detection as an operator identity

Let a denote the optical signal mode and b the local-oscillator mode. A balanced beam splitter acts as

$$c = \frac{a+b}{\sqrt{2}}, \qquad d = \frac{a-b}{\sqrt{2}}. \tag{17.1}$$

The photon-number difference measured by two ideal photodiodes is

$$n_c - n_d = c^\dagger c - d^\dagger d \tag{17.2}$$

$$= a^\dagger b + b^\dagger a. \tag{17.3}$$

If the local oscillator is strong and approximately coherent, replace b by its c-number amplitude $\beta = |\beta|\, e^{i\theta}$ to leading order. Then the difference current is proportional to

$$a^\dagger \beta + \overline{\beta} a = |\beta| \left(a e^{-i\theta} + a^\dagger e^{i\theta} \right) = |\beta|\; x_\theta. \tag{17.4}$$

The phase of the local oscillator selects the measured quadrature. At $\theta = 0$ the signal is $x = a + a^\dagger$; at $\theta = \pi/2$ the signal is $p = (a - a^\dagger)/i$ under the adopted convention. Finite detector efficiency and electronic noise require additional modeling, but the operator identity explains ideal balanced homodyne detection.

Example 17.1 (Homodyne mean and variance)**.** For a coherent state $|\alpha\rangle$ with $\alpha = 1 + i$, the means are $\langle x \rangle = 2$, $\langle p \rangle = 2$ and the ideal variances are both one. Rotating the local oscillator phase by $\pi/4$ gives

$$\langle x_{\pi/4} \rangle = 2\cos(\pi/4) + 2\sin(\pi/4) = 2\sqrt{2}, \tag{17.5}$$

while $\mathrm{Var}(x_{\pi/4}) = 1$. The readout phase changes the mean signal but not coherent-state vacuum noise.

17.2 Heterodyne detection and unavoidable added noise

Ideal heterodyne detection combines the signal with an ancillary vacuum on a balanced beam splitter, then homodynes complementary quadratures of the two output modes. With unit-gain rescaling, the jointly reported results can be modeled as

$$x_{\mathrm{het}} = x_{\mathrm{sig}} + x_{\mathrm{vac}}, \qquad p_{\mathrm{het}} = p_{\mathrm{sig}} - p_{\mathrm{vac}}, \tag{17.6}$$

where the two vacuum quadrature contributions each have variance one. The reported marginal variances are therefore

$$\mathrm{Var}(x_{\mathrm{het}}) = V_{xx} + 1, \qquad \mathrm{Var}(p_{\mathrm{het}}) = V_{pp} + 1 \tag{17.7}$$

for independent signal and ancillary vacuum. This extra noise is the operational price of measuring both noncommuting quadratures on the same optical signal through a joint POVM, rather than a single sharp quadrature per copy.

Example 17.2 (Coherent-state heterodyne outcome distribution)**.** For $V = I_2$ the two unit-gain heterodyne outcomes have marginal variances two. Their means remain $(2\Re\alpha, 2\Im\alpha)$. Consequently a coherent-state displacement can be estimated in both components, but the single-copy noise is larger than in an ideal homodyne measurement of just one chosen component.

17.3 Gaussian optics in the Heisenberg and Schrödinger pictures

A Gaussian unitary acts by an affine symplectic transformation on quadratures, while its Schrödinger-picture action maps Gaussian states to Gaussian states. A phase shift $U(\theta) = e^{-i\theta a^\dagger a}$ transforms annihilation operators by $U^\dagger a U = e^{-i\theta} a$, giving

$$\begin{pmatrix} x' \\ p' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ p \end{pmatrix} \tag{17.8}$$

for the corresponding chosen quadrature convention. A single-mode squeezing operation can implement $x' = e^{-r}x$, $p' = e^{r}p$. The determinant of its phase-space transformation is one, and it preserves $[x', p'] = 2i$. A displacement operator $D(\alpha) = e^{\alpha a^\dagger - \overline{\alpha} a}$ maps $a \mapsto a + \alpha$ in the appropriate conjugation convention and changes the quadrature means without changing V.

17.4 Covariance accounting through a lossy interferometer

Consider a signal mode with diagonal covariance $V_{\mathrm{in}} = \mathrm{diag}(v_x, v_p)$ that enters a beam splitter of transmissivity η with an independent vacuum environment. The signal output is

$$V_{\mathrm{out}} = \eta V_{\mathrm{in}} + (1-\eta) I_2 = \mathrm{diag}(\eta v_x + 1 - \eta, \eta v_p + 1 - \eta). \tag{17.9}$$

If $v_x < 1$ is squeezed, the output remains squeezed only when $\eta v_x + 1 - \eta < 1$, equivalently $v_x < 1$ and $\eta > 0$; however the achievable squeezing level is weaker. The output determinant exceeds one whenever loss entangles the signal with the environment and the input is nontrivially squeezed, so the reduced output becomes mixed.

Example 17.3 (Six-decibel squeezing with fifty-percent transmission)**.** Take an ideal input variance $v_x = 10^{-0.6} \approx 0.2512$, $v_p = 1/v_x \approx 3.9811$, and $\eta = 1/2$. Then

$$v_x' \approx 0.6256, \qquad v_p' \approx 2.4906. \tag{17.10}$$

The surviving squeezing is $-10\log_{10}(0.6256) \approx 2.04$ dB below vacuum, far weaker than the input six dB. The result demonstrates quantitatively that optical loss mixes unsqueezed environment fluctuations into the signal.

17.5 Two-mode squeezed resources and EPR correlations

Let A, B be modes of an ideal two-mode squeezed vacuum with squeezing parameter $r \geq 0$. Its covariance matrix is

$$V_{AB} = \begin{pmatrix} \cosh(2r) I_2 & \sinh(2r) Z \\ \sinh(2r) Z & \cosh(2r) I_2 \end{pmatrix}, \quad Z = \mathrm{diag}(1, -1). \tag{17.11}$$

The quadrature differences satisfy

$$\mathrm{Var}(x_B - x_A) = 2\cosh(2r) - 2\sinh(2r) = 2e^{-2r}, \tag{17.12}$$

$$\mathrm{Var}(p_B + p_A) = 2\cosh(2r) - 2\sinh(2r) = 2e^{-2r}. \tag{17.13}$$

The two EPR observables commute:

$$[x_B - x_A, p_B + p_A] = [x_B, p_B] - [x_A, p_A] = 2i - 2i = 0. \tag{17.14}$$

They may therefore be sharply correlated simultaneously in the ideal infinite-squeezing limit without violating a canonical commutation relation. Physical states at finite energy have finite r and strictly positive residual noise.

17.6 Continuous-variable teleportation as an additive-noise channel

Let Alice hold unknown input mode in and her half A of an EPR resource, while Bob holds mode B. Alice mixes in and A on a balanced beam splitter and measures commuting combinations

$$m_x = \frac{x_{\mathrm{in}} - x_A}{\sqrt{2}}, \qquad m_p = \frac{p_{\mathrm{in}} + p_A}{\sqrt{2}}. \tag{17.15}$$

She transmits the real outcomes (m_x, m_p) over a classical channel. With unit-gain feedback displacements $x_B \mapsto x_B + \sqrt{2} m_x$ and $p_B \mapsto p_B + \sqrt{2} m_p$, Bob's output quadratures are

$$\boxed{x_{\mathrm{out}} = x_{\mathrm{in}} + (x_B - x_A), \qquad p_{\mathrm{out}} = p_{\mathrm{in}} + (p_B + p_A).} \tag{17.16}$$

For an independent input and ideal two-mode squeezed resource, the additive noise components have zero means and variances $2e^{-2r}$ in each quadrature, with zero mutual covariance. The output covariance is

$$V_{\mathrm{out}} = V_{\mathrm{in}} + 2e^{-2r} I_2. \tag{17.17}$$

Perfect transfer occurs only in the formal $r \to \infty$ limit. For any finite squeezing, the teleported state experiences a noisy Gaussian channel rather than an exact identity channel.

Example 17.4 (Teleporting a coherent state with finite squeezing)**.** For an input coherent state $V_{\mathrm{in}} = I_2$ and $r = \frac{1}{2}\log 2$, one has $e^{-2r} = 1/2$. The output covariance becomes

$V_{\text{out}} = 2I_2$ while the mean displacement is preserved at unity gain. Thus the output is a displaced thermal Gaussian state with one vacuum unit of additional noise per quadrature, not a pure coherent state.

17.7 Gaussian-modulated coherent-state signal model

Let Alice sample a real modulation variable $X_A \sim \mathcal{N}(0, V_A)$ and encode it in a coherent-state quadrature mean. For an ideal homodyne receiver and a Gaussian channel with transmissivity η, an input-referred excess-noise convention is

$$X_B = \sqrt{\eta} X_A + Z, \qquad \mathrm{Var}(Z) = 1 + \eta\xi, \tag{17.18}$$

where Z is independent centered Gaussian noise and $\xi \geq 0$ is excess noise in shot-noise units. The observable covariance and total output variance are

$$\mathrm{Cov}(X_A, X_B) = \sqrt{\eta} V_A, \qquad \mathrm{Var}(X_B) = \eta V_A + 1 + \eta\xi. \tag{17.19}$$

If V_A is calibrated and nonzero, the covariance identifies $\sqrt{\eta}$ under this model, and the residual variance identifies the total excess noise. The Gaussian classical mutual information is

$$I(X_A : X_B) = \frac{1}{2} \log \left(1 + \frac{\eta V_A}{1 + \eta\xi} \right). \tag{17.20}$$

This quantity accounts for ideal homodyne observations of one real-valued signal; it is not automatically a secret-key rate.

Example 17.5 (Numerical modulation and noise estimation)**.** Suppose $V_A = 4$, $\eta = 1/4$, and $\xi = 0.2$. The modeled output is $X_B = 0.5X_A + Z$, with $\mathrm{Var}(Z) = 1.05$. Then

$$\mathrm{Cov}(X_A, X_B) = 2, \qquad \mathrm{Var}(X_B) = 2.05. \tag{17.21}$$

The signal-to-noise ratio is $\eta V_A/(1 + \eta\xi) = 1/1.05 \approx 0.95238$ and $I(X_A : X_B) \approx 0.4824$ bits per homodyne sample. An actual key rate also subtracts an adversary information term and includes reconciliation inefficiency and statistical uncertainty.

17.8 An optical experiment as a complete information-theoretic model

A security analysis must specify how the channel and detection models are estimated from finite data. The shot-noise calibration establishes the unit scale for quadrature variance. Saturation or nonlinearity of homodyne electronics violates an assumed Gaussian linear model, and uncharacterized leakage can carry information to an unintended environment. A public discussion channel must be authenticated because otherwise an adversary may impersonate an endpoint. These requirements are separate from the formal validity of $V + i\Omega \succeq 0$ or the calculation of Gaussian mutual information, but they determine whether those models actually apply to a physical communication device.

CHAPTER 18

Multitime Quantum Processes and Causal Constraints

18.1 Temporal experiments as multilinear maps

Consider a system S that evolves together with an environment E. At time t_0 an experimenter may apply a completely positive map $\mathcal{A}_0$ to S, at time t_1 another operation $\mathcal{A}_1$, and then measure an outcome at t_2. The probability of the outcome depends not only on the state of S at the beginning but on the entire sequence of interventions. A process tensor is the multilinear functional

$$\mathcal{T}_{2:0}(\mathcal{A}_1, \mathcal{A}_0) = \rho_{S_2}^{\text{out}} \tag{18.1}$$

that returns the generally unnormalized final operator. If each intervention is trace preserving, the output is normalized; if an intervention is a trace-nonincreasing branch of an instrument, the trace of the output gives its occurrence probability.

18.1.1 Derivation from unitary system-environment dynamics

Let $\rho_{S_0E_0}$ be a joint initial state and let $\mathcal{U}_{1:0}$,$\mathcal{U}_{2:1}$ denote joint unitary conjugations. The final state conditioned on interventions is

$$\rho_{S_2} = \mathrm{Tr}_E[\mathcal{U}_{2:1} \circ (\mathcal{A}_1 \otimes \mathrm{id}_E) \circ \mathcal{U}_{1:0} \circ (\mathcal{A}_0 \otimes \mathrm{id}_E)(\rho_{S_0E_0})] \,. \tag{18.2}$$

The dependence on $\mathcal{A}_0$ and $\mathcal{A}_1$ is linear in each argument. Crucially, $\mathcal{A}_0$ may change correlations between S and E even when a later reduced system density matrix is unchanged. Therefore knowing a single reduced state need not suffice to predict the outcome of later interventions.

18.2 A two-qubit model that remembers an earlier state

Let the system and environment be qubits, with environment initialized in $|0\rangle_E$. Take the first interaction to be $U_1 = \mathrm{CNOT}_{S\to E}$. If the system starts in $|+\rangle_S$, the joint state becomes $|\Phi^+\rangle_{SE}$, so the reduced system state is $I/2$. If the system instead starts in $|-\rangle_S$, the joint state becomes $|\Phi^-\rangle_{SE}$, and the reduced system is again $I/2$. Thus the local state at the intermediate time is identical for two distinct input histories. Now apply a second interaction $U_2 = \mathrm{CNOT}_{S\to E}$. Because $U_2U_1 = I$ on the joint space, the final joint state is $|\pm\rangle_S |0\rangle_E$, respectively. The final reduced states are $|+\rangle\langle+|$ and $|-\rangle\langle-|$, even though the intermediate reduced state was $I/2$ in both cases.

Proposition 18.1 (Failure of a history-independent intermediate channel)**.** *There is no single linear map $\mathcal{N}$ acting on the intermediate reduced state ρ_{S_1} alone that reproduces the final states of both experiments just described.*

Proof. Both experiments have exactly the same intermediate input $\rho_{S_1} = I/2$. A well-defined

linear map must send that operator to a unique output. Yet the actual final outputs are distinct orthogonal pure states. Therefore no such map exists. The difference is stored in $S : E$ correlations at the intermediate time. □

This is a concrete non-Markovian memory effect. It does not prove that every arbitrary two-time reduced evolution is non-CP; rather, it shows that one fixed channel from the intermediate local state cannot describe all relevant histories without access to their correlations.

18.3 Choi operators of sequential networks

Let a quantum channel $\mathcal{N} : A \to B$ have unnormalized Choi operator

$$\widetilde{J}_{AB}(\mathcal{N}) = \sum_{i,j} |i\rangle\langle j|_A \otimes \mathcal{N}(|i\rangle\langle j|). \tag{18.3}$$

It is positive, and trace preservation is equivalent to $\mathrm{Tr}_B \widetilde{J} = I_A$. A two-step deterministic quantum comb can similarly be represented by a positive operator W on an ordered collection of input and output spaces, satisfying nested trace constraints. In one conventional ordering A_0, B_0, A_1, B_1, the conditions are

$$W \succeq 0, \qquad \mathrm{Tr}_{B_1} W = I_{A_1} \otimes W^{(1)}, \qquad \mathrm{Tr}_{B_0} W^{(1)} = I_{A_0}. \tag{18.4}$$

The precise tensor ordering and transpositions in contraction formulas depend on the chosen Choi convention, but positivity and nested no-future-signalling normalization are invariant physical requirements. The causal trace constraints rule out a circuit in which the later choice of an intervention changes the marginal probability of an earlier registered outcome.

18.4 Passive observations and interventions

Observing a system at two times can reveal a temporal correlation, but it need not reveal the direction or source of causal influence. In a classical model, correlated outputs A and B may arise because A directly influences B or because a hidden variable C influences both. To test a causal hypothesis, an experimenter changes A under an intervention and observes the conditional response of B, with suitable assumptions about the other causal connections. In quantum mechanics, a measurement intervention may itself disturb the future dynamics. The operation is not generally described by a scalar random variable alone; an instrument branch is a completely positive map that specifies both the outcome probability and the output state. Consequently a causal model of a quantum process must account for the choice of instrument, not merely the marginal distributions of passive measurement records.

Example 18.2 (Two instruments with identical effects)**.** Consider the trivial one-outcome POVM $\{I\}$ on a qubit. One instrument may implement the identity map $\mathcal{I}_{\mathrm{id}}(\rho) = \rho$, while another implements complete dephasing $\mathcal{I}_{\mathrm{deph}}(\rho) = (\rho + Z\rho Z)/2$. Both produce the only outcome with probability one for every input, so their POVM effects are identical. Yet they generally lead to different later statistics: for input $|+\rangle$, a subsequent X measurement is deterministic after $\mathcal{I}_{\mathrm{id}}$ and unbiased after $\mathcal{I}_{\mathrm{deph}}$. Therefore a process cannot be fully characterized by POVM effects without state-update information.

18.5 A finite-memory state model

A finite Markovian model may enlarge the accessible system to include a memory register M and apply fixed CPTP transitions

$$\rho_{S_{j+1}M_{j+1}} = \mathcal{V}_j\left[(\mathcal{A}_j \otimes \mathrm{id}_M)(\rho_{S_jM_j})\right]. \tag{18.5}$$

If one traces out M after every step, the effective process seen on S can be non-Markovian. The memory dimension determines how much information about earlier interventions can persist. This distinction between Markovianity on an enlarged system and non-Markovianity on a reduced subsystem is central to the analysis of open quantum systems and temporally correlated noise.

18.6 Uncertainty in adaptive quantum experiments

Suppose an agent chooses between two instruments $\mathcal{A}$ and $\mathcal{B}$ and wishes to infer a hidden process parameter from the resulting multitime records. A rigorous uncertainty relation must state the allowed quantum comb, the set of admissible interventions, the decision or entropy functional being optimized, and whether the agent may adapt later choices to earlier outcomes. Such optimization can be formulated over tester effects constrained by a process normalization, analogous to a POVM optimization over effects constrained by $\sum M_x = I$.

18.6.1 Why static entropic bounds do not solve an adaptive task

The Maassen–Uffink inequality bounds the sum of entropies for two fixed projective measurements applied to a common state. An adaptive process problem permits an intervention that changes the state before the second readout. Its outcome distributions therefore need not be those of two fixed observables on the original input state. Applying a static uncertainty inequality without accounting for intervention dynamics can be invalid. A correct process uncertainty statement uses the admissible operations and causal constraints explicitly.

18.7 Resource costs for process reconstruction

A channel on a d-dimensional input and d-dimensional output is described by a $d^2 \times d^2$ Choi matrix, subject to positivity and trace constraints. A multitime process tensor acts on tensor products of input-output legs at every time step, so the ambient operator dimension grows exponentially with the number of times if the local dimension is fixed and no structural restriction is imposed. Process tomography therefore becomes quickly expensive, motivating methods that exploit bounded memory, locality, approximate Markov structure, or low-rank representations. Any such reduction requires explicit assumptions; merely observing a few temporal correlators does not identify an arbitrary general process tensor.

CHAPTER A

Fully Worked State, Measurement, and Channel Calculations

A.1 A matrix-to-ket consistency calculation

Let $|\psi\rangle = (|0\rangle + (1+i)\,|1\rangle)/\sqrt{3}$. Its conjugate bra is $\langle\psi| = (\langle 0| + (1-i)\,\langle 1|)/\sqrt{3}$. Multiplying the column by its conjugate transpose gives

$$|\psi\rangle\langle\psi| = \frac{1}{3}\begin{pmatrix} 1 & 1-i \\ 1+i & 2 \end{pmatrix}. \tag{A.1}$$

The off-diagonal entries are conjugates, so the projector is Hermitian. The trace is $(1+2)/3 = 1$. Its determinant is

$$\det\rho = \frac{1}{9}\big[2 - (1-i)(1+i)\big] = 0, \tag{A.2}$$

so one eigenvalue is zero. Since the trace is one, the other eigenvalue is one; equivalently $\rho^2 = \rho$.

Example A.1 (Expectation of the Pauli operators)**.** For this state,

$$\langle X\rangle = 2\Re\rho_{01} = \frac{2}{3}, \quad \langle Y\rangle = -2\Im\rho_{01} = \frac{2}{3}, \quad \langle Z\rangle = \rho_{00} - \rho_{11} = -\frac{1}{3}. \tag{A.3}$$

The Bloch vector length is $\sqrt{4/9 + 4/9 + 1/9} = 1$, again confirming purity by a third independent calculation.

A.2 Conditional probability without division by a null event

Let X, Y have joint probability table

	$Y=0$	$Y=1$
$X=0$	$1/2$	$1/4$
$X=1$	$1/4$	0

. (A.4)

The marginals are $p_X = (3/4, 1/4)$ and $p_Y = (3/4, 1/4)$. Conditional probabilities are

$$p(Y|X=0) = (2/3, 1/3), \qquad p(Y|X=1) = (1, 0). \tag{A.5}$$

Thus

$$H(Y|X) = \frac{3}{4}h_2(1/3) + \frac{1}{4}(0). \tag{A.6}$$

The joint entropy may be computed as $H(X,Y) = -(1/2)\log(1/2) - 2(1/4)\log(1/4) = 3/2$. The chain rule predicts

$$H(X) + H(Y|X) = h_2(1/4) + \frac{3}{4}h_2(1/3) = \frac{3}{2}, \tag{A.7}$$

which can be verified numerically or by exact logarithmic cancellation.

A.3 Pauli correlators and two-qubit state reconstruction

Consider

$$\rho = \frac{1}{4}\left[I \otimes I + aZ \otimes I + bI \otimes Z + cZ \otimes Z\right]. \tag{A.8}$$

It is diagonal in the computational basis, with entries

$$p_{00} = (1 + a + b + c)/4, \qquad p_{01} = (1 + a - b - c)/4, \tag{A.9}$$

$$p_{10} = (1 - a + b - c)/4, \qquad p_{11} = (1 - a - b + c)/4. \tag{A.10}$$

The state is physical precisely when all four probabilities are nonnegative. The local reduced states are $\rho_A = (I + aZ)/2$ and $\rho_B = (I + bZ)/2$. The correlator c is not determined by a and b; it measures joint information not recoverable from the marginals.

Example A.2 (Correlated but separable state)**.** Set $a = b = 0$ and $c = 1$. Then $p_{00} = p_{11} = 1/2$ and the other two probabilities vanish. The result is $\rho = (|00\rangle\langle 00| + |11\rangle\langle 11|)/2$. Both marginals are maximally mixed and $\langle Z \otimes Z\rangle = 1$, yet the state is separable.

A.4 Spectral decomposition of a nonorthogonal mixture

For $\rho = (|0\rangle\langle 0| + |+\rangle\langle +|)/2$, the characteristic equation is

$$\det(\rho - \lambda I) = \lambda^2 - \lambda + \frac{1}{8} = 0, \tag{A.11}$$

so $\lambda_\pm = (2 \pm \sqrt{2})/4$. The normalized eigenvectors may be parameterized by $\theta = \pi/8$:

$$|e_+\rangle = \cos\theta\,|0\rangle + \sin\theta\,|1\rangle, \qquad |e_-\rangle = -\sin\theta\,|0\rangle + \cos\theta\,|1\rangle. \tag{A.12}$$

The angle arises from the symmetric 2×2 diagonalization formula

$$\tan(2\theta) = \frac{2\rho_{01}}{\rho_{00} - \rho_{11}} = \frac{1/2}{1/2} = 1, \tag{A.13}$$

so $2\theta = \pi/4$. The spectral decomposition is then

$$\rho = \lambda_+ |e_+\rangle\langle e_+| + \lambda_- |e_-\rangle\langle e_-|. \tag{A.14}$$

A purification uses orthogonal environment flags for the two eigenspaces. Its Schmidt coefficients are $\sqrt{\lambda_+}$ and $\sqrt{\lambda_-}$.

A.5 An exact binary Helstrom calculation

Consider $\rho_0 = \mathrm{diag}(3/4, 1/4)$, $\rho_1 = \mathrm{diag}(1/4, 3/4)$ and equal prior. Their difference is $\rho_0 - \rho_1 = \mathrm{diag}(1/2, -1/2)$. Its trace norm is one, and the optimal success is

$$P_{\mathrm{succ}} = \frac{1}{2} + \frac{1}{4}\,\|\rho_0 - \rho_1\|_1 = \frac{3}{4}. \tag{A.15}$$

The effect $E_0 = |0\rangle\langle 0|$ selects the positive eigenspace and $E_1 = |1\rangle\langle 1|$ selects the negative eigenspace. The measurement is the ordinary computational basis. If the priors become very asymmetric, the optimum may instead make the same decision for both outcomes; the Bayesian posterior calculation determines the transition thresholds.

A.6 A degenerate measurement computed in full

Let $M = X \otimes X$ and $|\psi\rangle = (|00\rangle + 2\,|01\rangle + 3\,|10\rangle + 4\,|11\rangle)/\sqrt{30}$. Since $M^2 = I$, the spectral effects are $P_\pm = (I \pm M)/2$. Acting with M reverses both computational bits:

$$M\,|\psi\rangle = \frac{4\,|00\rangle + 3\,|01\rangle + 2\,|10\rangle + |11\rangle}{\sqrt{30}}. \tag{A.16}$$

Therefore the unnormalized plus and minus branches are

$$P_+\,|\psi\rangle = \frac{5}{2\sqrt{30}}\big(|00\rangle + |01\rangle + |10\rangle + |11\rangle\big), \tag{A.17}$$

$$P_-\,|\psi\rangle = \frac{1}{2\sqrt{30}}\big(-3\,|00\rangle - |01\rangle + |10\rangle + 3\,|11\rangle\big). \tag{A.18}$$

Their squared norms are $5/6$ and $1/6$, and the normalized states are $|{+}{+}\rangle$ and $(-3\,|00\rangle - |01\rangle + |10\rangle + 3\,|11\rangle)/\sqrt{20}$. These outcomes are eigenstates of M with eigenvalues ± 1, which can be checked by applying $X \otimes X$ again.

A.7 An entanglement witness without tomography

Let $\Pi = |\Phi^+\rangle\langle\Phi^+|$ and $W = I/2 - \Pi$. Expanding the Bell projector in Pauli strings,

$$\Pi = \frac{1}{4}(I \otimes I + X \otimes X - Y \otimes Y + Z \otimes Z). \tag{A.19}$$

If experimental estimates are $\langle X \otimes X\rangle = 3/5$, $\langle Y \otimes Y\rangle = -3/5$, $\langle Z \otimes Z\rangle = 2/5$, then

$$\mathrm{Tr}(W\rho) = \frac{1}{2} - \frac{1}{4}\left(1 + \frac{3}{5} + \frac{3}{5} + \frac{2}{5}\right) = -\frac{3}{20} < 0. \tag{A.20}$$

This certifies entanglement under the assumption that the correlators refer to the same well-defined two-qubit state and are estimated accurately enough for the negative value to survive statistical uncertainty.

A.8 A completely positive map checked from its Kraus operators

Take $K_0 = \sqrt{1-p}I$ and $K_1 = \sqrt{p}X$. Their completeness is

$$K_0^\dagger K_0 + K_1^\dagger K_1 = (1-p)I + pX^\dagger X = I. \tag{A.21}$$

The resulting map is $\mathcal{N}_p(\rho) = \sum_j K_j \rho K_j^\dagger = (1-p)\rho + pX\rho X$. It is trace preserving by the cyclicity of trace. To check complete positivity, extend each Kraus operator by the identity on a reference R:

$$(\mathrm{id}_R \otimes \mathcal{N}_p)(R_{RA}) = \sum_j (I_R \otimes K_j) R_{RA} (I_R \otimes K_j^\dagger). \tag{A.22}$$

Every summand is positive whenever $R_{RA} \succeq 0$, so the full extended map preserves positivity. The same argument works for any Kraus family in arbitrary finite dimensions.

A.9 Choi rank versus environment size

The bit-flip Choi state is $(1-p)\left|\Phi^+\right\rangle\!\left\langle\Phi^+\right| + p\left|\Psi^+\right\rangle\!\left\langle\Psi^+\right|$, hence has rank two when $0 < p < 1$. Thus a minimal isometric dilation needs a two-dimensional environment. For $p = 0$ or $p = 1$, the channel is a unitary conjugation, its Choi state is pure, and a one-dimensional environment suffices. The minimal-environment statement is not a statement that a laboratory noise source always has exactly that many degrees of freedom; it is the minimal dilation dimension required by channel theory.

A.10 Complete depolarization and loss of reference entanglement

The channel $\mathcal{D}_1(\rho) = \mathrm{Tr}(\rho)I/2$ maps every normalized qubit input to $I/2$. If a reference R is entangled with the input A in $\left|\Phi^+\right\rangle_{RA}$, the extended output is

$$(\mathrm{id}_R \otimes \mathcal{D}_1)(\left|\Phi^+\right\rangle\!\left\langle\Phi^+\right|) = \frac{I_R}{2} \otimes \frac{I_B}{2}. \tag{A.23}$$

The final state has no correlations, whereas the initial reference-input state was pure and maximally entangled. This shows why correct action on the local maximally mixed marginal is not enough to verify an identity channel: both identity and complete depolarization map $I/2$ to $I/2$.

CHAPTER B

Fully Worked Algorithmic and Information-Theoretic Calculations

B.1 A Boolean Fourier transform by direct summation

Let $f : \{0,1\}^2 \to \{\pm 1\}$ be specified by $f(00) = f(01) = f(10) = 1$ and $f(11) = -1$. Its Fourier coefficient at $s \in \{0,1\}^2$ is

$$\widehat{f}(s) = \frac{1}{4} \sum_{x \in \{0,1\}^2} f(x)(-1)^{s \cdot x}. \tag{B.1}$$

Compute them individually:

$$\widehat{f}(00) = \frac{1}{4}(1 + 1 + 1 - 1) = \frac{1}{2}, \tag{B.2}$$

$$\widehat{f}(01) = \frac{1}{4}(1 - 1 + 1 + 1) = \frac{1}{2}, \tag{B.3}$$

$$\widehat{f}(10) = \frac{1}{4}(1 + 1 - 1 + 1) = \frac{1}{2}, \tag{B.4}$$

$$\widehat{f}(11) = \frac{1}{4}(1 - 1 - 1 - 1) = -\frac{1}{2}. \tag{B.5}$$

Parseval gives $\sum_s \widehat{f}(s)^2 = 4(1/4) = 1$, so the phase-oracle Fourier sampling circuit has normalized output

$$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle). \tag{B.6}$$

All four measurement outcomes occur with probability $1/4$, while the negative coefficient remains as a relative phase in the state vector.

B.2 A three-bit Simon instance solved by elimination

Let $s = (1,0,1)$ and suppose independent Fourier samples are $y^{(1)} = (0,1,0)$ and $y^{(2)} = (1,0,1)$. The equations are

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ s_3 \end{pmatrix} = 0 \pmod 2. \tag{B.7}$$

The first row gives $s_2 = 0$, the second gives $s_1 = s_3$. The solution space is $\{000, 101\}$. The promise $s \neq 000$ selects 101 uniquely. The algorithm does not require the integer matrix inverse over real numbers; all arithmetic is performed in the finite field $\mathbb{F}_2$.

B.3 Grover rotation with two marked items

For $N = 8$ and $M = 2$, let $|w\rangle$ and $|r\rangle$ be normalized marked and unmarked superpositions. The initial state is

$$|s\rangle = \frac{1}{2}|w\rangle + \frac{\sqrt{3}}{2}|r\rangle = \sin(\pi/6)\,|w\rangle + \cos(\pi/6)\,|r\rangle\,. \tag{B.8}$$

The oracle reflection sends $(\sin\theta, \cos\theta)$ to $(-\sin\theta, \cos\theta)$, and the diffusion reflection about $|s\rangle$ produces a net rotation by 2θ. After one iteration the marked amplitude is $\sin(3\pi/6) = 1$, so the search succeeds with certainty. After two iterations the marked amplitude is $\sin(5\pi/6) = 1/2$, so the success probability is only $1/4$. Thus success is oscillatory, and a query bound must specify a chosen iteration count rather than assuming that repeated amplification increases probability monotonically.

B.4 Exact F_4 from Fourier matrix entries

Using $\omega_4 = i$ and the computational basis ordered $|0\rangle, |1\rangle, |2\rangle, |3\rangle$,

$$F_4 = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}. \tag{B.9}$$

Column $x = 1$ is $F_4|1\rangle = (|0\rangle + i|1\rangle - |2\rangle - i|3\rangle)/2$. The column norm is

$$\frac{1}{4}(1 + |i|^2 + 1 + |-i|^2) = 1. \tag{B.10}$$

The inner product of columns $x = 0$ and $x = 1$ is $(1 + i - 1 - i)/4 = 0$, confirming orthogonality on this pair. All column inner products follow from the geometric-series orthogonality identity.

B.5 Exact phase estimation on three control qubits

Suppose $U|u\rangle = e^{2\pi i(5/8)}|u\rangle$. The controlled powers of U imprint phase $e^{2\pi i(5/8)k}$ on the control state $|k\rangle$, producing

$$\frac{1}{\sqrt{8}}\sum_{k=0}^{7} e^{2\pi i 5k/8}|k\rangle|u\rangle = F_8|5\rangle|u\rangle\,. \tag{B.11}$$

Applying $F_8^\dagger$ gives $|5\rangle|u\rangle$, so the measured bits are 101 in most-significant-first order. If the QFT implementation omits its final output bit reversals, the raw physical register bit order must be reinterpreted; the mathematical inverse $F_8^\dagger$ always gives integer five in the standard computational order.

B.6 A Shor-style order computation

Choose $N = 21$ and base $a = 2$. Compute successive residues modulo 21:

$$2^1 \equiv 2, \quad 2^2 \equiv 4, \quad 2^3 \equiv 8, \quad 2^4 \equiv 16, \quad 2^5 \equiv 11, \quad 2^6 \equiv 1 \pmod{21}. \tag{B.12}$$

Thus the multiplicative order is $r = 6$. It is even, and $2^{r/2} = 8 \not\equiv -1 \pmod{21}$. The factor extraction gives

$$\gcd(8-1, 21) = 7, \qquad \gcd(8+1, 21) = 3. \tag{B.13}$$

A phase estimation trial may return a rational phase with denominator three rather than six; verification of the candidate order and repetition resolve such failures. The quantum circuit targets period information, while the gcd calculation produces factors classically.

B.7 The Hadamard test by complete state propagation

Let $|\psi\rangle$ be normalized and U unitary. Starting in $|0\rangle|\psi\rangle$, the first Hadamard produces

$$\frac{|0\rangle|\psi\rangle + |1\rangle|\psi\rangle}{\sqrt{2}}. \tag{B.14}$$

Controlled-U gives $(|0\rangle|\psi\rangle + |1\rangle U|\psi\rangle)/\sqrt{2}$. The final Hadamard produces

$$\frac{1}{2}|0\rangle(I+U)|\psi\rangle + \frac{1}{2}|1\rangle(I-U)|\psi\rangle. \tag{B.15}$$

Taking squared norms,

$$p(0) = \frac{1}{4}\langle\psi|(I+U^\dagger)(I+U)|\psi\rangle \tag{B.16}$$

$$= \frac{1}{4}\langle\psi|(2I+U+U^\dagger)|\psi\rangle \tag{B.17}$$

$$= \frac{1}{2}[1 + \Re\langle\psi|U|\psi\rangle]. \tag{B.18}$$

Similarly $p(1) = [1 - \Re\langle\psi|U|\psi\rangle]/2$. The bias is $p(0) - p(1) = \Re\langle\psi|U|\psi\rangle$.

B.8 The SWAP test for mixed states

Let S swap two equal-dimensional registers. For density operators ρ and σ, the controlled-SWAP interferometer yields

$$p(0) = \frac{1}{2}[1 + \mathrm{Tr}(S\rho\otimes\sigma)]. \tag{B.19}$$

Expand in matrix units to prove the swap trick:

$$\operatorname{Tr}[S(A \otimes B)] = \sum_{i,j} \langle ij| S(A \otimes B) |ij\rangle \tag{B.20}$$

$$= \sum_{i,j} \langle ji| (A \otimes B) |ij\rangle \tag{B.21}$$

$$= \sum_{i,j} \langle j| A |i\rangle \langle i| B |j\rangle = \operatorname{Tr}(AB). \tag{B.22}$$

Hence $p(0) = [1 + \operatorname{Tr}(\rho\sigma)]/2$. For $\rho = \sigma = I_d/d$, the overlap is $1/d$ and $p(0) = (1 + 1/d)/2$. A pair of identical maximally mixed states is therefore not accepted with certainty by the same criterion as two identical pure states.

B.9 A classical erasure channel computed from a distribution

Let X be binary with $\mathbb{P}(X = 0) = p$. For BEC(ϵ), the output distribution is

$$p_Y = ((1-\epsilon)p, (1-\epsilon)(1-p), \epsilon). \tag{B.23}$$

Its entropy is

$$H(Y) = h_2(\epsilon) + (1-\epsilon)h_2(p), \tag{B.24}$$

while $H(Y|X) = h_2(\epsilon)$. Subtracting gives

$$I(X:Y) = (1-\epsilon)h_2(p) \leq 1 - \epsilon, \tag{B.25}$$

with equality when $p = 1/2$. This derivation displays every conditional and marginal term, rather than quoting the channel capacity.

B.10 The Holevo quantity for a two-state ensemble

Take $\mathcal{E} = \{(1/2, |0\rangle\langle 0|), (1/2, |+\rangle\langle +|)\}$. The average state is

$$\bar{\rho} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}, \tag{B.26}$$

with eigenvalues $\lambda_\pm = (2 \pm \sqrt{2})/4$. As each letter state is pure,

$$\chi(\mathcal{E}) = S(\bar{\rho}) - \frac{1}{2}S(\rho_0) - \frac{1}{2}S(\rho_1) = h_2(\lambda_+). \tag{B.27}$$

The quantity is less than one bit. It bounds the mutual information accessible to any one-shot measurement on the ensemble's state. For a quantum channel, asymptotic coding capacity requires an optimization over ensembles and, in general, regularization over multiple uses.

CHAPTER C

Fully Worked Continuous-Variable and Uncertainty Calculations

C.1 Vacuum quadrature variance from ladder operators

With $[a, a^\dagger] = I$, define $x = a + a^\dagger$, $p = (a - a^\dagger)/i$, and $a\,|0\rangle = 0$. The vacuum mean is $\langle x \rangle = 0$. Expanding the square gives

$$x^2 = a^2 + (a^\dagger)^2 + aa^\dagger + a^\dagger a. \tag{C.1}$$

All vacuum expectations vanish except $\langle 0|aa^\dagger|0\rangle = \langle 0|(I + a^\dagger a)|0\rangle = 1$. Therefore $\mathrm{Var}_0(x) = 1$. The same argument gives $\mathrm{Var}_0(p) = 1$, and

$$[x, p] = \frac{1}{i}[a + a^\dagger, a - a^\dagger] = 2iI. \tag{C.2}$$

The uncertainty product is $\Delta x \Delta p = 1$, saturating the Robertson bound.

C.2 Squeezing and the decibel convention

If a squeezed vacuum has $V_x = e^{-2r}$ and $V_p = e^{2r}$, then $V_x V_p = 1$. The relative squeezing level in decibels is $L_x = 10 \log_{10} V_x$. Setting $L_x = -6$ dB gives

$$V_x = 10^{-0.6} \approx 0.2512, \qquad V_p = 1/V_x \approx 3.9811. \tag{C.3}$$

The standard deviation in the squeezed quadrature is $\sqrt{V_x} \approx 0.5012$, approximately one half the vacuum value. Thus reducing standard deviation by a factor of two corresponds approximately to a six-decibel *variance* suppression, not three decibels.

C.3 A thermal covariance spectrum

Consider the isotropic covariance $V = 3I_2$. Its symplectic eigenvalue is $\nu = \sqrt{\det V} = 3$. The equivalent thermal mean occupation is $\bar{n} = (\nu - 1)/2 = 1$. The von Neumann entropy is

$$S = g(1) = 2 \log 2 - 1 \log 1 = 2 \text{ bits}. \tag{C.4}$$

Its Gaussian purity is $1/\nu = 1/3$. In the Fock basis, the thermal eigenvalues for $\bar{n} = 1$ are $p_n = 2^{-(n+1)}$, and their Shannon entropy is

$$-\sum_{n=0}^{\infty} 2^{-(n+1)} \log 2^{-(n+1)} = \sum_{n=0}^{\infty} \frac{n+1}{2^{n+1}} = 2, \tag{C.5}$$

confirming the Gaussian entropy calculation independently.

C.4 A pure-loss transformation with finite thermal environment

Let the input quadrature covariance be V_{in} and the environment be thermal with covariance $V_E = (2\bar{n}_E + 1)I$. A beam splitter of transmissivity η gives the output covariance

$$V_{\text{out}} = \eta V_{\text{in}} + (1 - \eta)V_E. \tag{C.6}$$

For a coherent input $V_{\text{in}} = I$ this becomes

$$V_{\text{out}} = [1 + 2(1 - \eta)\bar{n}_E]\, I. \tag{C.7}$$

Vacuum environment ($\bar{n}_E = 0$) leaves the coherent state's variance one; a thermal environment adds noise even when the signal was initially coherent.

Example C.1 (Half transmission through a thermal bath)**.** For $\eta = 1/2$ and $\bar{n}_E = 2$, one obtains $V_{\text{out}} = [1 + 2(1/2)2]I = 3I$. The output is a displaced thermal state of mean occupation one, even though the input was a coherent state. Its entropy is two bits under the Gaussian convention.

C.5 Noise accounting for CV teleportation

Assume unit-gain continuous-variable teleportation with ideal EPR covariance. The output quadratures obey

$$x_{\text{out}} = x_{\text{in}} + n_x, \quad p_{\text{out}} = p_{\text{in}} + n_p, \qquad \text{Var}(n_x) = \text{Var}(n_p) = 2e^{-2r}. \tag{C.8}$$

For a coherent input $V_{\text{in}} = I$, the output covariance is $(1 + 2e^{-2r})I$. At $r = 0$, the resource is two independent vacuum modes and the added noise is two vacuum units per quadrature, giving output covariance $3I$. As $r \to \infty$, the noise approaches zero. The teleported mean displacement equals the input mean only because the feedback gain is assumed to be one; different gain choices yield different mean and noise transformations.

C.6 Gaussian mutual information from conditional variances

Let $X \sim \mathcal{N}(0, V_A)$, $Z \sim \mathcal{N}(0, V_Z)$ independently, and $Y = \sqrt{\eta}X + Z$. Then Y is Gaussian with variance $V_Y = \eta V_A + V_Z$, whereas $Y|X$ has variance V_Z. Using the differential entropy of a Gaussian,

$$h(Y) = \frac{1}{2}\log(2\pi e V_Y), \quad h(Y|X) = \frac{1}{2}\log(2\pi e V_Z), \tag{C.9}$$

one obtains

$$I(X : Y) = h(Y) - h(Y|X) = \frac{1}{2}\log\left(1 + \frac{\eta V_A}{V_Z}\right). \tag{C.10}$$

This formula is in bits because the logarithm is base two. It describes a classical Gaussian observation channel; in quantum key distribution the adversary's information is an additional distinct quantity.

C.7 Robertson–Schrödinger from covariance positivity

Let $\widetilde{A} = A - \langle A\rangle I$, $\widetilde{B} = B - \langle B\rangle I$. Define the 2×2 Gram matrix

$$G = \begin{pmatrix} \langle \widetilde{A}^2\rangle & \langle \widetilde{A}\widetilde{B}\rangle \\ \langle \widetilde{B}\widetilde{A}\rangle & \langle \widetilde{B}^2\rangle \end{pmatrix}. \tag{C.11}$$

For every $(u, v) \in \mathbb{C}^2$, positivity of $\langle (u\widetilde{A} + v\widetilde{B})^\dagger (u\widetilde{A} + v\widetilde{B})\rangle$ implies $G \succeq 0$. Therefore $\det G \geq 0$. Since

$$\langle \widetilde{A}\widetilde{B}\rangle = \mathrm{Cov}(A, B) + \frac{1}{2}\langle [A, B]\rangle, \tag{C.12}$$

with the covariance real and commutator contribution purely imaginary,

$$\mathrm{Var}\, A\, \mathrm{Var}\, B \geq \mathrm{Cov}(A, B)^2 + \frac{1}{4}\left|\langle [A, B]\rangle\right|^2. \tag{C.13}$$

This proof reveals the common algebraic source of both the covariance and commutator terms.

C.8 A spin-j variance bound from the Casimir identity

For an irreducible spin-j representation,

$$\sum_{k=x,y,z} J_k^2 = j(j+1)I. \tag{C.14}$$

Subtracting squared expectation values gives

$$\sum_k \mathrm{Var}(J_k) = j(j+1) - \left\|\langle \boldsymbol{J}\rangle\right\|^2. \tag{C.15}$$

Because every component along a unit direction has eigenvalues in $[-j, j]$, the mean-spin vector has length at most j. It follows that

$$\sum_k \mathrm{Var}(J_k) \geq j(j+1) - j^2 = j. \tag{C.16}$$

Spin-coherent states polarized along a direction have mean-spin length j and saturate this bound. A maximally mixed spin state has zero mean spin and total variance $j(j+1)$.

CHAPTER D

Notation and Conventions

Symbol	Meaning
$\mathcal{H}_A$, $\mathcal{L}(A)$, $\mathcal{D}(A)$	Hilbert space, operator space, density operators on A
$\|\psi\rangle$, $\langle\psi\|$, $\|\psi\rangle\langle\phi\|$	Dirac notation for vectors, duals, and rank-one operators
Tr_B	Partial trace over subsystem B
$\|M\|_1$, $\|M\|_\infty$	Trace norm and operator norm
$S(A)$, $H(X)$	Von Neumann entropy and classical Shannon entropy
$I(A:B)$, $I(A:C \mid B)$	Mutual information and conditional mutual information
$D(\rho\|\sigma)$	Quantum relative entropy; distinct from trace distance $T(\rho,\sigma)$
χ, C, C_E, Q	Holevo information; unassisted/assisted classical and quantum capacities
$J(\mathcal{N})$	*Normalized* Choi operator, of trace one for channels
F_N	Positive-phase QFT, $F_N \|x\rangle = N^{-1/2}\sum_y e^{2\pi i xy/N}\|y\rangle$
$[x,p] = 2i$	Continuous-variable chapter convention; vacuum covariance I_2

Logical qualifications

Unless otherwise stated, Hilbert spaces in finite-register chapters are finite-dimensional; logs for information measures are base two; zero-probability entropy terms are zero; and asymptotic statements presume explicitly specified access and accuracy models. Infinite-dimensional Gaussian systems are treated separately and require physical moment and energy constraints for operational statements.